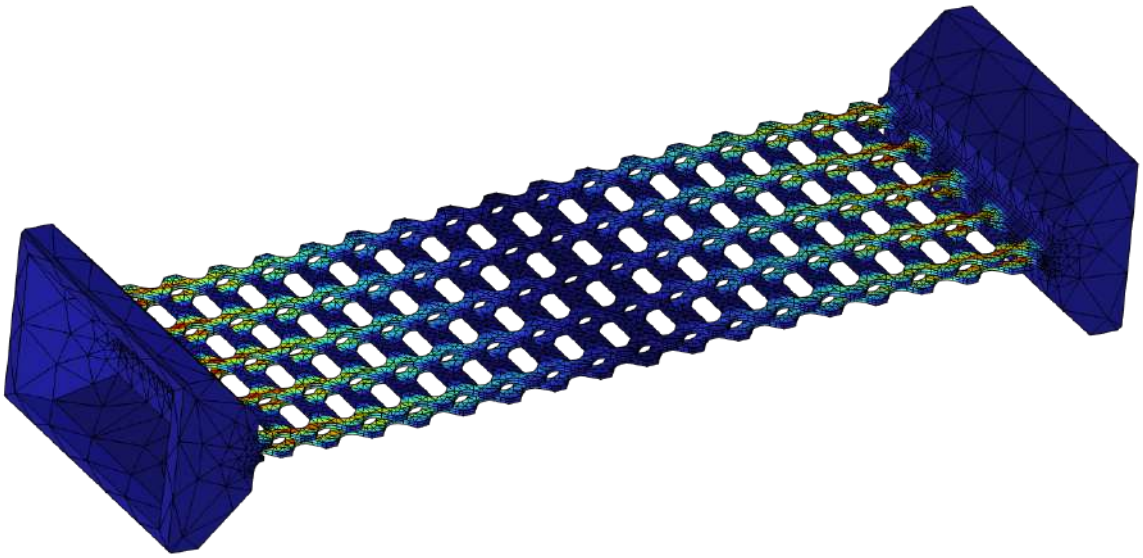
EXPERIMENTAL INVESTIGATION AND NUMERICAL VALIDATION OF ADDITIVELY MANUFACTURED DAMPED FLEXURES



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A Papà...

When you grow up you tend to get
told that the world is the way it is
and your life is just to live your life
inside the world. Try not to bash into
the walls too much. Try to have a nice
family, have fun, save a little money.
That's a very limited life. Life can be
much broader once you discover one
simple fact: everything around you
that you call life was made up by
people that were no smarter than you
and you can change it, you can
influence it... Once you learn that,
you'll never be the same again.

- Steve Jobs

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List of Symbols

Abbreviation	Meaning
AM	Additive Manufacturing
FEA	Finite Elements Analysis
FEM	Finite Elements Method
SLM	Selective Laser Melting
LPDM	Laser Powder Bed Fusion
DMA	Dynamic Mechanical Analysis
PTFE	PolyTetraFluoroEthylene
PDMS	PolyDiMethylSiloxane
LVR	Linear Viscoelastic Region
CMF-C	Chemical Mechanical Finishing - Carbon
DC	Direct Current
PSD	Power Spectral Density
SDOF	Single Degree of Freedom
FRF	Frequency Response Function
FFT	Fast Fourier Transform
SNR	Signal-to-Noise Ratio
RMSE	Root Mean Square Error
DAQ	Data Acquisition System

Symbol	Meaning	Unit
E	Young's Modulus	$[MPa]$
E'	Storage (Elastic) Modulus	$[MPa]$
E''	Loss (Viscous) Modulus	$[MPa]$
G	Shear Modulus	$[MPa]$
G'	Shear Storage Modulus	$[MPa]$
G''	Shear Loss Modulus	$[MPa]$
ν	Poisson Ratio	$[]$
$\tan\delta$	Loss Factor	$[]$
K	Stiffness	$[N/m]$
b	Blade's Width	$[m]$
t	Blade's Thickness	$[m]$
l	Blade's Length	$[m]$
s	Stroke	$[m]$
σ_{adm}	Admissible Stress	$[MPa]$
m	Mass	$[kg]$
f_n	Natural Frequency	$[Hz]$
f_s	Sampling Frequency	$[Hz]$
ω	Angular Frequency	$[rad/s]$
Q	Quality Factor	$[]$
ζ	Damping Ratio	$[]$
τ	Relaxation Time	$[s]$

Abstract

Flexure-based compliant mechanisms have become state-of-the-art solutions for designing high-precision systems that are free from wear, backlash, and lubrication. However, these benefits often come with the drawback of very high quality factors, which can lead to adverse vibrations during highly dynamic motions. In space applications, this can lead to damage during launch or to degraded performance in operation. To address these issues, a novel passive damping concept is explored in which titanium lattice blades produced by Laser Powder Bed Fusion (LPBF) are impregnated with a space-qualified silicone elastomer, enabling energy dissipation through shear deformation of the elastomer. Material characterization via Dynamic Mechanical Analysis reveals frequency-dependent viscoelastic behavior and significant aging effects, with stiffness and loss factor evolving over several months after curing. Free oscillation experiments on multiple blade pairs demonstrate highly repeatable modal responses, with a consistent linear viscous damping mechanism confirmed by both half-power bandwidth and time-domain logarithmic-decrement methods. Forced oscillation tests further validate the damping performance across a broad excitation bandwidth up to 750 Hz. A fully three-dimensional finite element model incorporating a generalized Maxwell viscoelastic material law is developed to replicate the experimental setup and capture the frequency-dependent behavior of the elastomer. Correlation between FEM predictions and measurements shows strong agreement in resonance frequency and damping trends, confirming the model's suitability for predictive design. Overall, this work establishes the first integrated framework combining experimental results and validated FEM simulations for damped flexures. The results demonstrate that elastomer-impregnated lattice blades are a viable and tunable solution for applications relying on compliant mechanisms such as isolation structures or precision positioning mechanisms. The design freedom provided by additive manufacturing allows the main structural characteristics (stiffness and damping) to be tuned according to the specifications of such applications.

§ 1 Introduction

Compliant mechanisms have gained considerable attention in recent years as an alternative to conventional jointed systems. Unlike mechanisms based on rolling or sliding contacts, they rely on the elastic deformation of slender members to transmit motion and force. This structural principle eliminates friction and wear, while offering high repeatability and long-term stability. For applications where reliability and cleanliness are paramount, such as aerospace instrumentation or precision engineering, compliant mechanisms are therefore particularly attractive.

A factor that has further accelerated the adoption of compliant mechanisms is the development of additive manufacturing (AM). By fabricating parts layer by layer, AM enables the realization of complex geometries that would be extremely difficult, if not impossible, to machine conventionally. This freedom is particularly advantageous for flexure-based designs, where integrated features, reduced assembly steps, and material-efficient structures can be achieved. In the present work, Selective Laser Melting (SLM) is used to manufacture the flexure blades. SLM belongs to the powder bed fusion family of AM processes and produces fully dense metallic parts by selectively melting fine powder layers with a high-power laser. Compared to polymer-based AM techniques, SLM offers the possibility to produce structural components with the mechanical strength, thermal stability, and durability required for demanding applications such as space mechanisms. It also allows lattice-like designs with internal voids or channels, which are ideally suited for advanced flexure concepts.

While compliant mechanisms offer many advantages, they remain sensitive to external disturbances. In high-precision environments, vibrations and shocks can degrade performance, introduce unwanted displacements, or shorten operational lifetimes. The problem becomes more critical when internal modes of compliant elements interact with external excitations, as these modes are generally not controllable through active means. To mitigate such effects, passive damping strategies are required. A promising concept, recently explored at CSEM, is to enhance flexure blades with embedded viscoelastic materials (Figure 1.1). Instead of using plain flexures, lattice structures produced by SLM can be impregnated with elastomers in liquid form, which subsequently cure and fill the voids between blade elements. By impregnating lattice flexures with elastomers, energy dissipation occurs through shear deformation of the polymer, providing additional damping capacity without compromising the mechanism's compliance.

Initial prototypes demonstrated that this concept is feasible, and preliminary tests revealed measurable damping effects. However, these first demonstrations were primarily exploratory. The underlying damping mechanisms require deeper investigation to move beyond empirical observations. In particular, the lack of correlation between finite element simulations and experimental data limits the use of numerical models as a reliable design aid. Moreover, while a polyurethane elastomer was used for the proof of concept, the selection of a suitable viscoelastic material for long-term and space-application-specific performance has yet to be systematically addressed.

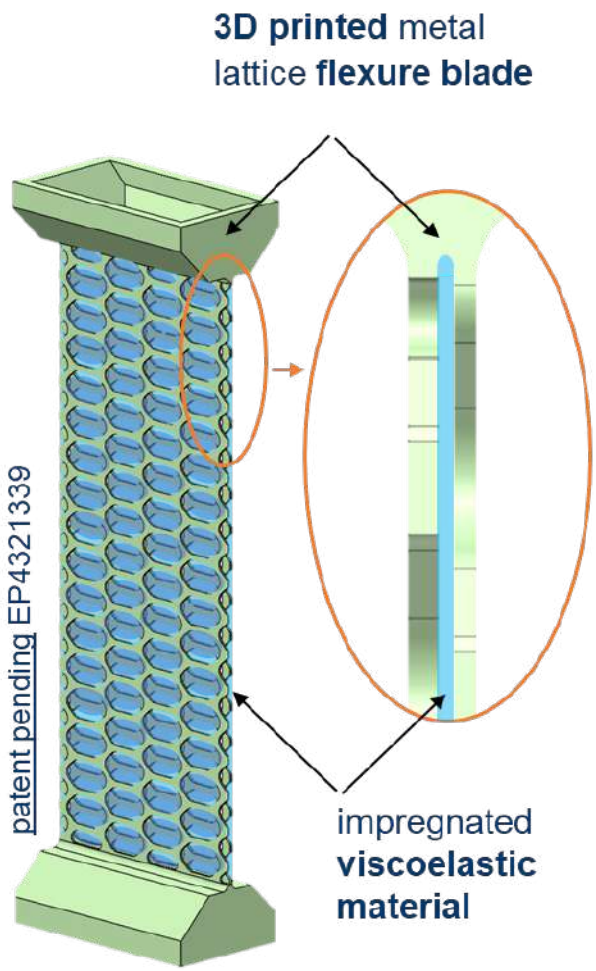

Figure 1.1: 3D printed metal lattice flexure blade impregnated with viscoelastic material

The present Master's thesis seeks to advance this concept from a proof of feasibility toward a validated engineering solution. The work combines theoretical modeling, numerical simulation and experimental validation in order to provide a comprehensive understanding of damped flexures. Particular emphasis is placed on the correlation of finite element simulations with measured data to enable predictive design and the commissioning of an experimental test bench dedicated to characterizing vibration and damping properties.

By uniting modeling, design optimization and systematic experimental validation, this thesis aims to provide the first integrated framework for the development of damped flexures produced by additive manufacturing. Beyond demonstrating their feasibility, the results are expected to establish practical guidelines and predictive tools that enable the adoption of such components in precision mechanisms, with a particular relevance for vibration and shock isolation in aerospace and other high-performance domains.

§ 2 Material Selection

This chapter presents the material selection process adopted for the development of the lattice flexures. Both the structural material of the blades and the viscoelastic filling material play a key role in the overall dynamic behavior of the system, influencing stiffness, damping capacity, manufacturability, and long-term stability.

2.1 *Blades*

In the previous project [15], stainless steel 17-4PH was used for the lattice blades. However, in several projects, this material proved problematic, as it tends to deform during heat treatment, not only during stress relief but also during hardening. Consequently, stainless steel 316L was initially considered as an alternative. Discussions with the additive manufacturing provider, AM Solutions, revealed that they were more confident working with titanium. They print titanium on a smaller machine (Trumpf TruPrint 1000), which offers higher manufacturing precision. Additionally, the lower elastic modulus of titanium allows for thicker blades without significantly increasing the natural frequency. Table 2.1 summarizes the criteria used to select the lattice blade material.

Table 2.1: Summary of trade-offs between lattice blade materials

Material	**17-4PH**	**316L**	**Ti–6Al–4V (grade 23)**
Elastic modulus (AM) [GPa]	165	190	110
Risk of distortion during heat treatment	High	Low	Low
AM precision @ AM Solutions	Medium	Medium	High
CMF-C post-treatment	Aggressive	Aggressive	Less aggressive

2.2 *Elastomer*

In the previous project [15], the selected viscoelastic material was not compatible with space applications. For this reason, MAPSIL® QS 1123 was chosen in the present work as the material to be impregnated between the blades, as it effectively satisfies the required constraints [6]. This low-outgassing silicone elastomer was specifically developed for the space industry, where it is used to support thermal management between adjacent components [16].

2.2.1 Impregnation

The process flow for impregnation was based on the preliminary trials of the previous project [15] and adapted to the specifications of the silicone. It consisted in the following three main steps:

- Surface preparation of the part
- Casting
- Curing

For surface preparation, the parts were first cleaned using isopropanol and ultrasonication. Since silicone generally exhibits poor adhesion to most surfaces, a primer supplied by MAP-Space Coating was applied in the next step. The primed surface was then used within a few hours. The casting process began with the preparation of MAPSIL QS 1123,

which involved mixing it with an hardener and degassing the mixture. For the first batch of blades, a custom PTFE (Teflon) support was fabricated to hold the blades in place and produce test specimens for DMA measurements (Figure 2.1). The silicone was manually dispensed, and a glass cover was placed on top. Curing was carried out by leaving the parts overnight at room temperature, followed by 24 hours at 40°C.

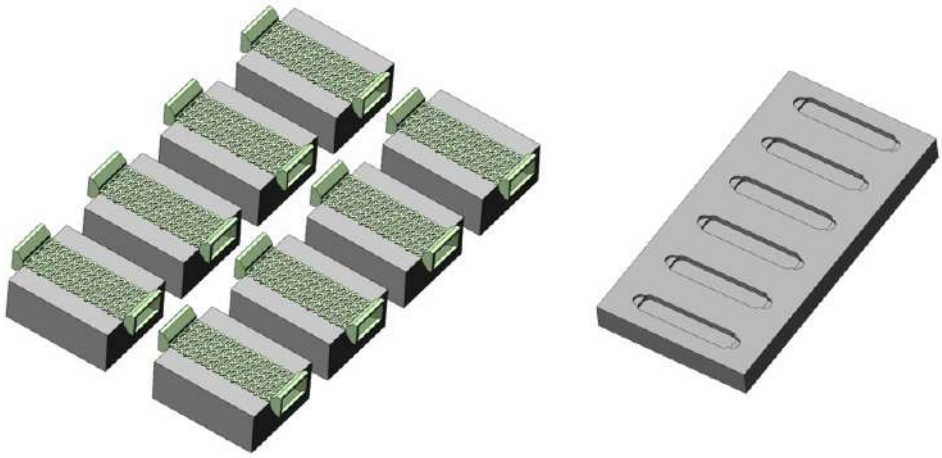

Figure 2.1: Teflon holder used for the casting step (left) and mold for the DMA specimens (right)

Figure 2.2 shows the resulting impregnated parts. While the silicone successfully filled all cavities, some parts exhibited trapped bubbles. This issue was traced to the application of the glass slide: uneven silicone layers on the parts caused air pockets to form.

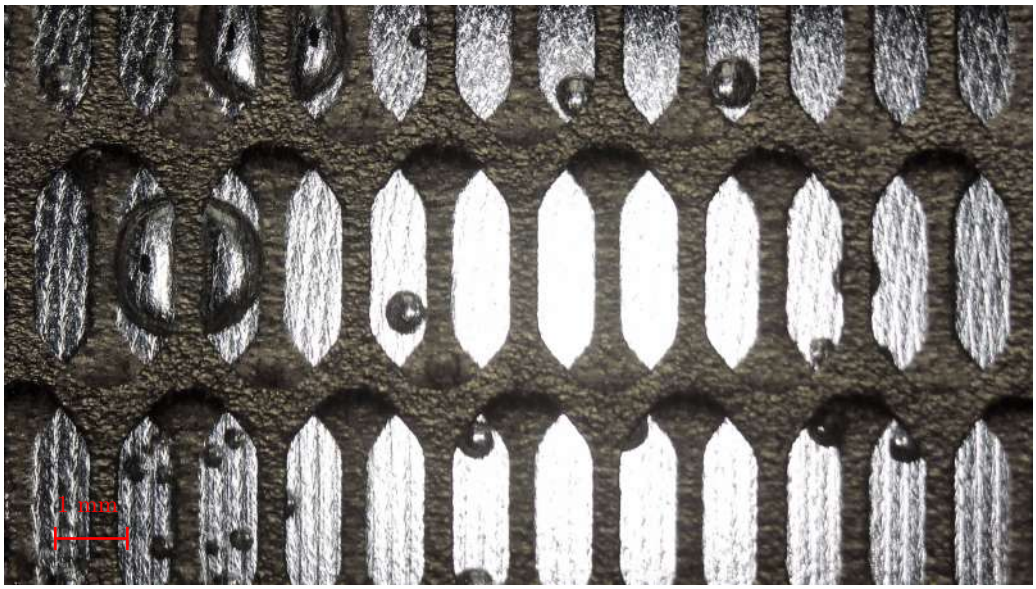

Figure 2.2: Bubbles in the first impregnation parts

To address this, a revised Teflon holder was designed (Figure 2.3) to improve control over the impregnation process. The updated design encloses the blade within a cavity, allowing silicone to be injected through a dedicated port. These preliminary trials also revealed challenges during demoulding. Despite not using a primer on the glass, the

silicone adhered strongly, occasionally causing cohesive rupture in some cells. To facilitate removal, the samples were heated to 80 °C, yielding two successfully impregnated blades.

Figure 2.3: New Teflon holder with polycarbonate glass

The redesigned Teflon holder was developed to improve the hermeticity of the impregnation system. In the previous configuration, direct application of silicone increased the risk of air entrapment, which in turn led to bubble formation within the cavities. To address this limitation, a dedicated inlet port was integrated into the new design, allowing silicone to be injected in a controlled manner without contact with the surrounding air. In addition, two outlet ports were incorporated to provide escape paths for residual air and trapped bubbles during filling. This configuration ensures a more uniform impregnation, reduces the likelihood of defects, and enhances the overall reproducibility of the process (Figure 2.4).

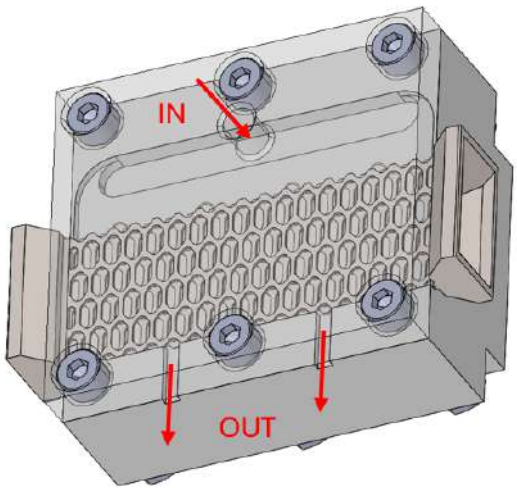

Figure 2.4: Inlet/Outlet ports of the Teflon holder

Apart from the modifications to the holder design, the impregnation process itself remained unchanged. The results obtained with the new configuration are significantly more convincing than those of the preliminary batch. No bubbles were observed at the center of the blades, although some were still present near the edges (Figure 2.5). However, an inspection performed with a Microqubic microscope revealed the presence of defects (Figure 2.6).

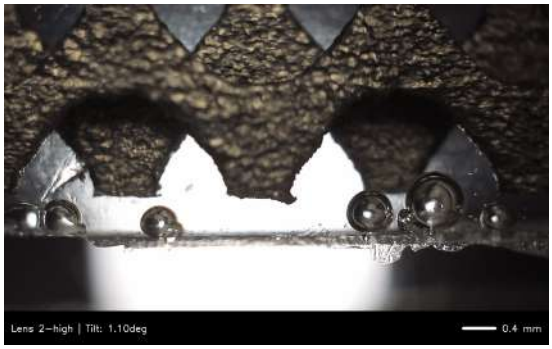

Figure 2.5: Sides Bubbles

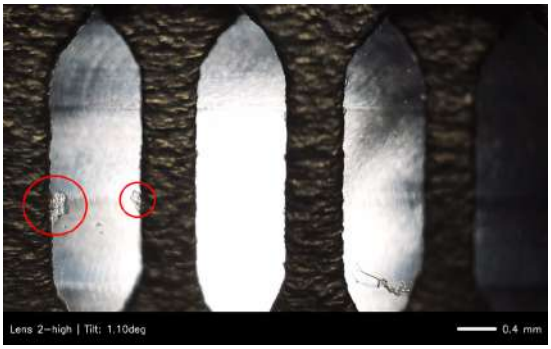

Figure 2.6: Internal Defects

Interestingly, the first batch of blades, which underwent the same inspection process, did not exhibit defects of this type. This observation suggests that the newly observed defects are not inherent to the material, design, or general process, but rather are likely linked to variations in the processing conditions. Among the potential contributing factors, the use of expired primer could have altered the chemical behavior during curing, promoting undesired reactions with the PDMS. In addition, deviations from the recommended environmental conditions, such as temperature fluctuations or excessive humidity during the impregnation and curing steps, may have further influenced the chemical interactions and the curing kinetics, increasing the likelihood of defect formation. It is therefore plausible that a combination of these factors, rather than a single cause, led to the observed defects.

A second batch of impregnated blades was produced following the same procedure used for the first batch. The overall setup and process parameters remained unchanged, with the only modification being the addition of side tape on the Teflon holder to limit bubble formation along the edges. Nevertheless, upon inspection, the same defects identified in the first batch were once again present. Their recurrence indicates that the issues initially suspected, such as expired primer or inadequate temperature and humidity control, were not responsible for their formation. Since neither the environmental conditions nor the primer condition nor the improved sealing strategy influenced the outcome, the defects must originate from another, yet unidentified, aspect of the process. Further investigation will therefore be needed to determine the underlying cause, potentially involving a more detailed assessment of the primer–PDMS interaction, the curing protocol, or the surface properties of the components involved.

A third batch of blades was impregnated under vacuum to further reduce bubble formation and improve the uniformity of the silicone filling. Overall, the results were better than those obtained in the previous batches, with no bubbles and a more consistent impregnation across the cavities. However, the same defects observed in earlier batches were still present, indicating that the root cause remains unrelated to bubble formation or en-

vironmental conditions. The vacuum process also introduced additional challenges: due to the necessary connections through pipes, there was increased leakage of PDMS during impregnation, making the procedure more complex and costly. Despite these drawbacks, the vacuum setup demonstrates potential for improving impregnation quality, although further optimization and investigation are required to fully address the persistent defects.

2.2.2 DMA Measurements

The Teflon holder depicted in Figure 2.1 (right) was specifically designed to manufacture viscoelastic specimens with controlled geometry and uniform thickness for subsequent Dynamic Mechanical Analysis (DMA) testing. Its purpose was to ensure repeatable sample dimensions and surface quality during the curing process, while avoiding adhesion between the silicone material and the mold. Rectangular samples measuring 18 mm × 5 mm × 2 mm were prepared, with dimensions selected to provide sufficient length for accurate strain measurement while maintaining a homogeneous stress state and limiting edge or bending effects. DMA measurements were subsequently carried out at EPFL–LPAC1 using a Q800 apparatus from TA Instruments.

Frequency sweep tests were performed to measure the storage and loss moduli of the material over 14 frequencies ranging from 0.1 Hz to 30 Hz on a logarithmic scale. The applied force ranged from 0.05 N to 0.1 N depending on frequency, chosen to generate measurable responses while keeping the material within the linear viscoelastic regime. To investigate potential nonlinearity, these frequency sweeps were conducted at three strain levels: 0.5%, 1%, and 2%, allowing comparison of the viscoelastic response under different deformation amplitudes. All tests were performed at ambient temperature (20 °C) to ensure reproducibility and relevance to typical operating conditions.

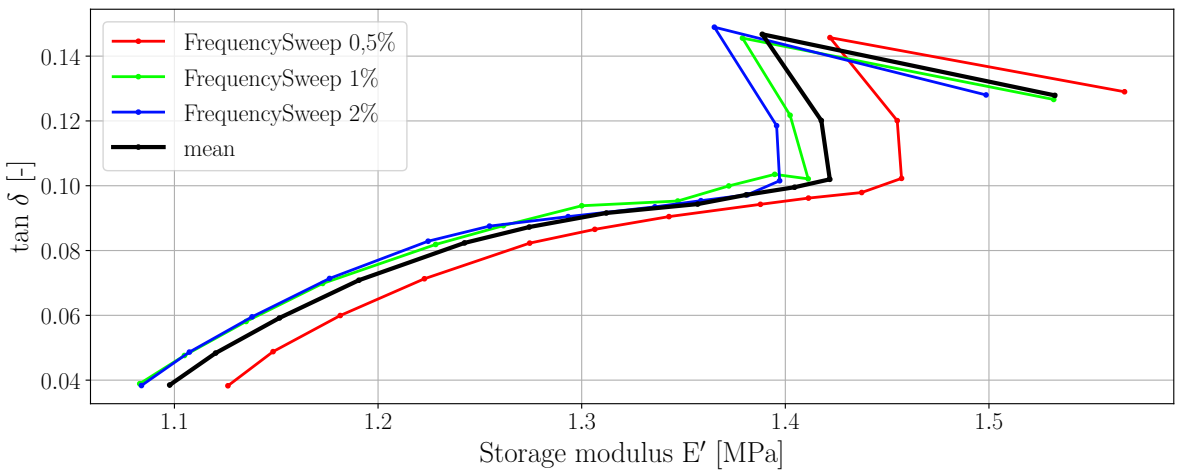

Figure 2.7: Frequency Sweep Test: Variation of tan δ with respect to the Storage Modulus

In a DMA frequency sweep at fixed strain amplitude the general behavior should be that as frequency increases, E′ often rises (material behaves more like an elastic solid at higher deformation rates). Tan δ often peaks around a transition region (like glass transition, secondary relaxations, or other dissipative processes). At very high frequencies, tan δ can drop because viscous contributions diminish. So, in the Linear Viscoelastic Region (low strain, small deformations), smooth monotonic curves without sudden jumps

are expected. Possible causes for the jump after 1.4 MPa could be:

1. **Molecular relaxation**: a β-relaxation (secondary molecular mobility) where damping increases sharply, followed by stiffening as the molecular segments "freeze" again.
2. **Instrument or data processing artifact**: In DMA, phase angle (δ) is derived from force and displacement. Near very low damping, a small error in δ can make tan δ swing unnaturally.
3. **Strain-dependent nonlinearity**: At higher moduli, the material might be in a non-linear viscoelastic regime, especially at 2% strain. This can cause E′ and E″ (and therefore tan δ) to change abruptly with frequency if the material's microstructure is rearranging.

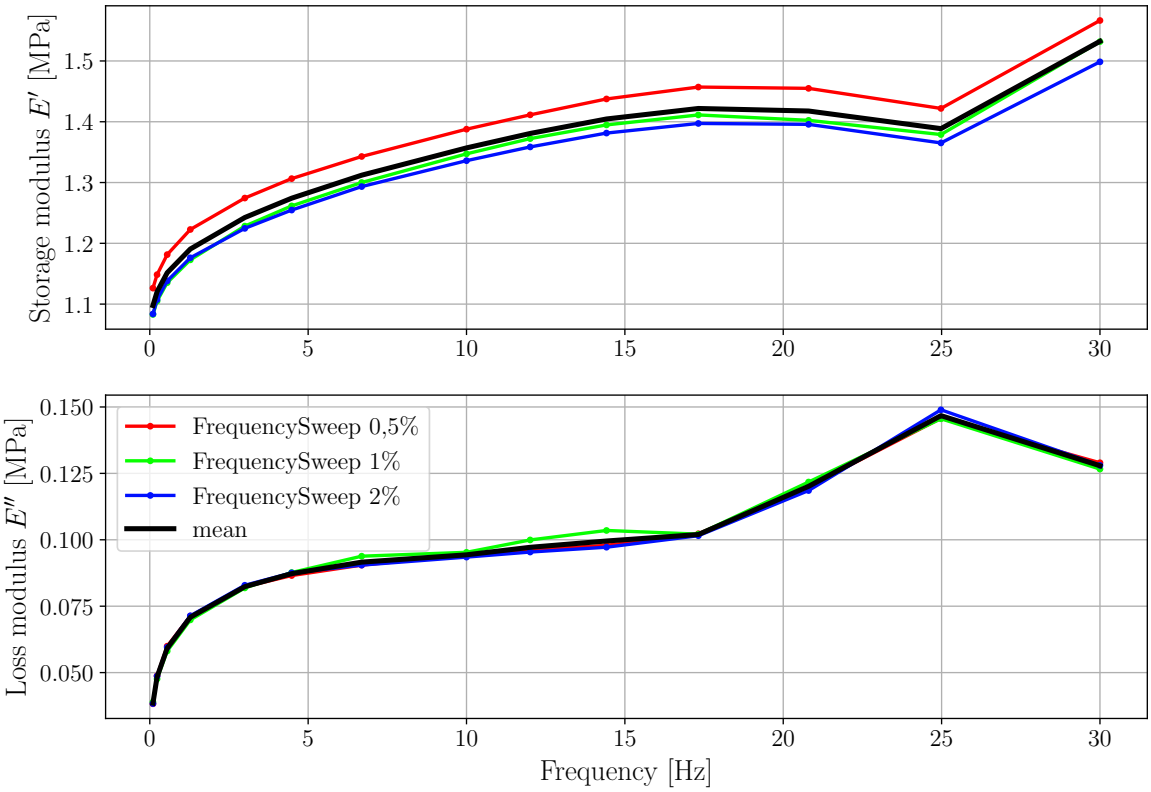

Figure 2.8: Frequency Sweep Test: Variation of Storage and Loss Moduli with respect to the Frequency

This second plot makes the "weird turn" more explainable:

- E′ (top): climbs quickly at low frequency, then flattens; around 20–25 Hz there's a small dip; by 30 Hz it rises again.
- E″ (bottom): increases with f and shows a clear peak at 25 Hz, then falls slightly at 30 Hz.
- The three strain levels (0.5%, 1%, 2%) are very close (0.5% gives the highest E′ and E″ while 2% the lowest), so the material is mostly in the Linear Viscoelastic Region with mild softening at higher strain.

When the data are parametrized by frequency and the frequency axis is removed (as in Figure 2.7), the path first moves to higher E′ and modest tan δ, then, near the E″ peak, tan δ shoots up while E′ drops, and finally tan δ falls while E′ recovers. That creates the bend/zig-zag we saw near E′≈1.4 MPa. In other words, it could be the signature of a relaxation process centered near 25 Hz at our test temperature (20 °C), but MAPSIL® QS1123 at 20 °C is deep in the rubbery plateau (Glass Transition Temperature (T_g) is far below room temperature). In that regime no pronounced molecular relaxation is expected within the 1–30 Hz frequency range. E′ should rise gently with frequency and tan δ should be fairly flat or only slowly varying. Given that, the observed feature (E″ peaking and E′ dipping around 25 Hz (and the matching kink in Figure 2.7)), is most likely instrumental/inertial rather than a true molecular relaxation. With a soft, low-modulus sample in tension, the drive + fixture + specimen mass can approach a resonance at higher frequencies. Because it appears at the same frequency for 0.5%, 1%, and 2% strain, that also points to a setup/fixture effect rather than a strain-dependent material process.

Further considerations can be done looking at the strain sweep tests which were conducted to further explore the limits of linear viscoelastic behavior. Strain was varied from 0.1% to 2%, and the tests were performed at three representative frequencies: 0.1 Hz, 1 Hz, and 5 Hz. This approach allowed evaluation of how the material's storage and loss moduli changed with increasing strain at different timescales, providing a comprehensive understanding of both amplitude and frequency-dependent viscoelastic behavior.

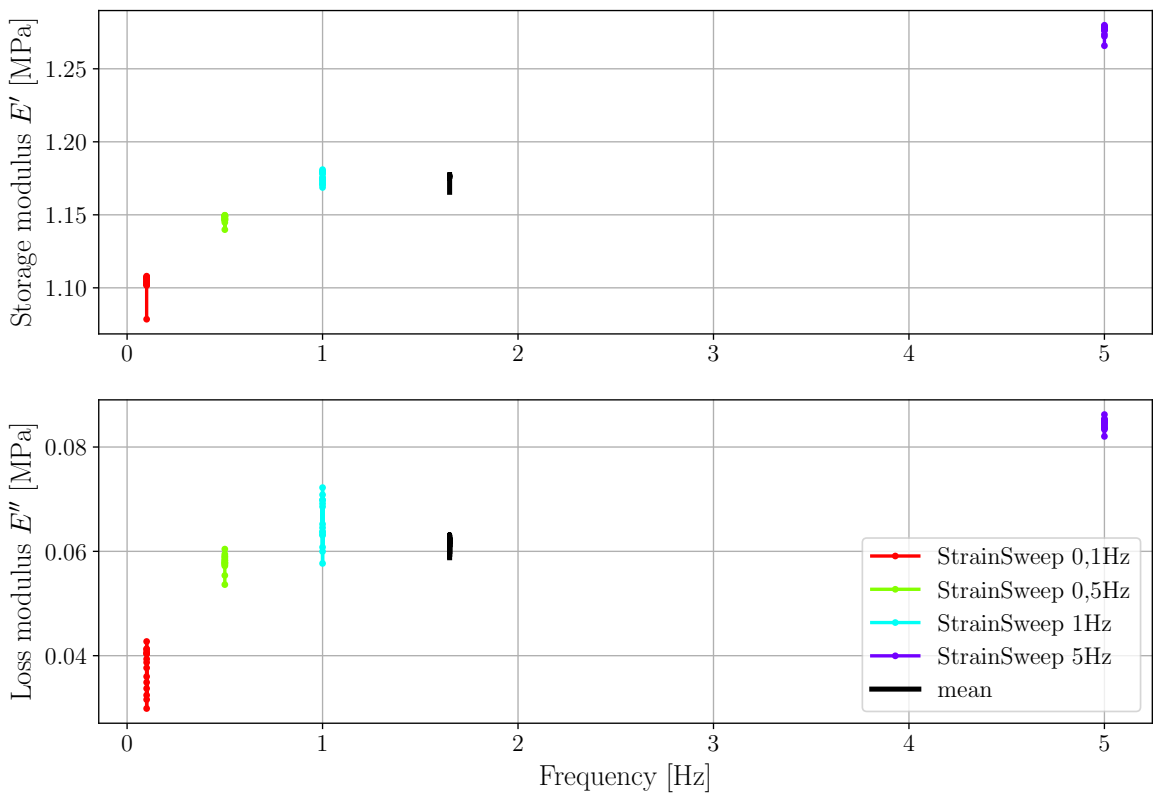

Figure 2.9: Strain Sweep Test: Variation of Storage and Loss Moduli with respect to the Frequency

The storage modulus E′, remains nearly constant across the tested strain levels at each frequency, indicating minimal strain dependence within the measured range. As expected

for a viscoelastic material, E′ increases with frequency, rising from approximately 1.6 MPa at 0.1 Hz to around 1.85 MPa at 5 Hz. No significant drop in E′ is observed within the tested strain range, suggesting that the material has not yet been pushed beyond its Linear Viscoelastic Region (LVR). The loss modulus E″, shows a similar trend, increasing with frequency while remaining essentially flat across the different strain levels at each frequency. This confirms that the material response is still within the LVR for the applied strains.

Since all tests up to 2% strain remain well within the LVR, the previously observed feature at 25 Hz cannot be attributed to strain-induced nonlinearity. This supports the interpretation that the 25 Hz behavior is likely due to instrument or fixture inertia, or some form of resonance, rather than a true material relaxation phenomenon [3].

The mean plots of the different formulations (Fig. 2.10 and 2.11) highlight systematic differences between the N-series (New) and O-series (Old) samples. The New (cured in August) and Old (cured in April) MAPSIL samples reveal clear evidence of aging-related changes in viscoelastic properties. The Old samples exhibit consistently higher storage modulus (E′ ≈ 1.6–2.1 MPa) compared to the New ones (E′ ≈ 1.1–1.4 MPa), indicating a stiffer network. At the same time, their damping factor (tan δ) is lower, reflecting a more elastic and less dissipative response. This evolution can be attributed to post-curing and aging reactions in silicone elastomers: over time, additional crosslinking occurs within the polysiloxane network, and interactions between the polymer matrix and the reinforcing fumed silica filler are strengthened.

The presence of fumed silica is essential in commercial silicone elastomers, as it provides mechanical reinforcement and damping through polymer–filler and filler–filler interactions. While the filler content remains constant, the interfacial interactions are known to evolve after curing. The increase in stiffness and reduction in relative damping observed in the Old samples is therefore consistent with a gradual densification of the network combined with enhanced filler networking. Such behavior has been reported for silica-filled silicones, where physical aging leads to a progressive reduction in chain mobility and increased modulus [10, 5].

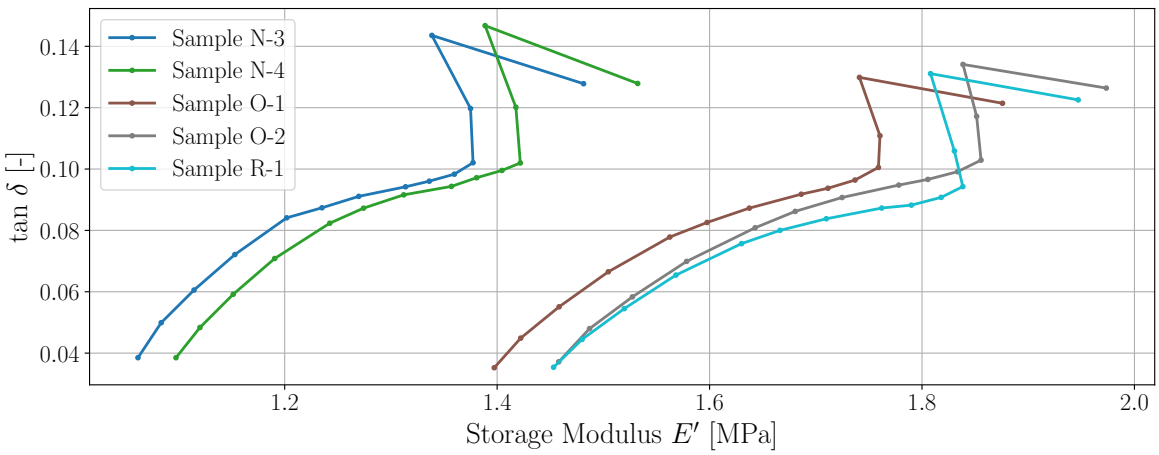

Figure 2.10: Frequency Sweep Test: Variation of Means of tan δ with respect to the Storage Modulus[1]

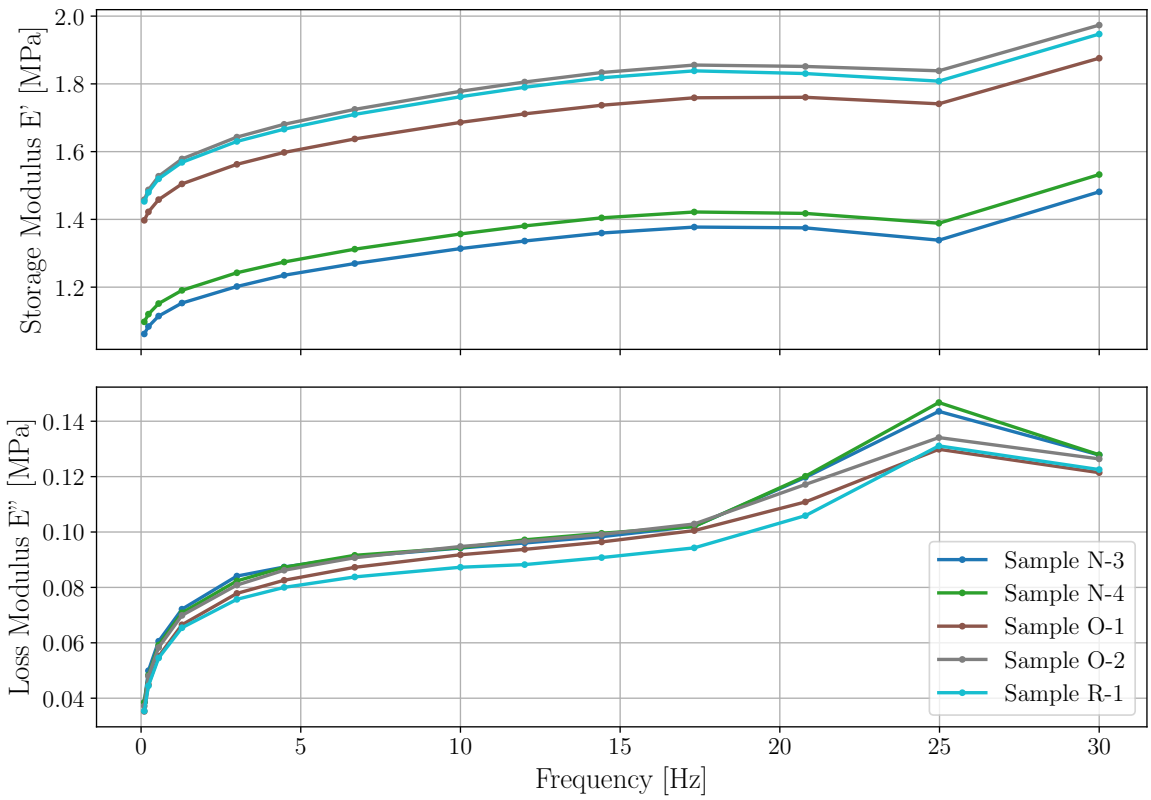

Figure 2.11: Frequency Sweep Test: Variation of Means of Storage and Loss Moduli with respect to the Frequency

Overall, the results demonstrate that MAPSIL undergoes significant changes in mechanical response during the first months after curing: New samples are softer and relatively more dissipative, while Old samples are stiffer and more elastic, reflecting the combined influence of ongoing crosslinking and filler interaction development. Reliable frequency-dependent properties can be reported up to 17 Hz, while data above this limit should be treated cautiously due to instrumental artifacts.

[1]Sample R-1 corresponds to a remake of the dynamic mechanical analysis test, conducted after an experimental error affected the initial measurement of that sample. The re-test was performed under identical conditions and does not represent a different formulation or curing history.

§ 3 Flexures Design

In the development of the new blade configuration (Figure 3.1), particular attention was paid to achieving a balance between manufacturability, structural integrity and dynamic performance. Experience from the previous project [15] highlighted several limitations, especially in relation to blade geometry and surface quality, which informed the improvements implemented here.

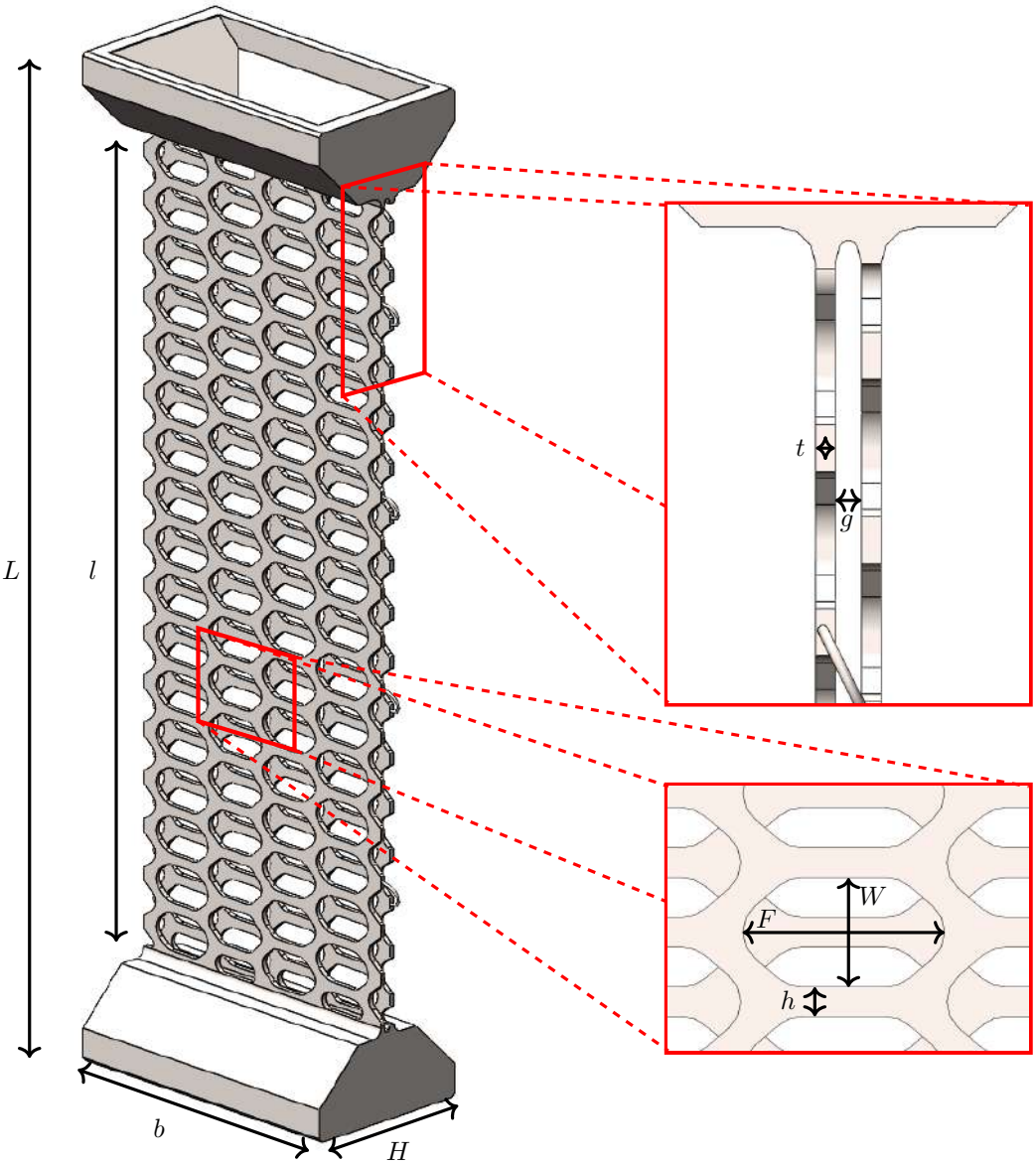

Figure 3.1: Flexure Design

Table 3.1 reports the geometrical parameters defining the flexure blade configuration used in this study. The table provides a complete overview of the nominal dimensions adopted for the current design, together with the corresponding values from the previous configuration for reference.

Table 3.1: Flexure Dimension Parameters

Parameter	Old Value [mm]	Actual Value [mm]
Thickness slave blade (t_s)	0.16	0.3
Thickness master blade (t_m)	0.22	0.3
Gap (g)	0.125	0.3
Blade Length (l)	47.6	47.6
Total Length (L)	58.6	58.5
Width (b)	18.2	18.2
Total Thickness (H)	10.5	10.4
Lattice edge-to-edge Length (W)	1.4	2.2
Lattice corner-to-corner Length (F)	4.1	4.057
Lattice Thickness (h)	0.5	0.5

One major modification concerns the thickness of the metal blades. In the earlier design, the slave and master blades had thicknesses of 0.16 mm and 0.22 mm, respectively, values that were already at the lower limit of the manufacturing process. Operating so close to these limits made fabrication difficult and increased the likelihood of defects or inconsistencies between specimens. To address this, the blade thicknesses were increased in the new version to improve manufacturability. This change, however, comes with important trade-offs. Thicker blades inevitably exhibit higher stiffness, which shifts the eigenfrequencies of the system to higher values, potentially moving them away from the desired operating range. In addition, higher thickness results in higher stresses, raising the risk of fatigue damage over time. The chosen configuration therefore represents a compromise: while a thicker blade simplifies fabrication and improves reliability, it must remain sufficiently compliant to avoid excessive stress and to keep the eigenfrequency within the targeted range.

A second improvement relates to the surface quality of the blades. Additive manufacturing often leaves partially melted particles attached to the surfaces, which can compromise both mechanical performance and dimensional accuracy. To overcome this, a special surface treatment (CMF-C) from AM Solutions was selected. This process removes a thin material layer (typically a few hundredths of a millimeter across all surfaces), thereby improving surface finish and reducing the likelihood of stress concentrations. For the present application, AM Solutions was asked to tune their process parameters, in particular treatment time, to achieve a target material removal of 0.05 mm. To anticipate this removal, an additional 0.05 mm was included on all blade surfaces in the design phase. The actual configuration also incorporates adjustments to the bridge geometry to account for the surface treatment (Fig. 3.2 and 3.3). Specifically, the bridge diameter was increased from 0.16 mm to 0.26 mm, compensating for the expected material removal during CMF-C. In addition, the number of bridges was reduced to one every four cells, which leaves more space to verify gap dimensions with gauges during quality control. These changes together ensure that the surface treatment achieves the desired finish without compromising dimensional tolerances, while also facilitating easier inspection and measurement of the final parts.

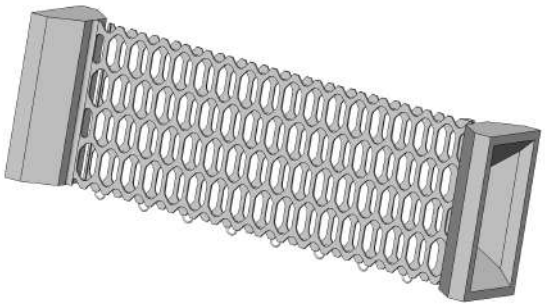

Figure 3.2: Old Flexure (8 bridges)

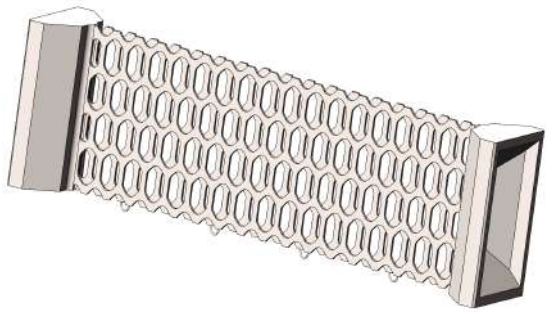

Figure 3.3: Actual Flexure (4 bridges)

3.1 *Analytical Calculations*

The stiffness of the blades in their unimpregnated state is estimated by modeling each blade as a cantilever beam, clamped at one end and free at the other. Since the system uses two identical blades arranged in parallel, the total stiffness can be expressed as:

$$K = 2 \cdot \frac{E \cdot b \cdot t^3}{l^3} \cdot \frac{1}{f} = 401 \text{ N/m}$$

where $E = 110$ GPa is the Young's modulus of Ti–6Al–4V and $f = 2.5$ is a correction factor introduced to account for the presence of the internal lattice structure, which reduces the effective stiffness compared to a solid blade. The factor of 2 accounts for the two blades working in parallel.

The maximum allowable stroke of the blades, limited by the admissible fatigue stress of the material, can be estimated using classical beam theory:

$$s = \frac{\sigma_{\text{adm}} \cdot l^2}{3 \cdot E \cdot t} = 4.58 \text{ mm}$$

where $\sigma_{\text{adm}} = 200$ MPa is taken as a conservative fatigue strength for Ti–6Al–4V.

The force required to achieve this stroke can then be obtained directly from Hooke's law:

$$F = K \cdot s = 1.84 \text{ N}$$

In summary, using an admissible fatigue stress of 200 MPa gives a maximum theoretical stroke of approximately 4.58 mm. However, under the current design configuration, reaching this stroke would cause stresses exceeding the material's fatigue limit. This implies that the blades would operate outside their safe elastic regime, posing a risk of damage or failure over repeated cycles. To ensure mechanical safety, either the blade geometry, the material properties, or the allowed stroke must be revised. A detailed discussion of potential solutions is presented in Section 4.1.1.

The out-of-plane stroke is derived just like for in-plane motion, but switching the roles of b and h, so the expression becomes:

$$s = \frac{\sigma_{\text{adm}} \cdot l^2}{3 \cdot E \cdot b} = 0.075 \text{ mm}$$

§ 4 Experimental Procedures

This chapter presents the experimental procedures carried out to investigate the damping characteristics of the flexure blades. A dedicated test bench was designed and manufactured to enable accurate characterization of their dynamic behavior under controlled and repeatable conditions. The experimental setup allows for precise excitation of the structure and reliable acquisition of response data, ensuring consistency across repeated measurements.

Both free and forced oscillation tests were conducted to comprehensively assess the dynamic response of the flexure blades. Non-contact laser vibrometry was employed to measure the displacement response, eliminating any influence of sensor mass or mechanical interaction with the system. In the free oscillation tests, the flexure blades were subjected to impulse excitation, and the resulting time-domain responses were analyzed to extract key dynamic parameters, including the natural frequency, damping ratio, and amplitude decay. Forced oscillation tests were performed by applying controlled harmonic excitation to the structure, enabling the investigation of steady-state response characteristics and resonance behavior. Complementary frequency-domain analyses were conducted on the measured signals to examine the spectral content of the response and to identify resonance peaks associated with the dynamic behavior.

4.1 *Free Oscillation Test*

The objective of this test is to excite the impregnated blades into free oscillation by imposing a prescribed displacement and subsequently releasing the structure. During the resulting transient response, the velocity of the system is recorded as a function of time. The measurements are performed using the experimental test bench originally developed in the previous project [15], which is adopted here as a reference platform to ensure consistency with previous investigations. The experimental data obtained from this test are subsequently used for the correlation and validation of the finite element model developed in the present work.

4.1.1 Test Bench Design Optimization

The test bench shown in Figure 4.1 was used to perform the free oscillation tests, following the methodology outlined in the previous project [15], with some modifications to the design and the sensing strategy. Both blades are mounted on a dedicated support to form a parallelogram stage. A DC motor, positioned at the top of the setup, displaces the parallelogram stage by a fixed amount via a lever and then releases it, initiating free oscillations (Figure 4.1).

During the assembly of the test bench, several issues were identified, leading to the redesign of some components to improve both mechanical performance and measurement reliability. Among these, the end-stops, which serve to limit the maximum stroke of the moving element, required particular attention. In the initial configuration, the available stroke range was insufficient for some experimental conditions, thereby limiting the flexibility of the setup. To overcome this limitation, the end stops were redesigned to provide an extended stroke capacity, enabling the system to support a broader range of measurement scenarios.

Furthermore, alignment guides were incorporated into the end-stop plate to ensure precise and repeatable positioning of the end stops. This addition allows for easier assembly and guarantees that the end stops remain properly aligned even after repeated mounting and dismounting operations. As a result, the end-stop plate itself was also redesigned to integrate these improvements and to ensure structural compatibility with the new components. The updated configuration ensures consistent alignment for both in-plane and out-of-plane tests. In this last configuration, the end stops allow a maximum stroke of 85 μm. This modification increases the versatility of the system, making it suitable for a broader set of experimental conditions without requiring further mechanical adjustments.

In addition to the end-stops, the moving component of the test bench also underwent a redesign. During preliminary assembly and testing, it was observed that once all fastening screws were tightened, the part exhibited a bending deformation along its longitudinal axis. This deformation compromised the overall alignment of the system and introduced undesired internal stresses, which could potentially affect the accuracy and repeatability of the measurements. Such behavior was attributed to the insufficient stiffness of the original design, likely due to the combination of material selection and geometric slenderness. To mitigate these issues, the moving component was re-engineered with an increased thickness of 2 mm and manufactured in steel, the same material used for the other two redesigned parts. This change in material provided a significant improvement in mechanical rigidity and dimensional stability, reducing the likelihood of deformation under operational loads or tightening forces.

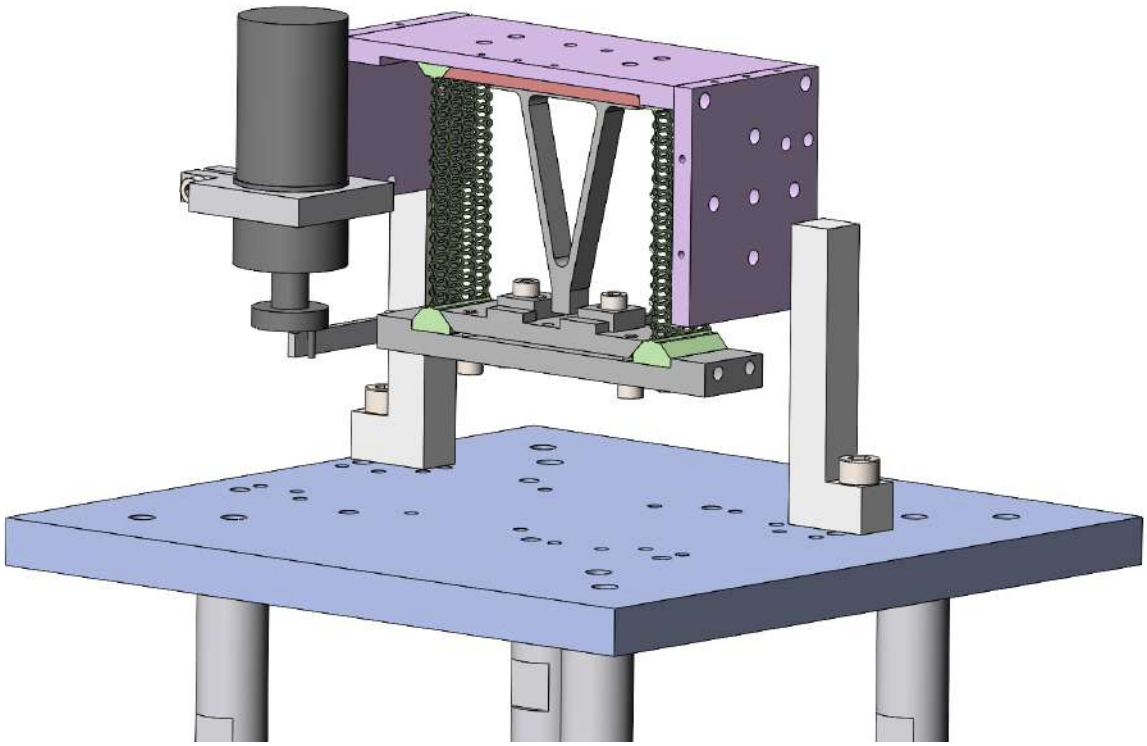

Figure 4.1: Test Bench Design Setup - Isometric View

The choice of steel was made not only to reduce bending but also to address dynamic

performance concerns identified in previous experiments. In the earlier setup, developed for the preceding project [15], the first eigenfrequency of the system was measured at 21.88 Hz, which is excessively high to ensure accurate and stable measurements. As discussed in Section 2.2.2, operating at frequencies above 17 Hz can lead to significant issues due to the viscoelastic behavior of the material involved, introducing non-linearities and compromising measurement reliability.

By manufacturing the redesigned components in steel instead of aluminum, it was possible to lower the natural frequency of the system, bringing it closer to the desired operational range and thus improving measurement quality. The target additional mass required to achieve this frequency reduction can be determined from the fundamental relationship between angular frequency, stiffness, and mass:

$$\omega = 2\pi f_{\text{target}} = \sqrt{\frac{k}{m_{\text{target}}}}$$

where k is the stiffness of the system and m_{target} is the equivalent mass required to reach the desired target frequency f_{target}. Rearranging the expression gives:

$$m_{\text{target}} = \frac{k}{(2\pi f_{\text{target}})^2}$$

Since the stiffness k of the system can also be expressed in terms of the original frequency f and mass m as:

$$k = (2\pi f)^2 \cdot m$$

substituting this last expression into m_{target} yields:

$$m_{\text{target}} = \frac{(2\pi f)^2}{(2\pi f_{\text{target}})^2} \cdot m = \left(\frac{f}{f_{\text{target}}}\right)^2 \cdot m$$

This relationship allows the computation of the additional mass required to achieve a specific reduction in the natural frequency, assuming that the system stiffness k remains constant.

In this case, the objective was to obtain a system eigenfrequency within the range of 13 Hz to 17 Hz, where the viscoelastic effects remain negligible and measurement accuracy is ensured. A target frequency of 14 Hz was therefore selected as a suitable compromise between dynamic stability and system sensitivity. To achieve this target, the components were manufactured in AISI 316L stainless steel, whose material properties provide the necessary density and mechanical stiffness to bring the eigenfrequency within the desired range.

All technical drawings of the redesigned components are provided in Appendix B for reference.

Unlike the original methodology, which relied on a Renishaw linear encoder to record the position of the mobile part, the current setup uses a contactless Vibrometer from Polytec as the primary sensor (Figure 4.2). This device measures the velocity of a single point on the mobile part without physical contact, reducing noise and avoiding potential artifacts caused by mechanical coupling or limited encoder resolution.

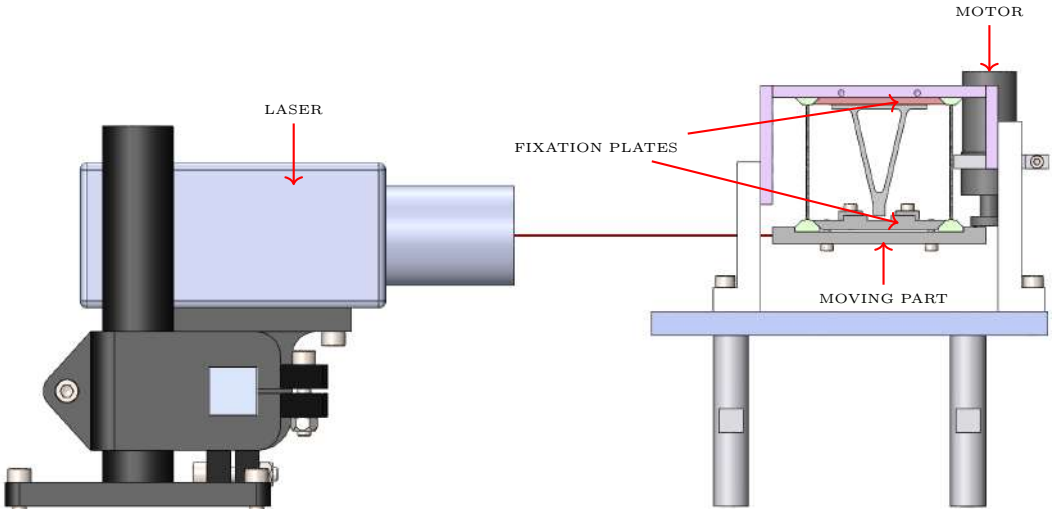

Figure 4.2: Test Bench Design Setup - Front View

To ensure the precise alignment of the Vibrometer's laser during measurements, a dedicated laser holder was designed and 3D-printed (Figure 4.3). The holder features a bi-axial adjustment mechanism that enables fine movement of the laser along two perpendicular axes. This configuration allows the operator to accurately direct the laser beam toward the desired point on the moving component, ensuring optimal signal reflection and measurement accuracy. One of the key advantages of this design is its modular mounting system, which allows the holder to be mounted and dismounted without disturbing the alignment. Each axis is equipped with an independent adjustment screw system, meaning that once the correct position is set, it remains fixed. To remove the holder, the operator only needs to release the base screws, leaving the alignment settings on both axes untouched. When reassembled, the laser automatically returns to its previous calibrated position without the need for realignment. This feature significantly improves the repeatability and efficiency of measurement operations enabling the operator to perform measurements at different times or in separate experimental sessions while maintaining consistent alignment conditions.

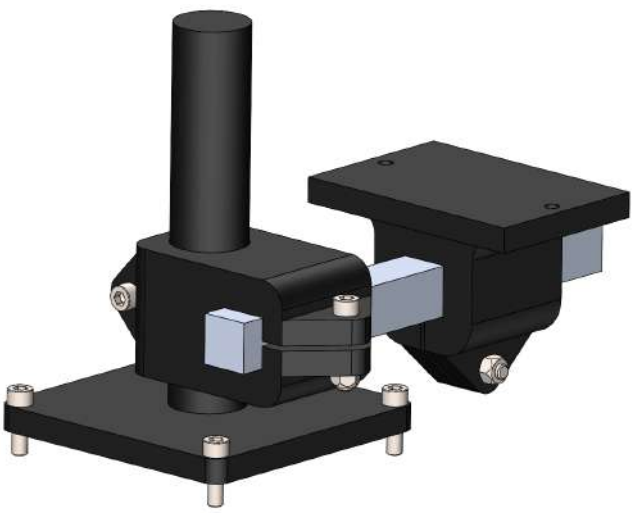

Figure 4.3: Laser Holder

4.1.2 Results

All tests were conducted under laboratory conditions, maintaining consistent ambient temperature, pressure, and humidity. Five pairs of blades were tested to ensure an adequate dataset for meaningful correlation analysis. For each pair of blades, five measurements were performed to ensure repeatability and to account for any potential variability in the results. The key experimental parameters are summarized in Table 4.1.

Table 4.1: Free Oscillations test parameters

Parameter	**Value**
Motor Voltage	2.5 V
In-plane stroke	4 mm
Blades Orientation	Vertical

The results of the free oscillation test provide a detailed characterization of the dynamic response and damping behavior of the tested system. Figure 4.4 presents the time-domain displacement responses obtained from repeated measurements following a small mechanical perturbation. Each subplot exhibits a decaying oscillatory waveform, characteristic of a lightly damped system, in which the vibration amplitude progressively decreases as energy is dissipated. The strong consistency observed in both oscillation frequency and decay rate across the different datasets (distinguished by color) confirms the repeatability of the measurements and indicates that the structural and material properties of the system remain stable throughout the experiments.

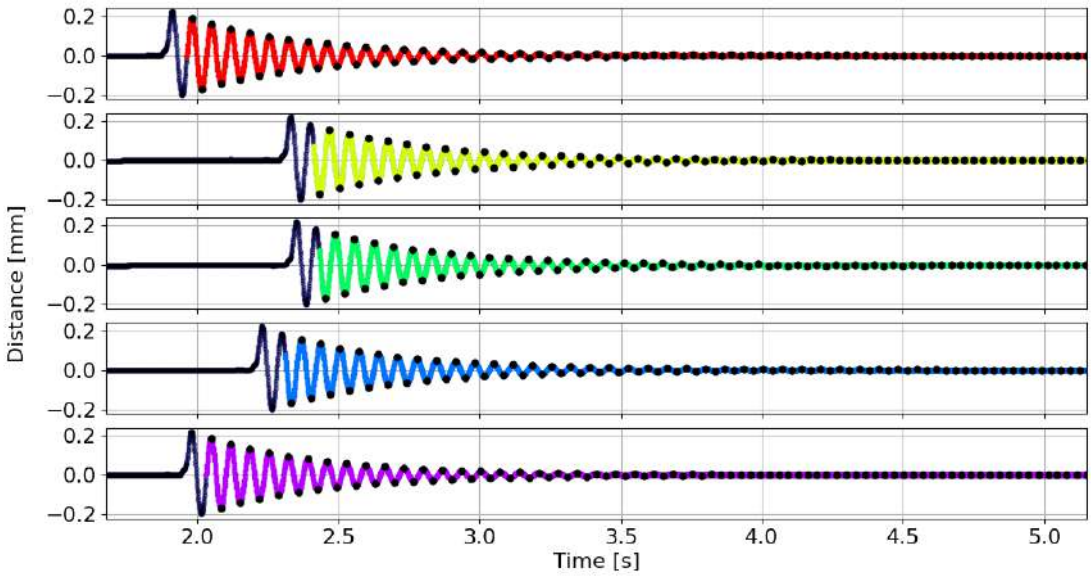

Figure 4.4: Time-domain displacement responses showing exponential decay of free oscillations for repeated measurements

To further investigate the dynamic behavior, the frequency-domain representation of the same signals was obtained through power spectral density (PSD) analysis, as shown in Figure 4.5. The PSD provides a quantitative description of how vibrational energy is distributed over frequency and allows direct identification of the system's natural frequency and damping characteristics.

All spectra display a pronounced resonance peak centered at approximately 14 Hz, in excellent agreement with the theoretical prediction reported in Section 4.1.1. The vertical dashed line indicates the identified resonance frequency, which remains invariant across all measurements, thereby confirming the linearity and stability of the system response. The symmetric decay of spectral amplitude around the resonance and the approximately $1/f^2$ roll-off at higher frequencies are consistent with the expected behavior of a viscously damped single-degree-of-freedom oscillator.

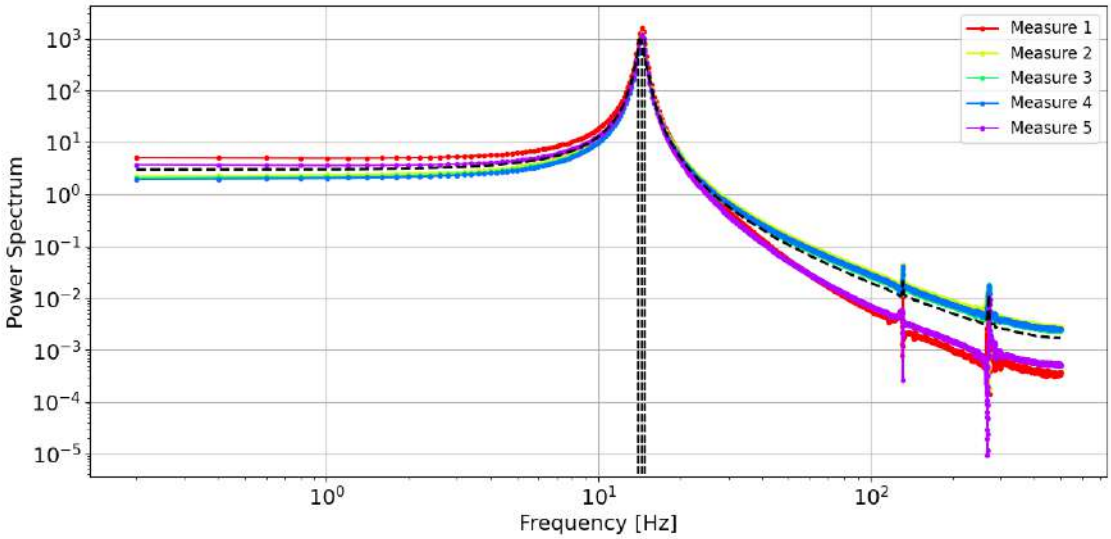

Figure 4.5: Power spectral density of the free oscillation signal for repeated measurements

A quantitative frequency-domain estimate of the damping was obtained through evaluation of the quality factor Q using the half-power bandwidth method. The frequencies f_1 and f_2 corresponding to the -3 dB points on either side of the resonance peak were identified numerically from the interpolated spectral data. The quality factor was then computed as

$$Q = \frac{f_n}{f_2 - f_1}$$

where f_n denotes the resonance frequency. This approach provides an estimate of damping that is largely insensitive to amplitude variations and measurement noise.

To complement and validate the frequency-domain analysis, an independent time-domain investigation of the damping behavior was performed using the free decay response. A peak-tracking algorithm was applied to each displacement signal to identify successive local maxima and minima of the oscillatory motion. From the temporal spacing of consecutive extrema, the mean oscillation frequency was computed and found to be in excellent agreement with the resonance frequency obtained from the PSD. For each oscillation cycle, the amplitude loss per period was evaluated as

$$\Delta A_k = A_k - A_{k+1}$$

together with the corresponding mean amplitude

$$A_k = \frac{A_k + A_{k+1}}{2}$$

These quantities provide a direct measure of the energy dissipated during each oscillation cycle. For a lightly damped linear oscillator, the logarithmic decrement formulation leads to the following relationship between amplitude decay and quality factor:

$$Q = \pi \frac{A}{\Delta A}$$

This expression highlights the physical interpretation of the quality factor as the ratio between stored and dissipated energy per oscillation cycle.

Figure 4.6 shows the relationship between oscillation amplitude A and amplitude loss per period ΔA for all datasets. A strong linear correlation is observed across the entire amplitude range, indicating that the amplitude loss scales proportionally with the oscillation amplitude. This behavior is a hallmark of linear viscous damping, for which the dissipative force is proportional to velocity.

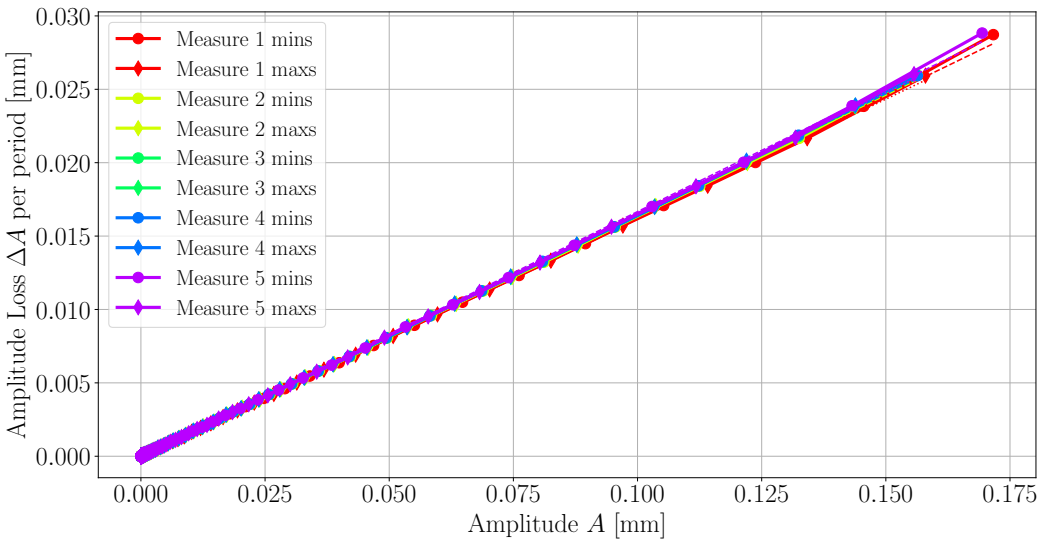

Figure 4.6: Amplitude loss per period as a function of oscillation amplitude for repeated measurements

The minimal scatter between datasets and the collapse of all curves onto a common linear trend demonstrate the high repeatability and precision of the experimental procedure. Physically, this implies that the energy dissipated per cycle, proportional to $A\Delta A$, scales quadratically with amplitude, in agreement with theoretical predictions.

4.1.2.1 Q-factor from linear fit

To obtain a more robust estimate of the damping, the amplitude–loss data were fitted using a linear regression model of the form

$$\Delta A = mA + b$$

For all measurements, the fitted intercept b was found to be negligible, as expected for a physical damping mechanism: when the oscillation amplitude approaches zero, the dissipated energy per cycle must also vanish.

The slope m represents the relative amplitude loss per oscillation cycle,

$$m = \frac{\Delta A}{A}$$

By combining this result with the logarithmic decrement formulation, the quality factor can be expressed as

$$\delta = \frac{1}{n} \ln\left(\frac{A_k}{A_{k+n}}\right) \quad \Longrightarrow \quad Q = -\frac{\pi}{\ln(1-m)}$$

which, for lightly damped systems ($m \ll 1$), reduces to the commonly used approximation

$$Q \approx \frac{\pi}{m}$$

This formulation is mathematically equivalent to the relation $Q = \pi A/\Delta A$, but provides increased robustness by averaging the damping behavior over multiple oscillation cycles.

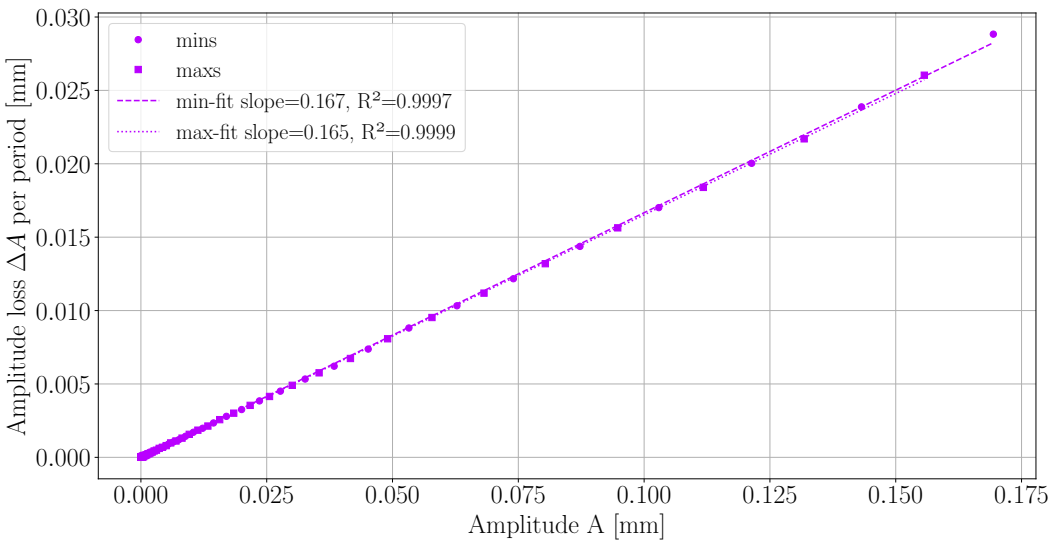

Figure 4.7: Linear regression of amplitude loss per cycle versus oscillation amplitude for positive and negative peaks

The excellent linearity of the fitted curve shown in Figure 4.7, together with the high coefficients of determination reported in Table 4.2, confirms that the damping coefficient remains constant over the full amplitude range. The close agreement between fits obtained from maxima and minima further indicates a symmetric system response about the equilibrium position, with no detectable nonlinear effects or directional bias.

4.1.2.2 Q-factor from Hilbert envelope

As a final validation of the time-domain damping analysis, the quality factor was independently estimated using the Hilbert transform to extract the instantaneous oscillation envelope [13]. For a lightly damped single-degree-of-freedom oscillator, the displacement response can be expressed as

$$x(t) = A_0 e^{-\alpha t} \cos(\omega_d t + \phi)$$

where A_0 is the initial amplitude, α is the exponential decay constant, ω_d is the damped angular frequency, and ϕ is a phase offset. The quality factor is related to the damping coefficient by

$$Q = \frac{\omega_n}{2\alpha}$$

with ω_n denoting the undamped natural angular frequency. The analytic signal associated with the measured displacement $x(t)$ is defined as

$$z(t) = x(t) + i\,\mathcal{H}\{x(t)\}$$

where $\mathcal{H}\{x(t)\}$ denotes the Hilbert transform. The instantaneous amplitude envelope is then obtained as

$$A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x(t)\}^2}$$

For a linear viscously damped system, the envelope follows an exponential decay law,

$$A(t) = A_0 e^{-\zeta\omega_n t}$$

which can be rewritten in logarithmic form as

$$\ln[A(t)] = \ln(A_0) - \zeta\omega_n t$$

Therefore, the damping ratio ζ can be obtained directly from the slope of a linear regression applied to the logarithm of the amplitude envelope.

In practice, the regression was performed over the time interval in which the decay appeared approximately exponential. Specifically, the interval between 1.7 s and 4.5 s was selected, as this portion of the signal exhibited a clear linear trend in the semi-logarithmic representation (Figure 4.8). Outside this region, the envelope shows small oscillations and eventually deviates upward as the signal amplitude approaches the noise floor. These effects are expected and arise from numerical artifacts of the Hilbert transform and from the increasing influence of measurement noise at low amplitudes [2]. Consequently, only the initial linear section was considered for quantitative slope estimation.

For clarity, only one representative dataset among the five repeated measurements is reported here, since all tests exhibited qualitatively identical decay behavior and yielded consistent damping estimates.

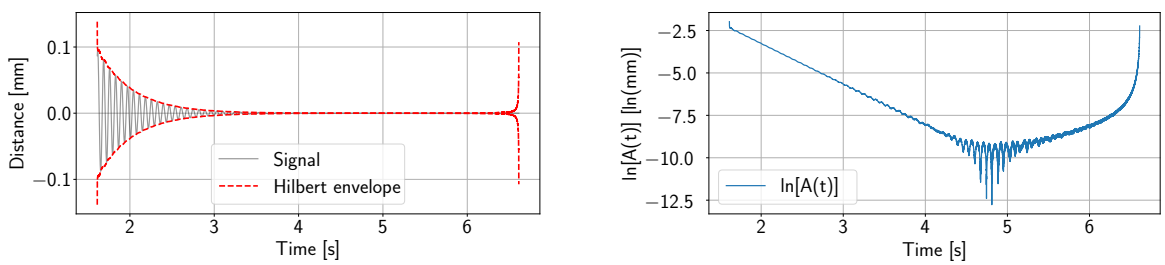

Figure 4.8: Hilbert envelope of the free oscillation response and corresponding logarithmic decay over time

For visualization purposes, the fitted regression line was superimposed on the logarithmic envelope together with a shaded region indicating the selected fitting window, as shown in Figure 4.9.

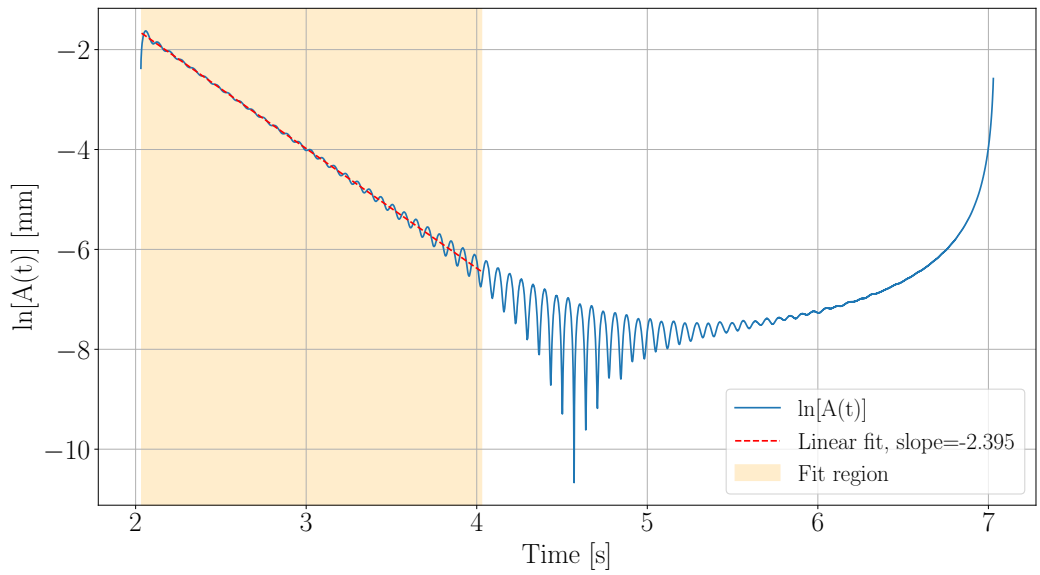

Figure 4.9: Linear regression of $\ln[A(t)]$ performed between 1.7 s and 4.5 s

The following table summarizes the values of all previously mentioned parameters:

Table 4.2: Parameters derived from free oscillation signal analysis of one pair of blades

Measurement	**Power spectrum**		**Linear Fit**				**Hilbert**
	f_n (Hz)	Q	R^2(min)	R^2(max)	Q(min)	Q(max)	Q
1	14.00	18.05	0.9998	0.9998	17.897	18.406	18.551
2	14.00	17.85	0.9998	0.9998	18.136	18.193	18.524
3	14.00	17.80	0.9998	0.9997	18.036	17.906	18.504
4	14.00	17.79	0.9998	0.9996	18.209	18.142	18.498
5	14.00	17.86	0.9998	0.9998	17.770	18.267	18.458
Average	**14.00**	**17.87**	**0.9998**	**0.9997**	**18.010**	**18.183**	**18.507**

When these results are considered alongside the spectral analysis and the amplitude-decay curves, they reinforce the conclusion that the system behaves as a stable, linear, single-degree-of-freedom oscillator. The natural frequency remains sharply defined at 14 Hz, and the damping quality factor Q remains nearly constant across measurements, with only minor variation between methods. The linear fits of the peak amplitudes confirm a strictly proportional amplitude-loss relationship with negligible scatter, while the Hilbert-transform-based Q-factor provides a slightly higher but more robust estimate, reflecting its improved sensitivity to the full oscillation envelope. Altogether, these observations substantiate the applicability of the linear viscous damping model and demonstrate that the experimental setup and analysis pipeline yield accurate, repeatable, and physically consistent modal parameters, suitable for model validation, parameter identification, or benchmarking of numerical simulations.

4.1.2.3 Mean behavior across blade pairs

All the previously shown results were about only one pair of blades; the comparison across all five blade pairs provides a broader perspective on the consistency and repeatability of

the system response. The amplitude loss curves in Figure 4.10 exhibit an exceptionally tight clustering of linear trends, with both minimum and maximum extracted amplitudes aligning almost perfectly across all blade combinations.

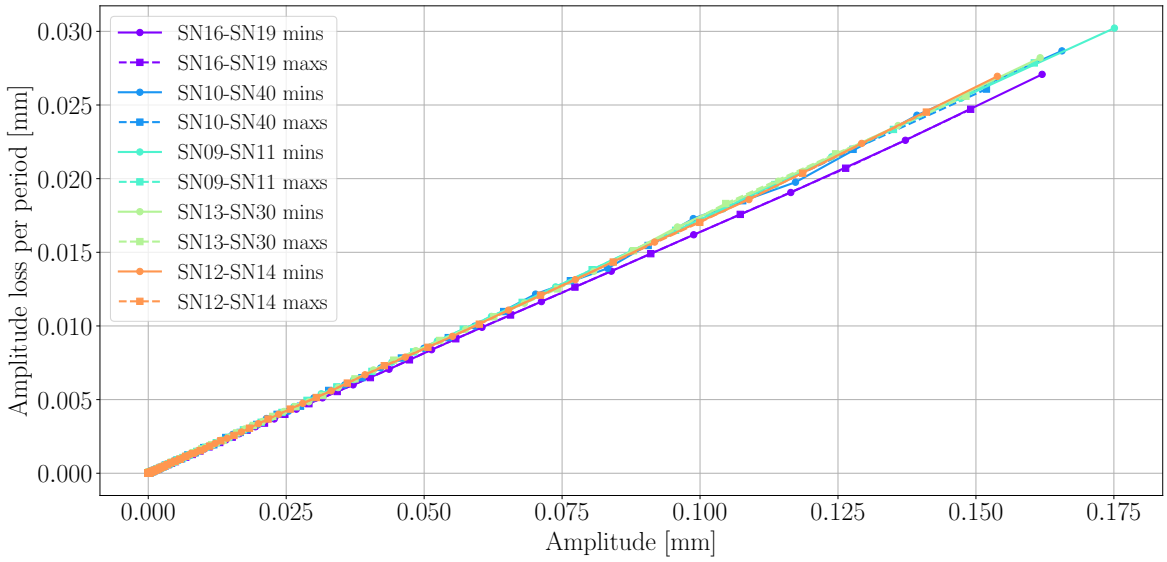

Figure 4.10: Comparison of Mean Amplitude vs Amplitude Loss for Five Blade Pairs

This uniform behavior indicates that the damping mechanism is highly consistent, independent of the specific blade pair, and that any geometric or material variability among the blades introduces only negligible deviations in the energy-dissipation rate. The near-identical slopes confirm that the amplitude decay per cycle scales proportionally with oscillation amplitude for all pairs, reinforcing the suitability of a linear viscous damping model.

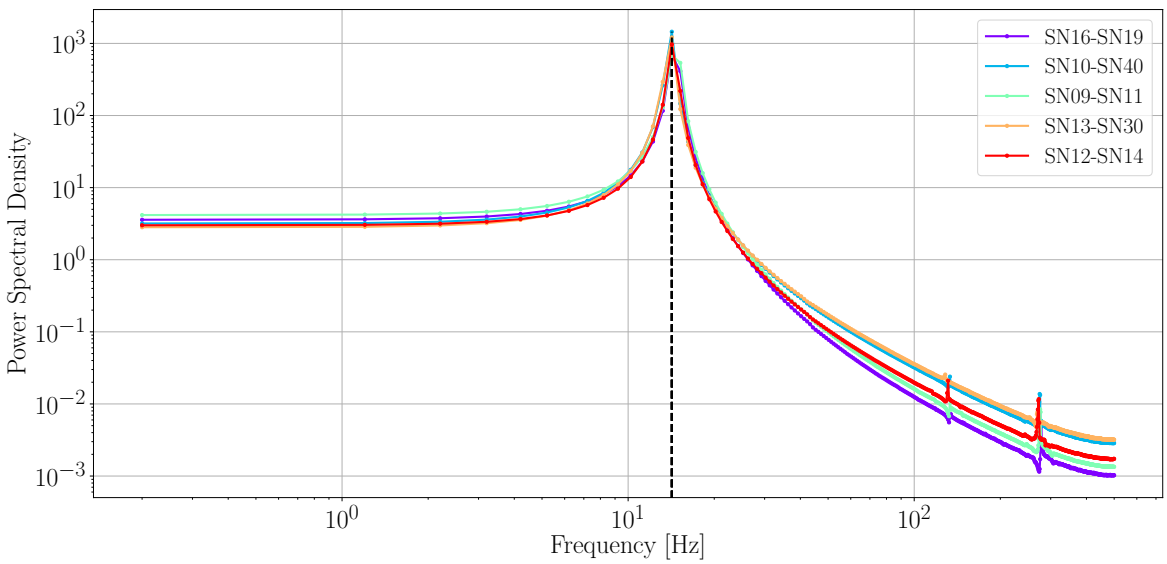

Figure 4.11: Comparison of Mean Power Spectra for Five Blade Pairs

The averaged power spectra in Figure 4.11 exhibit a well-defined resonance peak at

approximately the same frequency for all blade pairs, confirming the consistency of the identified natural frequency and damping behavior. The consistent collapse of the spectral tails further indicates uniform stiffness and mass properties across the blades. Small differences in the absolute peak magnitude are observed, with some spectra reaching values on the order of 10^2 and others approaching 10^3. These variations are primarily due to differences in initial excitation amplitude and measurement noise. Since the power spectral density scales with the square of the signal amplitude, modest variations in the time-domain response can lead to noticeable differences in spectral magnitude, without affecting the location or sharpness of the resonance peak. As modal parameters are determined from the relative spectral shape rather than its absolute level, these differences do not influence the identified natural frequency or damping ratio.

Overall, the comparison across all blade pairs demonstrates a high level of reproducibility in the measured modal parameters. Table 4.3 reports the natural frequency and quality factor obtained from each method, together with blade-pair averages and a global mean value representative of the system damping characteristics.

Table 4.3: Parameters derived from free oscillation signal analysis of five pairs of blades

Blades	**Meas.**	**Power spectrum**		**Linear Fit**				**Hilbert**
		f_n (Hz)	Q	$R^2_{\min}$	$R^2_{\max}$	$Q_{\min}$	$Q_{\max}$	Q
SN09–SN11	1	14.80	16.96	1.0000	1.0000	18.097	18.307	18.563
	2	14.80	17.07	1.0000	0.9999	18.351	17.985	18.574
	3	14.80	17.09	1.0000	0.9999	18.265	18.048	18.536
	4	14.80	17.03	1.0000	0.9999	18.344	17.955	18.531
	5	14.80	16.92	0.9999	1.0000	17.987	18.230	18.497
	Avg.	14.80	17.01	1.0000	0.9999	18.208	18.105	18.540
SN10–SN40	1	14.20	17.25	0.9995	0.9995	17.870	18.812	18.769
	2	14.20	17.43	0.9996	0.9996	18.574	17.992	18.710
	3	14.20	17.40	0.9996	0.9998	18.578	18.212	18.702
	4	14.20	17.41	0.9995	0.9998	18.447	18.111	18.684
	5	14.20	17.19	0.9993	0.9997	17.399	18.342	18.635
	Avg.	14.20	17.34	0.9997	0.9997	18.173	18.294	18.700
SN12–SN14	1	14.40	18.14	0.9996	0.9999	17.734	18.177	18.617
	2	14.40	18.07	0.9998	0.9997	18.017	17.937	18.643
	3	14.40	18.03	0.9998	0.9998	18.084	17.886	18.653
	4	14.40	18.10	0.9999	0.9998	18.123	18.043	18.674
	5	14.40	18.20	0.9997	0.9999	17.797	18.283	18.668
	Avg.	14.40	18.11	0.9998	0.9998	17.951	18.065	18.651
SN13–SN30	1	14.00	18.05	0.9998	0.9998	17.897	18.406	18.551
	2	14.00	17.85	0.9998	0.9998	18.136	18.193	18.524
	3	14.00	17.80	0.9998	0.9997	18.036	17.906	18.504
	4	14.00	17.79	0.9998	0.9996	18.209	18.142	18.498
	5	14.00	17.86	0.9998	0.9998	17.770	18.267	18.458
	Avg.	14.00	17.87	0.9998	0.9997	18.010	18.183	18.507
SN16–SN19	1	14.60	18.93	0.9997	0.9999	18.774	19.141	19.491
	2	14.60	18.52	0.9999	0.9999	18.918	18.968	19.339
	3	14.60	18.42	0.9999	0.9999	18.912	18.932	19.255
	4	14.60	18.29	1.0000	0.9999	18.943	18.909	19.213
	5	14.60	18.43	0.9997	0.9999	18.458	18.786	19.157
	Avg.	14.60	18.52	0.9998	0.9999	18.801	18.947	19.291
Global Avg.		**14.40**	**17.77**	**0.9998**	**0.9998**	**18.23**	**18.32**	**18.74**

4.2 *Forced Oscillation Test*

A new series of forced oscillation tests is carried out to evaluate the dynamic behavior and damping performance of the test bench equipped with double-blade elements. The bench is mounted on the shaker through decoupling rods and suspended by three threads to allow free motion in the targeted direction (Figure 4.12). During the tests, a Polytec laser Vibrometer measures the velocity of the moving part, while an accelerometer captures the base excitation, enabling accurate characterization of the system response.

For further details on the cabling procedure, hardware connections and software setup refer to Appendix C.

4.2.1 Excitation Strategy and Test Configuration

The forced oscillation test was performed using a sine sweep excitation subdivided into multiple frequency blocks rather than a single continuous sweep over the full bandwidth (5-750 Hz). This choice was driven by both physical and experimental considerations related to signal-to-noise ratio, shaker limitations, and the dynamic behavior of the tested structure [9]. The 5-750 Hz bandwidth was selected to encompass the relevant eigen-frequencies of the parallelogram mechanism, the blades, and the shaker, while remaining well below the thread resonance and within the shaker's operating bandwidth, which extends up to 9000 Hz. This range ensures that all frequencies of interest are properly captured. To ensure accurate measurements without reaching the mechanical end-stops, the excitation amplitude is tuned prior to each test.

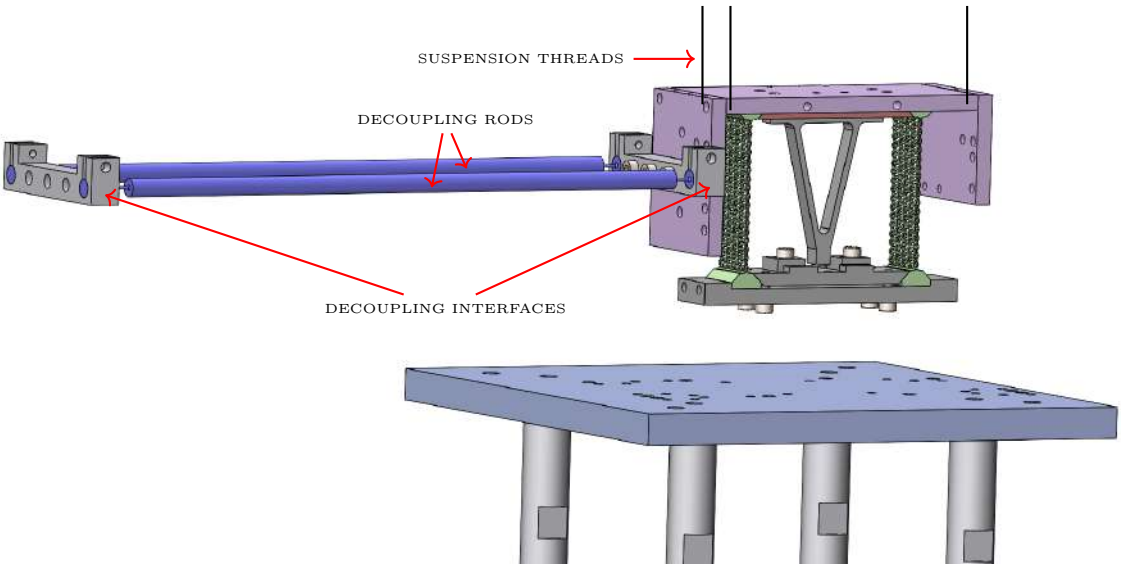

Figure 4.12: Forced Oscillation Test Bench - In-Plane Configuration

4.2.1.1 Low-frequency excitation block (5–30 Hz)

The first excitation block, covering the frequency range from 5 to 30 Hz, was dedicated to the characterization of the system's low-frequency dynamic behavior and, in particular, of the first resonance occurring at approximately 14 Hz. This frequency range was treated

separately because the response amplitude increases significantly near resonance, making this region especially sensitive to both excitation parameters and mechanical constraints of the experimental setup. For this reason, the excitation amplitude was limited to 0.23 g, which corresponds to the maximum acceleration level that could be safely applied without causing the moving components of the test bench to reach their mechanical end-stops. Furthermore, a slow sweep rate was used in 5–30 Hz because quasi-steady response is hardest to achieve at low frequency and especially near the resonance peak. If the sweep is too fast, the response cannot fully settle, producing potential distorted peak height, and phase errors (i.e., a dynamic tracking error rather than the true steady FRF). A slower sweep increases the effective frequency resolution around the resonance and improves repeatability of the peak shape, which is critical when the block will be later used for modal parameters extraction. The sampling frequency for this block was selected to be high enough to avoid aliasing, preserve phase accuracy, and support stable digital filtering and numerical integration when deriving velocity from acceleration. Although the band limit is only 30 Hz, using an f_s comfortably above the Nyquist requirement reduces sensitivity to filter edge effects and provides cleaner time-domain signals for detrending/band-pass filtering before integration. The key is not having f_s as high as possible, but to have it high enough to be numerically stable and well-conditioned while keeping file sizes manageable. Ultimately, the number of samples (equivalently, the acquisition duration at a given f_s) was chosen to ensure sufficient cycles at the lowest frequencies and adequate spectral resolution for the FFT-based FRF estimate. At 5-30 Hz, a too short record leads to poor resolution and higher variance in the FRF, while longer records improve SNR by providing more data through the resonance and more consistent averaging in the frequency domain (even when only a single run is used). In practice, the record length is driven by the sweep rate and the need to include enough time around the resonance for the response to stabilize, ensuring that the peak frequency and bandwidth (used later for damping calculation) are not artifacts of insufficient data length [14].

4.2.1.2 Mid-frequency excitation block (24–100 Hz)

The second excitation block, spanning the frequency range from 24 to 100 Hz, overlaps with the first one in the 24–30 Hz region, allowing consistency checks between the two datasets and ensuring a smooth transition in the assembled frequency response function. In this frequency range, the dynamic amplification associated with the first mode has already decayed, and the structural response becomes progressively less sensitive to small variations in excitation force, making it possible to relax some of the constraints required in the resonance-dominated band. Since the response levels above the first resonance are inherently lower, the same value of amplitude with respect to the previous block remains sufficient to achieve an adequate signal-to-noise ratio while continuing to respect the mechanical limitations of the test bench. Compared to the low-frequency block, a substantially higher sweep rate was adopted. In this frequency range the system reaches steady-state more rapidly, and the response is less susceptible to transient tracking errors. Increasing the sweep rate therefore improves test efficiency without compromising the accuracy of the measured FRF. The chosen sweep rate represents a balance between minimizing test duration and preserving sufficient frequency resolution for reliable identification of the response shape in the transition region between low-frequency resonance-dominated behavior and higher-frequency stiffness-controlled response. The sampling frequency was increased relative to the first block to maintain adequate tem-

poral resolution over the wider frequency range and to ensure accurate representation of both magnitude and phase of the response. A higher sampling rate also supports stable digital filtering and FFT processing, particularly when combining data from multiple blocks acquired under different sweep conditions. Finally, the number of samples acquired in this block was adjusted in accordance with the higher sweep rate and increased sampling frequency. While the total acquisition duration is shorter than in the low-frequency block, the resulting record length remains sufficient to provide adequate frequency resolution and low variance in the estimated FRF across the entire band. This ensures that the measured response in the 24–100 Hz range can be reliably merged with adjacent blocks and used for subsequent analysis without introducing discontinuities or bias.

4.2.1.3 High-frequency excitation block (85–500 Hz)

The third excitation block, spanning the frequency range from 85 to 500 Hz, was defined following preliminary tests in which a single wide-band sweep resulted in a frequency response function heavily contaminated by noise, particularly in the mid-to-high frequency range. In those initial measurements, the combination of a uniform sweep rate and uniform excitation level across the entire bandwidth led to suboptimal signal-to-noise ratio (SNR) at higher frequencies, where the structural response amplitude is inherently lower. The adoption of a dedicated block for this frequency range was therefore motivated by the need to tailor the excitation parameters specifically to reduce noise and improve the reliability of the measured FRF. In this band, the system response is no longer dominated by the first resonance and exhibits significantly reduced dynamic amplification. As a consequence, the excitation amplitude was reduced compared to the lower-frequency blocks. This choice avoids unnecessary mechanical loading while still ensuring that the response remains sufficiently above the sensor noise floor. Importantly, lowering the amplitude also helps to reduce broadband noise injection from the shaker and control system, which was observed to be a major contributor to the noisy FRF obtained with the initial full-band sweep approach. A substantially higher sweep rate was employed in this block to further mitigate noise while keeping the measurement efficient. At higher frequencies, the structural response reaches steady-state rapidly, and long dwell times are not required. The sweep rate was therefore selected based on the requirement that multiple oscillation cycles occur within the effective frequency resolution used for FRF estimation, ensuring quasi-steady behavior without excessive time spent at frequencies where the response amplitude is small. By increasing the sweep rate, the total acquisition time is reduced, which in turn limits the accumulation of low-level noise and drift effects that were evident in earlier broadband measurements.

4.2.1.4 Upper-frequency excitation block (425–750 Hz)

The final excitation block, covering the frequency range from 425 to 750 Hz, was introduced to extend the measured frequency response function to the upper limit required for the analysis while explicitly accounting for both the increased noise level observed in preliminary broadband sweeps and the operational limitations of the shaker system at high frequencies. Initial full-band tests demonstrated that applying uniform excitation parameters across this region resulted in unstable shaker behavior and a highly noisy FRF tail. In this high-frequency band, the structural response amplitude is inherently small and the dynamics are predominantly stiffness-controlled. Increasing excitation amplitude in this

region does not lead to a proportional increase in useful response, but instead increases the risk of exciting broadband noise and compromising shaker control. For this reason, the excitation amplitude was deliberately reduced to 0.05 g, which represents a level that ensures stable shaker tracking throughout the sweep while still producing a measurable response above the noise floor. The sweep rate for this block was selected based on empirical evaluation of shaker stability rather than structural settling considerations. A sweep rate of 433 Hz/min was identified as the maximum rate that allowed reliable tracking of the sine excitation without controller saturation or loss of synchronization, particularly near the upper end of the frequency range. Faster sweeps were found to induce instability or aborted runs, while slower sweeps provided no significant improvement in measurement quality relative to the increased test duration. The chosen sweep rate therefore represents an optimal compromise between measurement efficiency and operational stability.

To provide a clear overview of the experimental setup, Table 4.4 summarizes the frequency ranges and excitation parameters adopted for each sine-sweep block, including excitation amplitude, sweep rate, sampling frequency, and number of acquired samples.

Table 4.4: Excitation parameters adopted for each sine-sweep block

Parameter	Block 1	Block 2	Block 3	Block 4
Frequency range [Hz]	5–30	24–100	85–500	425–750
Excitation amplitude [g]	0.23	0.23	0.15	0.05
Sweep rate [Hz/min]	4	50	340	433
Sampling frequency [kHz]	0.6	1.2	6.0	9.6
Number of samples	262144	131072	524288	1048576

4.2.1.5 Limitations above 750 Hz

The original objective of the forced-oscillation campaign was to extend the measured frequency response function up to 1500 Hz in order to fully characterize the high-frequency behavior of the system. However, practical limitations of the shaker and control system prevented reliable testing beyond approximately 750 Hz. Preliminary attempts to extend the sweep to higher frequencies revealed a strong trade-off between sweep rate, excitation stability, and measurement noise. In order to reduce noise in the high-frequency response, faster sweep rates were desirable, as longer dwell times at frequencies with very low structural response tended to amplify broadband noise and drift effects. Nevertheless, when sweep rates higher than the final selected value were applied, the shaker repeatedly lost tracking or aborted the test, particularly in the upper frequency region. Conversely, reducing the sweep rate to maintain shaker stability resulted in excessively long acquisition times and a significant degradation of signal quality due to increased noise. This behavior indicates that, in the high-frequency range, shaker controllability rather than structural dynamics became the dominant limiting factor. As a result, a practical upper frequency limit of 750 Hz was adopted, representing the highest frequency at which a stable and repeatable sweep could be performed while maintaining an acceptable balance between noise level, sweep duration, and shaker tracking reliability. This compromise ensured the integrity and usability of the measured FRF while avoiding test interruptions or unstable operating conditions.

4.2.2 Results

The post-processing of the forced-oscillation measurements was structured to reflect the block-wise acquisition strategy adopted during testing and to ensure consistency, numerical robustness, and physical interpretability of the resulting FRF. Rather than treating the dataset as a single uniform time record, the processing pipeline was explicitly organized around the individual excitation blocks, with each block processed independently using parameters consistent with its acquisition conditions, and subsequently merged into a single FRF covering the full range from 5 to 750 Hz.

For each excitation block, the raw time histories of acceleration (shaker reference) and velocity (measured response) were first imported and synchronized using the common time vector. A detrending operation was applied to both signals to remove low-frequency offsets and slow drift, which is particularly important when dealing with long records and low-frequency excitation. Block-specific band-pass filtering was then applied, with cutoff frequencies matched to the excitation bandwidth of each block. This filtering step serves multiple purposes: it suppresses out-of-band noise, limits the amplification of low-frequency drift during numerical integration, and ensures that the processed signals contain only frequency content relevant to the current block. The acceleration signal was then numerically integrated in the time domain to obtain a velocity contribution associated with the base excitation. This integrated velocity was subtracted from the directly measured velocity signal to compute the relative velocity response of the structure. Performing this operation in the time domain, after filtering and detrending, minimizes integration drift and ensures a physically meaningful relative motion signal. The resulting input and output velocity signals were then transformed into the frequency domain using the Fast Fourier Transform (FFT), and the complex FRF for each block was computed as the ratio between output and input spectra.

Once the individual block FRFs were computed, the full-spectrum FRF was assembled through a controlled merging process. A linearly spaced frequency grid consisting of 6000 points spanning the full investigated bandwidth was defined as a common reference. Each block FRF was interpolated onto this grid in the complex domain in order to preserve both magnitude and phase continuity. At each frequency point, the merged FRF was obtained by averaging the available block contributions while ignoring missing or invalid values. As a result of this procedure, 5998 valid frequency points out of the 6000 defined grid points were retained in the final merged FRF, indicating excellent frequency coverage and strong consistency across the overlap regions between adjacent blocks. From the merged FRF, the magnitude and phase responses were then extracted and represented in Bode form for visualization and analysis.

4.2.2.1 Frequency Response Function Estimation

Although frequency response functions can in principle be expressed in terms of displacement, velocity, or acceleration, the FRFs presented in this work are formulated in terms of velocity. This choice was made based on both physical and numerical considerations related to the experimental setup and the characteristics of the measured signals. From a theoretical standpoint, the modal parameters of interest, including resonance frequency and quality factor, are intrinsic properties of the system and are invariant with respect

to the chosen kinematic representation. However, from a practical experimental perspective, the use of velocity-based FRFs offers significant advantages in terms of numerical robustness and noise sensitivity. In particular, obtaining displacement responses from acceleration or velocity measurements requires additional integration, either in the time domain or through division by frequency in the frequency domain, both of which amplify low-frequency noise, bias, and drift. These effects are especially pronounced outside the resonance region and can obscure the baseline behavior of the FRF, making peak identification and 3 dB bandwidth estimation less reliable. Furthermore, velocity is directly related to dissipative forces in viscously damped systems, which enhances the physical interpretability of damping-related features such as resonance bandwidth and quality factor.

To isolate the blade's true vibration, the difference between the absolute blade velocity measured by the Vibrometer and the integrated bench velocity derived from the accelerometer signal has been calculated during post-processing of the data. This subtraction yields the relative velocity of the blade with respect to its moving support, effectively removing the rigid-body component of the motion. The resulting relative-motion signal is critical for accurate modal identification, transmissibility analysis, and interpretation of the blade's dynamic response under shaker excitation, as it reflects only the deformation-induced motion of the structure rather than the motion imposed by the test bench.

Figure 4.13 presents the frequency response function estimated using the relative blade velocity, computed by subtracting the base motion from the absolute velocity measured by the laser vibrometer. This representation emphasizes the motion of the blade relative to the base and highlights dynamic features associated with deformation and relative amplification. The resulting FRF exhibits distinct resonance and anti-resonance patterns, reflecting the interaction between the blade dynamics and the base excitation.

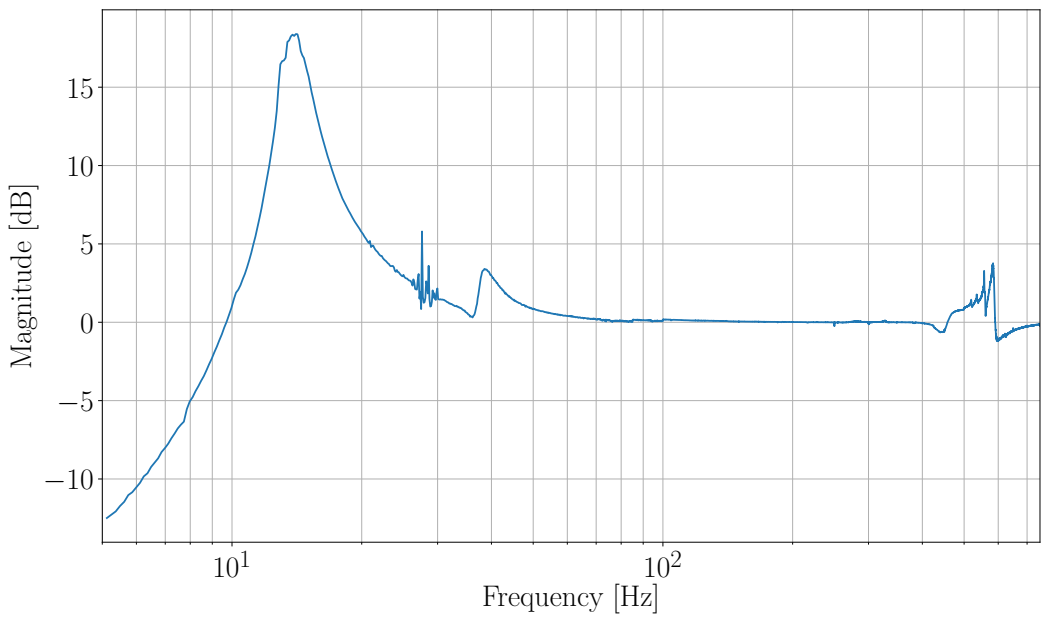

Figure 4.13: FRF obtained from relative velocity

While the relative velocity provides valuable information on the deformation of the blade with respect to the base, its computation inherently includes the input signal within the output definition. This kinematic coupling introduces artificial zeros and phase distortions in the resulting transfer function, which may obscure the intrinsic dynamic characteristics of the structure. By using the absolute velocity response, the input and output signals remain independent, allowing the FRF to more directly represent the physical transfer behavior of the system. The absolute velocity-based FRF therefore provides a clearer and more robust basis for modal interpretation, while the relative response is reserved for complementary analyses focused on deformation and transmissibility.

The FRF was initially estimated using a direct FFT-based approach, computed as the ratio between the Fourier transforms of the output and input velocity signals. While this method provides a straightforward estimation of the system dynamics, it is known to be highly sensitive to noise, spectral leakage, and non-stationarities, particularly when applied to swept-sine excitations. In such conditions, the excitation energy is not uniformly distributed in time across frequencies, leading to significant distortions in the estimated FRF magnitude.

To obtain a more robust and statistically reliable estimate, the FRF was subsequently computed using the Welch method with an H1 estimator. The Welch approach consists of dividing the time-domain signals into partially overlapping segments, applying windowing to each segment, and averaging the resulting spectral estimates [2]. This procedure significantly reduces the variance of the power spectral density (PSD) estimates and mitigates spectral leakage effects. The H1 estimator, defined as the ratio between the cross-power spectral density of input and output and the auto-power spectral density of the input, is particularly well suited for forced excitation tests, as it minimizes the influence of output noise on the FRF estimation.

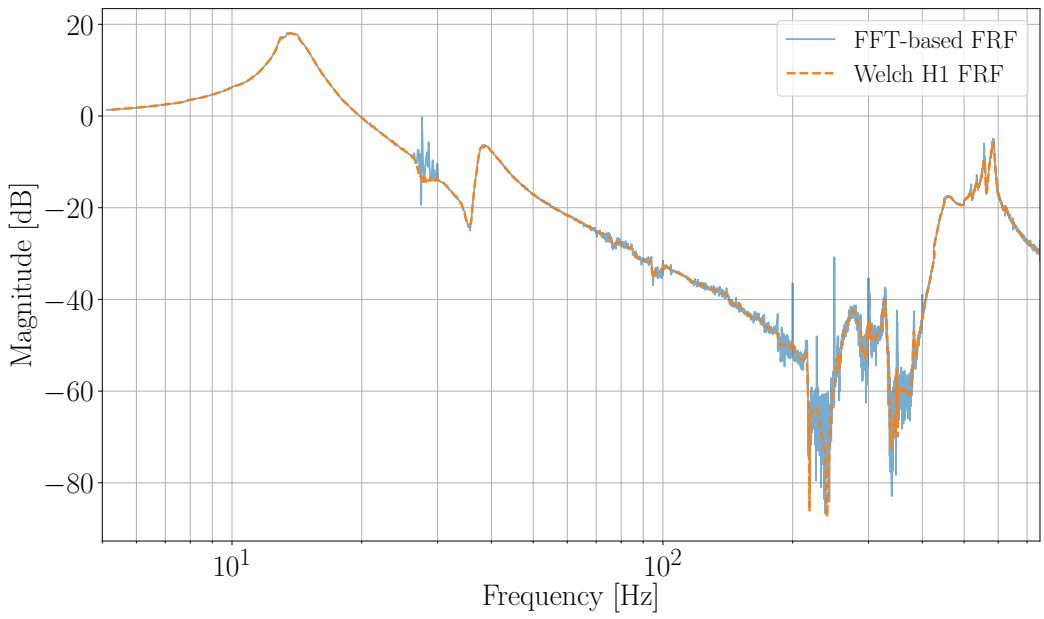

Figure 4.14: FRF Comparison: FFT-based vs Welch-based

The comparison between the FFT-based and Welch-based FRFs, shown in Figure 4.14, highlights the advantages of the Welch approach. While both methods correctly identify the main resonance frequencies, the FFT-based FRF exhibits pronounced high-frequency fluctuations and a significantly higher noise floor, especially in frequency regions with lower excitation energy. In contrast, the Welch-based FRF presents a smoother and more physically meaningful magnitude response, with well-defined resonances and anti-resonances and a substantially reduced noise level. This improved clarity is particularly evident in the mid-to-high frequency range, where the FFT-based estimate is dominated by broadband noise, whereas the Welch estimate preserves the underlying dynamic features of the system. Based on these observations, the Welch-based H1 FRF was adopted as the reference representation of the system dynamics for all subsequent analyses, including modal identification and damping estimation. The FFT-based FRF is retained solely for qualitative comparison purposes, as it does not provide the same level of interpretability under swept-sine excitation conditions.

Figure 4.15 presents a comparison of the velocity-based frequency response function magnitudes obtained for five different blade pairs using Welch's method over the investigated frequency range.

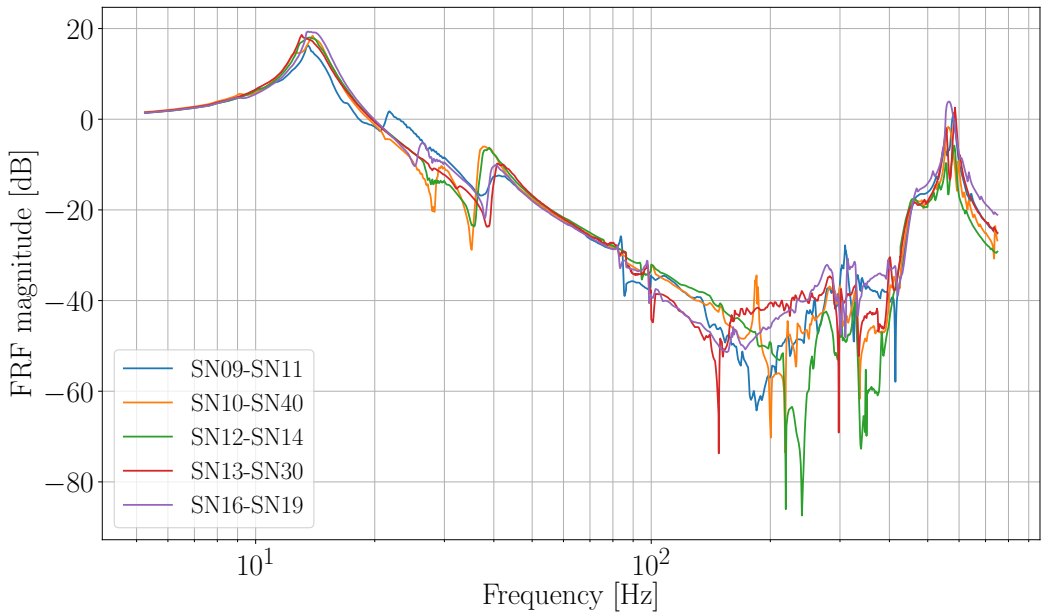

Figure 4.15: FRF magnitude comparison for five different blade pairs

At low frequencies, below approximately 10 Hz, all curves exhibit a similar increasing trend in magnitude, corresponding to the stiffness-dominated response of the system. In this region, the FRF magnitude rises smoothly with frequency and shows only minor variability between blade pairs, indicating a consistent global structural behavior and good repeatability of the measurements. Small deviations observed in this range can be attributed to low-frequency noise and slight differences in boundary conditions or assembly tolerances, which are known to have a greater influence when the dynamic response amplitudes are relatively small. At higher frequencies, distinct resonance peaks appear,

although their amplitudes and exact frequencies vary among specimens, reflecting the increasing sensitivity of the system to local structural differences, damping distribution, and coupling effects.

4.2.2.2 Spectral analysis of the dynamic response (5–30 Hz)

The dynamic response in the 5–30 Hz frequency range was first assessed using raw spectral quantities in order to evaluate data quality and establish the validity limits of the measurement prior to any modal parameter identification [2, 12].

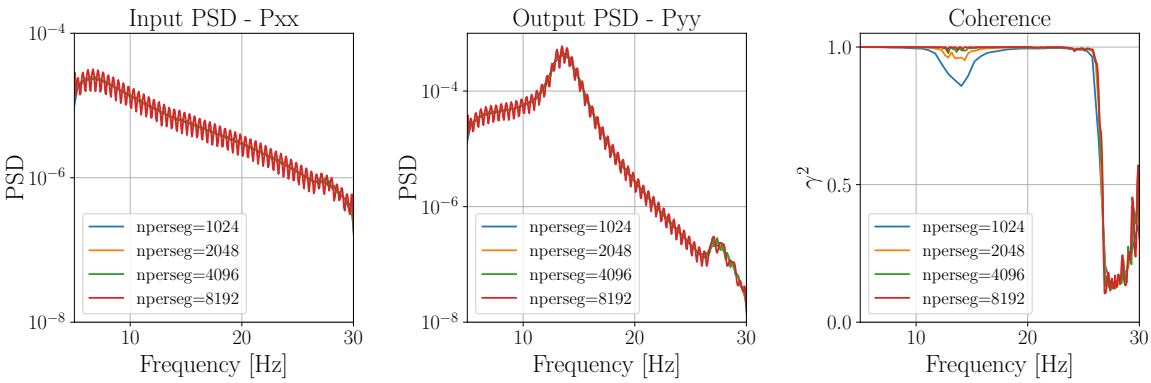

Figure 4.16: Input/Output PSD - `nperseg` sensitivity (5-30 Hz)

Figure 4.16 (left) shows the input power spectral density (P_{xx}) computed using the Welch method for different segment lengths (`nperseg`). The PSD exhibits a smooth and monotonic decay with frequency, with a consistent spectral envelope across all segment lengths. The increasing oscillatory pattern observed for larger `nperseg` values is attributed to the interaction between the sweep excitation and the Welch averaging procedure and does not affect the underlying energy distribution. The invariance of the PSD envelope with respect to the estimation parameters confirms that the excitation level is sufficient and well distributed throughout the 5–30 Hz band, and that the input signal is robust against changes in spectral resolution.

Figure 4.16 (center) presents the output PSD (P_{yy}), which is dominated by a single, well-defined peak at approximately 14 Hz, corresponding to the fundamental dynamic mode of the system. The close agreement of the output PSD curves for all `nperseg` values up to approximately 25 Hz indicates that the measured response in this frequency range is governed by the physical system dynamics rather than by estimation artifacts or noise. Beyond approximately 25 Hz, the output PSD approaches the noise floor and exhibits increased irregularity, suggesting a reduction in the signal-to-noise ratio and a progressive loss of usable dynamic information.

The coherence function

$$\gamma^2 = \frac{P_{xy}^2}{P_{xx} \cdot P_{yy}}$$

shown in Figure 4.16 (right), remains close to unity over the entire frequency range up to approximately 25 Hz for all considered values of `nperseg`, demonstrating a strong linear

correlation between input and output and confirming the reliability of the FRF estimation in this region. A small local reduction in coherence near the resonance is observed for the smallest segment length, which is attributed to insufficient frequency resolution rather than to nonlinearity or measurement errors. A sharp and repeatable drop in coherence occurs beyond approximately 26–27 Hz, independent of the spectral estimation parameters, indicating a genuine experimental limitation rather than a numerical artifact. This behavior is consistent with a decreasing excitation level and an output response approaching the noise floor.

Based on this comprehensive raw-signal analysis, the dynamic behavior in the 5–25 Hz range can be reliably characterized as dominated by a single, isolated mode with high signal quality and linear response. The consistent identification of an upper validity limit at approximately 25 Hz across all spectral quantities defines the effective frequency range of this dataset. Frequencies above this limit are not suitable for modal parameter extraction and are excluded from subsequent analyses.

4.2.2.3 Spectral analysis of the dynamic response (24–100 Hz)

On the 24-100 Hz band, the output PSD (P_{yy}) in Figure 4.17 reveals a pronounced minimum around approximately 34–36 Hz, followed by a local maximum at approximately 38–40 Hz. This characteristic dip-and-peak pattern is consistently observed for all values of `nperseg` and therefore cannot be attributed to numerical artifacts or estimation noise. Instead, it is indicative of a dynamic anti-resonance followed by a resonance, a behavior that is incompatible with a single-degree-of-freedom system and strongly suggests the presence of multiple interacting degrees of freedom. Beyond approximately 45 Hz, the output PSD exhibits a smooth monotonic decay with frequency, approaching the noise floor only near the upper end of the analyzed range.

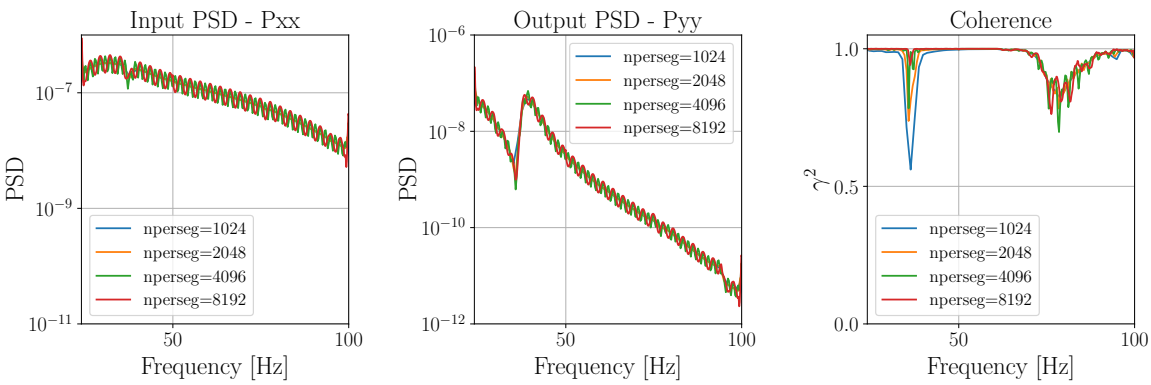

Figure 4.17: Input/Output PSD - `nperseg` sensitivity (24-100 Hz)

The coherence function remains close to unity over most of the frequency range, confirming a strong linear correlation between input and output and validating the use of FRF-based analysis in this band. However, two localized and repeatable reductions in coherence are observed: the first centered around the PSD minimum near 35 Hz, and the second spanning approximately 70–85 Hz. The coherence dip near 35 Hz coincides with the anti-resonance observed in the output PSD and is a well-known feature of multi-degree-of-freedom systems, where destructive interference between modal contributions

reduces the correlation between input and output [12]. The higher-frequency coherence reduction is broader and shallower, suggesting a gradual decrease in signal-to-noise ratio rather than a sharp loss of linearity.

The phase of the FRF, shown in Figure 4.18, exhibits an abrupt phase inversion in correspondence with the anti-resonance region, followed by a recovery toward a nearly constant phase at higher frequencies. This behavior is consistent with the presence of a zero in the transfer function and indicates destructive interference between multiple dynamic contributions providing further confirmation that the observed dip in magnitude is a true anti-resonance rather than a consequence of noise or insufficient excitation.

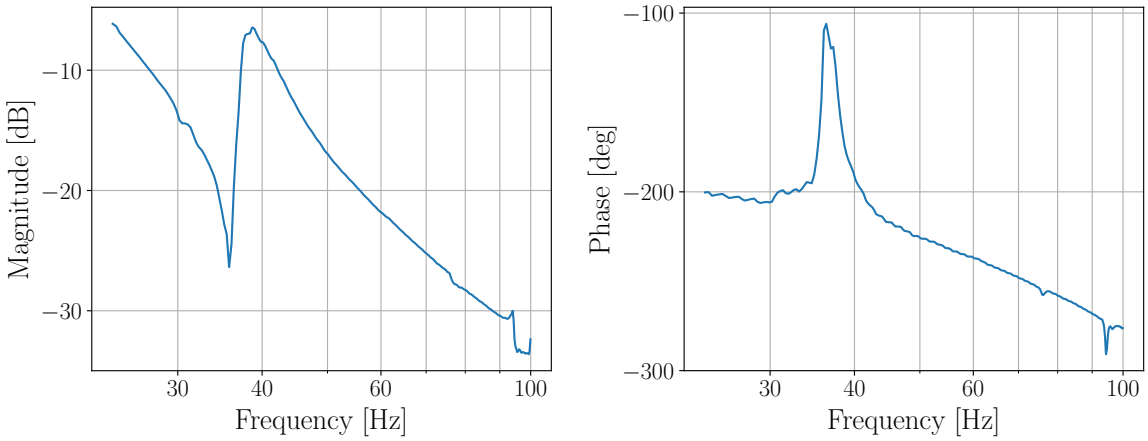

Figure 4.18: FRF Magnitude and Phase (24-100 Hz)

Based on the combined analysis of PSDs, coherence, FRF magnitude and phase, the dynamic response in the 24–100 Hz range cannot be described by a single-degree-of-freedom model. The presence of a clear anti-resonance followed by a resonance indicates modal interaction and suggests the influence of additional dynamic degrees of freedom, potentially associated with local structural modes, attachment compliance, or shaker–structure coupling. The robustness of these features with respect to the spectral estimation parameters and the generally high coherence confirm that the data are suitable for further modal analysis. However, any subsequent identification procedure must employ a multi-degree-of-freedom model capable of capturing both poles and zeros of the system, as local single-mode fitting would be insufficient to represent the observed dynamics.

4.2.2.4 Spectral analysis of the dynamic response (85–500 Hz)

For the 85–500 Hz frequency range, the spectral analysis was performed using a fixed set of Welch parameters, as preliminary investigations showed that variations in segment length no longer have a significant influence on the estimated spectral quantities. At these higher frequencies, the excitation sweep provides a sufficiently dense frequency coverage and the averaging inherent to the Welch method ensures stable estimates. Consequently, further sensitivity analysis with respect to `nperseg` was deemed unnecessary, allowing the focus to shift toward the physical interpretation of the measured response.

Figure 4.19 shows the input power spectral density, which exhibits a smooth and

monotonic decay over the analyzed frequency range. The absence of abrupt drops or irregularities indicates that the excitation remains present throughout the band, although its energy progressively decreases with increasing frequency. This behavior is typical of shaker-driven sweep excitations and does not, by itself, limit the validity of the measurements. The output PSD, displays a markedly different behavior compared to the lower-frequency blocks. Between approximately 150 Hz and 350 Hz, the output energy drops by several orders of magnitude and approaches the noise floor, with multiple narrow minima and isolated spikes. These features suggest that the measured response in this region is strongly attenuated, either due to low structural receptance or due to destructive interference between multiple dynamic contributions. At higher frequencies, above approximately 400 Hz, a partial recovery of the output PSD is observed, indicating the presence of localized dynamic activity, albeit at very low amplitude levels.

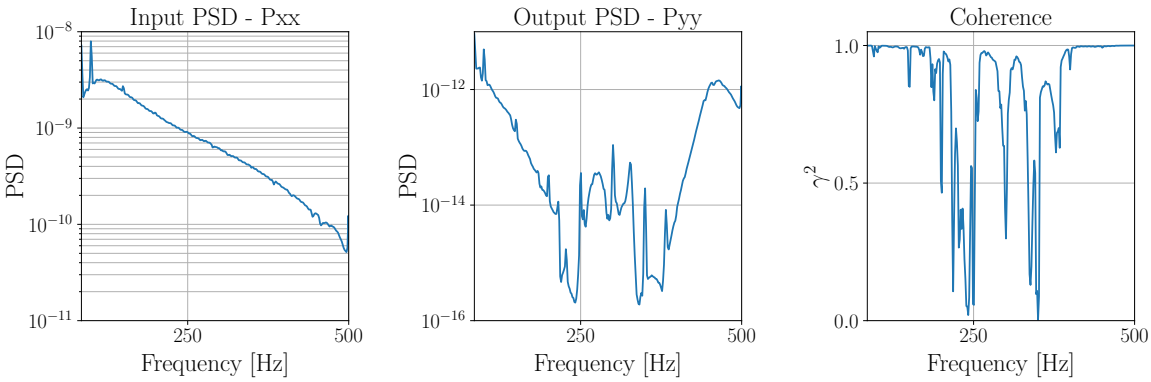

Figure 4.19: Input/Output PSD (85-500 Hz)

The coherence function provides critical insight into the reliability of the FRF estimation in this frequency range. While the coherence remains close to unity at the lower and upper ends of the band, it exhibits multiple sharp and deep drops between approximately 180 Hz and 380 Hz. These coherence losses coincide with the regions where the output PSD approaches the noise floor and indicate a reduced signal-to-noise ratio. Importantly, the coherence degradation is not broadband but occurs in narrow frequency intervals, suggesting that it is primarily driven by local dynamic cancellations and low response levels rather than by a global loss of linearity or measurement issues.

The magnitude of the frequency response function, reported in Figure 4.20, reflects this behavior. The FRF magnitude decreases steadily with frequency, reaching extremely low levels in the mid-band region where several deep and narrow minima are observed. These minima are characteristic of anti-resonant behavior or zeros in the transfer function and are consistent with the presence of multiple interacting degrees of freedom. Above approximately 400 Hz, the FRF magnitude increases again, forming a broad high-frequency feature that may be associated with a local structural mode or with dynamics of the shaker–structure–sensor assembly. However, given the low response levels and partial coherence degradation in this region, this feature should be interpreted with caution.

The phase response of the FRF, exhibits large and rapid variations over the analyzed frequency range, including multiple abrupt phase rotations. While such behavior is expected in the presence of closely spaced poles and zeros, the strong phase fluctuations

observed in regions of low coherence and low output PSD indicate that the phase information is increasingly dominated by noise and numerical sensitivity. As a result, the phase in this frequency band is not considered suitable for detailed dynamic interpretation and is used only to qualitatively support the identification of regions affected by anti-resonances and modal interactions.

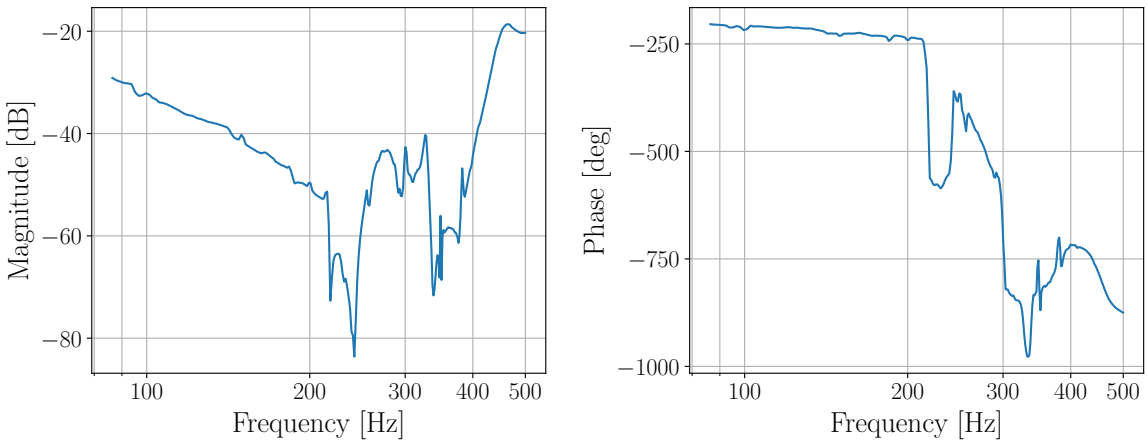

Figure 4.20: FRF Magnitude and Phase (85-500 Hz)

Overall, the analysis of the 85–500 Hz range indicates a transition toward a highly complex and weakly excited dynamic regime, characterized by multiple interacting degrees of freedom, pronounced anti-resonant behavior, and a progressively decreasing signal-to-noise ratio. While the data remain physically meaningful in selected sub-bands, the widespread coherence degradation and extremely low FRF levels suggest that this frequency range is not well suited for robust modal parameter identification. Any subsequent modeling effort in this band should therefore be supported by complementary analyses, such as numerical simulations or targeted narrow-band excitations, to improve confidence in the identified dynamic features.

4.2.2.5 Spectral analysis of the dynamic response (425–750 Hz)

The dynamic response in the 425–750 Hz frequency range was analyzed to investigate the presence of high-frequency dynamic features and to assess the reliability of the measured data in a regime characterized by low response amplitudes and increased system complexity.

Figure 4.21 shows the input power spectral density, which exhibits a broadband excitation over the analyzed frequency range, with moderate fluctuations and a gradual reduction in spectral energy toward higher frequencies. Sufficient excitation energy is maintained across the frequency band of interest, ensuring proper stimulation of the system dynamics and excluding insufficient input as the primary cause of the observed response attenuation. The output PSD exhibits a generally low baseline level over most of the frequency range, superimposed with a clear and localized increase in energy centered around approximately 580–600 Hz. This pronounced energy peak indicates the activation of a high-frequency dynamic contribution, while the surrounding frequency regions show significantly lower response levels. Beyond approximately 600 Hz, the output

PSD decreases steadily toward higher frequencies, suggesting a progressive attenuation of the system response and a reduction in effective receptance. It is noted that the blades are impregnated with viscoelastic material, which is expected to introduce increased and frequency-dependent energy dissipation. This effect provides a physically consistent explanation for the strong attenuation of the response observed at higher frequencies. The coherence function remains close to unity over most of the analyzed band, confirming that the measured input–output relationship is predominantly linear despite the low response amplitudes. Localized drops in coherence are observed in proximity to the energy peak around 580–600 Hz and at a limited number of neighboring frequencies. These coherence reductions coincide with regions of rapid variation in the response and are attributed to local dynamic interactions and increased sensitivity of the spectral estimates rather than to a global degradation of measurement quality.

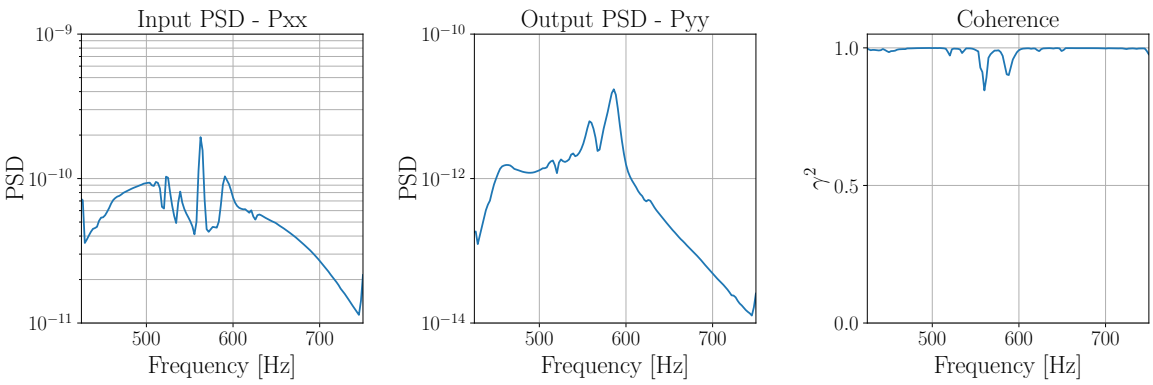

Figure 4.21: Input/Output PSD (425-750 Hz)

The magnitude of the frequency response function, reported in Figure 4.22, reflects a complex high-frequency behavior. A dominant peak is observed around approximately 580–600 Hz, consistent with the energy increase detected in the output PSD. This peak is accompanied by several sharp magnitude reductions in its vicinity, indicating the presence of closely spaced poles and zeros in the transfer function.

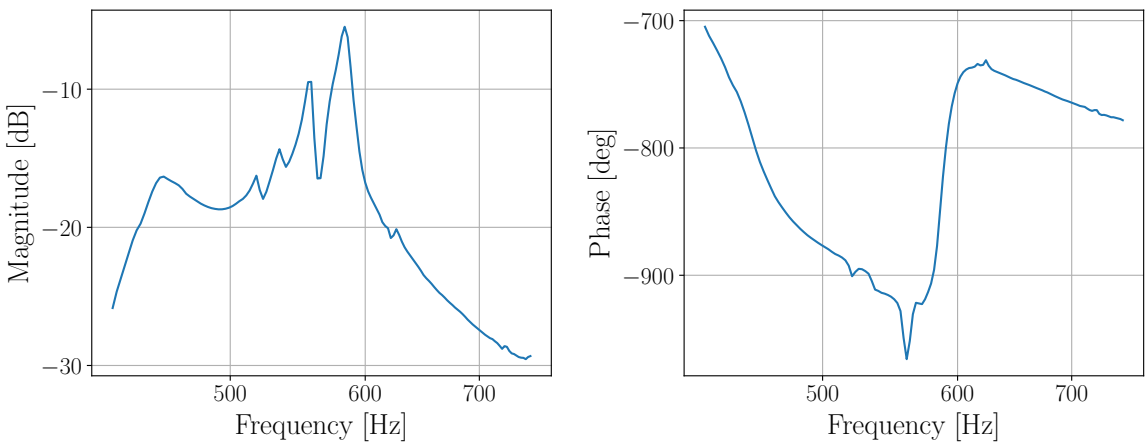

Figure 4.22: FRF Magnitude and Phase (425-750 Hz)

Such behavior is characteristic of a multi-degree-of-freedom dynamic regime, potentially involving local structural modes or interactions with the excitation and measurement interfaces. Beyond the dominant peak, the FRF magnitude decreases smoothly with increasing frequency, in agreement with the decay observed in the output PSD. The phase response of the FRF exhibits large and rapid variations in correspondence with the dominant magnitude peak and the surrounding minima. These phase excursions are consistent with strong dynamic interactions and anti-resonant effects in the high-frequency range. However, given the low response levels outside the dominant peak region, the phase becomes increasingly sensitive to noise and numerical effects. Consequently, phase information in this frequency band is interpreted only qualitatively and is not used for detailed dynamic characterization.

Overall, the analysis of the 425–750 Hz frequency range reveals the presence of a localized high-frequency dynamic feature around 580–600 Hz embedded in a generally weak response regime. While this feature is consistently observed across all spectral quantities and therefore represents a physical characteristic of the system, the combination of low response amplitudes, localized coherence degradation, and complex FRF structure limits the robustness of modal parameter identification. At the same time, the limited sensitivity of the 580 Hz feature to blade-to-blade variations suggests a reduced degree of blade-specific participation compared to the first bending mode. This behavior is consistent with either a higher-order local blade mode with weak modal participation or a coupled blade–fixture mode involving both the test specimen and the experimental setup. Since experimental FRFs inherently reflect the dynamics of the complete coupled system (including blade, fixture, and shaker) it is not always possible to unambiguously distinguish between purely blade-dominated modes and rig-influenced modes based on experimental data alone. For this reason, an eigenfrequency study in COMSOL Multiphysics 6.3 was identified as a valuable complementary tool to support the experimental findings (see Section 5.3 for further details).

In this context, a numerical eigenfrequency analysis provides a complementary perspective by identifying the intrinsic natural frequencies and mode shapes of the structure independently of excitation and measurement constraints. While the eigenfrequency study is discussed in detail in a dedicated chapter, its role within the present work is to support the physical interpretation of the experimentally observed resonances and anti-resonances. In particular, the numerical results are used to assess whether the dynamic features identified in the forced response correspond to structural modes, local components, or coupling effects introduced by the test configuration. The comparison between experimentally observed frequency bands and numerically predicted eigenfrequencies allows the experimental findings to be interpreted within a physically consistent framework, while acknowledging that not all numerical modes are expected to be observable in the measured FRFs due to excitation, sensor placement, and boundary condition effects.

4.2.2.6 Multi-modal Analysis

To further interpret the experimentally obtained frequency response function and to identify the dynamic contributions of multiple resonant phenomena, a multimodal fitting procedure was implemented directly in the frequency domain. The starting point of the analysis is the complex-valued velocity-based frequency response function $H(\omega)$, obtained

from the merged experimental data and expressed in decibels to facilitate peak detection:

$$H_{\mathrm{dB}}(\omega) = 20 \log_{10} \left(|H(\omega)| \right)$$

A preliminary identification of candidate resonant frequencies is performed by detecting prominent peaks in the FRF magnitude using a prominence-based criterion. This step provides an initial estimate of the modal frequencies and supports the initialization of the nonlinear optimization process.

Starting from the equation of motion of a linear single-degree-of-freedom oscillator with viscous damping,

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = f(t)$$

and introducing the natural frequency $\omega_n = \sqrt{k/m}$ and the damping ratio $\zeta = c/2m\omega_n$, the steady-state harmonic response can be expressed in the frequency domain. The displacement-to-force frequency response function (receptance) is given by

$$\frac{X}{F} = \frac{1/m}{\omega_n^2 - \omega^2 + j2\zeta\omega_n\omega}$$

Since the measured quantity is velocity, the frequency response function is obtained by multiplying the receptance by $j\omega$, yielding the velocity-to-force transfer function

$$\frac{V}{F} = \frac{j\omega/m}{\omega_n^2 - \omega^2 + j2\zeta\omega_n\omega}$$

In practical experimental conditions, the measured frequency response function may include additional scaling factors, phase shifts, and kinematic contributions arising from the specific definition of input and output quantities. These effects are therefore incorporated into a complex-valued modal constant. As a result, the velocity-based frequency response function is modeled in the general form

$$H(\omega) = \sum_{k=1}^{N} \frac{A_k}{\omega_{n,k}^2 - \omega^2 + j2\zeta_k\omega_{n,k}\omega}$$

where $\omega_{n,k}$ and ζ_k are the natural frequency and damping ratio of the k-th mode, and A_k is a complex modal constant accounting for modal participation and phase effects. This formulation provides a compact and flexible representation suitable for multi-modal curve fitting of experimentally measured frequency response functions.

The fitting is performed over the tested frequency range by minimizing the discrepancy between the measured FRF $H_{\mathrm{meas}}(\omega)$ and the analytical model. This discrepancy is quantified through a residual vector defined in the complex domain. To enable the use of real-valued nonlinear least-squares solvers, the complex residual is decomposed into its real and imaginary components:

$$\mathbf{r} = \begin{bmatrix} \Re\left(H_{\mathrm{model}}(\omega) - H_{\mathrm{meas}}(\omega)\right) \\ \Im\left(H_{\mathrm{model}}(\omega) - H_{\mathrm{meas}}(\omega)\right) \end{bmatrix}$$

The optimization problem thus consists of finding the set of modal parameters $\{\omega_{n,k}, \zeta_k, A_k\}$ that minimizes the Euclidean norm of the residual vector:

$$\min_{\theta} \|r(\theta)\|_2$$

where θ denotes the vector collecting all unknown modal parameters.

Initial guesses for the modal frequencies are obtained from the detected FRF peaks, while initial damping ratios are chosen conservatively to reflect lightly damped structural behavior. The initial modal constants are estimated from the local FRF amplitude in the vicinity of each resonance and scaled by the square of the corresponding natural frequency to ensure physically meaningful starting values. Bound constraints are imposed on the optimization variables to enforce stability and physical admissibility, in particular by restricting damping ratios to positive values and preventing nonphysical negative frequencies.

The nonlinear least-squares problem is solved using an iterative trust-region-based algorithm, which updates the modal parameters until convergence is achieved or a maximum number of function evaluations is reached. Once the optimal parameter set is obtained, the fitted multimodal FRF is reconstructed over the full frequency range and compared directly to the measured FRF in both magnitude and phase. This comparison allows for a clear assessment of the ability of the identified modes to reproduce not only the dominant resonance but also secondary dynamic features observed at higher frequencies [4].

The final selection of the modal frequencies used for the fitting procedure was not based on an automatic prominence-based peak detection applied directly to the FRF magnitude. Although such methods are useful for an initial screening, they were found to be sensitive to local fluctuations, anti-resonances, and noise-dominated regions of the spectrum. Instead, the final choice of the modal frequencies was guided by the comprehensive spectral analysis presented in the previous paragraphs, which combined FRF magnitude, phase behavior, power spectral densities, and coherence. This approach allowed the identification of frequency regions characterized by high coherence, stable phase evolution, and physically meaningful resonant behavior. As a result, three modes were selected at approximately 14 Hz, 38 Hz, and 580 Hz, corresponding to dominant and consistently observable dynamic features of the system.

An important outcome of the fitting process is the robustness of the identified modal parameters. Starting from physically reasonable initial guesses for the damping ratios (0.1, 0.05, and 0.01 for the first, second, and third mode, respectively), the optimization converges repeatedly toward the same natural frequencies and damping values, even when the initial frequency estimates are slightly perturbed. This convergence behavior indicates that the identified modes are well separated, structurally significant, and not artifacts of local noise or numerical conditioning. In particular, the stability of the estimated damping ratios suggests that the data contain sufficient information to reliably characterize the energy dissipation mechanisms associated with the dominant modes.

The comparison between measured and fitted FRF magnitude shown in Figure 4.23 further supports these conclusions. The first mode is reproduced with excellent accuracy in terms of resonance frequency, peak amplitude, and bandwidth, which is essential for the reliable estimation of damping. The second mode is also reasonably captured, with correct localization of the anti-resonance and an acceptable representation of its overall contribution to the response. Minor discrepancies in peak sharpness and surrounding slopes are observed, but they remain limited and do not compromise the physical inter-

pretation of the mode. These deviations are consistent with the simplified nature of the reduced-order model and with the increasing influence of neighboring dynamics at higher frequencies.

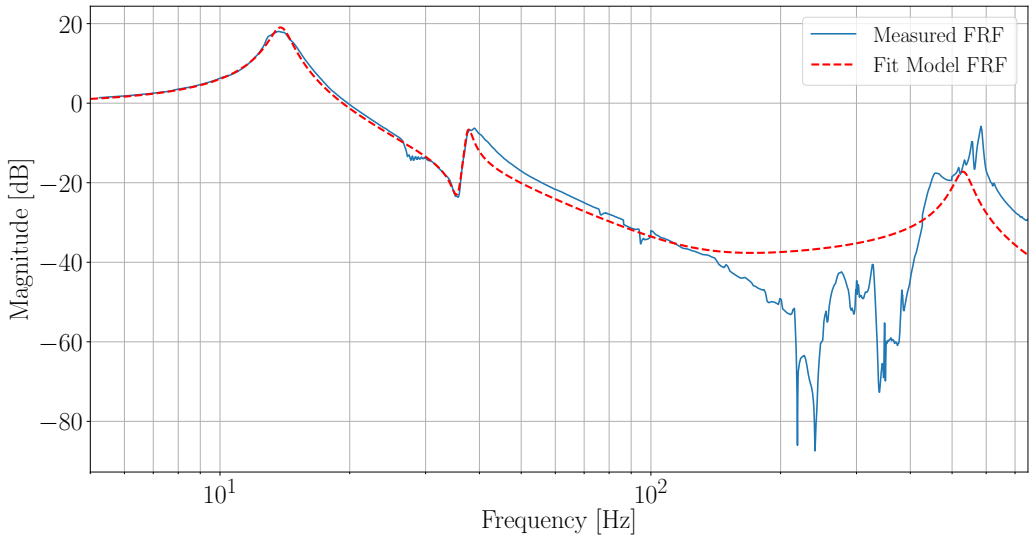

Figure 4.23: FRF Comparison: Measured vs Modal Fit

In contrast, the model is intentionally not forced to reproduce fine-scale features such as deep anti-resonances or narrow local irregularities visible in the measured magnitude. These features are primarily associated with modal interactions, spatial effects, and the specific input–output configuration, and cannot be represented without explicitly introducing zeros or additional modes.

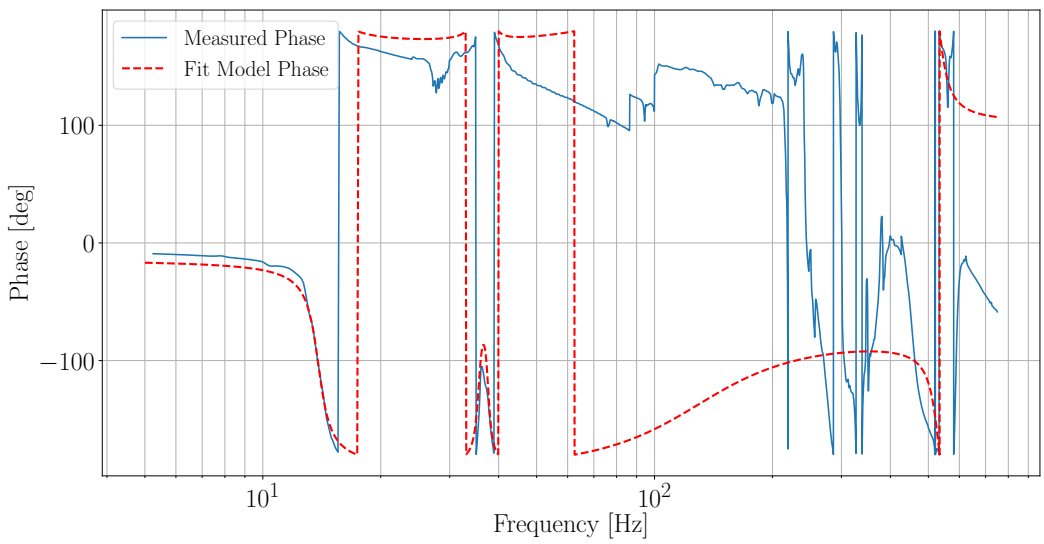

Figure 4.24: Phase Comparison: Measured vs Modal Fit

The phase response shown in Figure 4.24 provides further insight into the validity

and limitations of the model. The main phase rotations associated with the identified modes are well aligned between the measured and fitted responses, confirming that the pole locations are correctly captured. However, the measured phase exhibits multiple abrupt jumps and irregular trends in frequency regions characterized by low coherence and complex dynamic behavior. Consequently, the phase is used here primarily as a qualitative validation tool, rather than as a quantity to be matched point by point.

At higher frequencies (approximately above 50 Hz), the quality of the fit is further limited by the characteristics of the measured data itself. The raw-signal analysis has already shown a progressive reduction in coherence in this frequency range, indicating a lower signal-to-noise ratio and a weaker linear relationship between input and output. Under these conditions, an optimal fit cannot be expected, regardless of model complexity. The observed mismatch between measured and fitted FRFs in this region is therefore consistent with the intrinsic limitations of the experimental data rather than with deficiencies in the identification approach.

Although the reduced-order modal model provides an accurate representation of the dominant system dynamics, several strategies could, in principle, be adopted to further improve the agreement between the measured and fitted frequency response function (FRF). Each strategy, however, involves a trade-off between local accuracy, robustness of the identified parameters, and physical interpretability of the model. The main available options are discussed below.

1. Increasing the number of modes

A straightforward approach to improve the fit consists in increasing the number of retained modes in the global modal expansion. In the frequency domain, the velocity FRF can be expressed as a superposition of modal contributions,

$$H(\omega) = \sum_{r=1}^{N} \frac{A_r \, i\omega}{\omega_r^2 - \omega^2 + 2i\zeta_r\omega_r\omega}$$

where ω_r and ζ_r are the natural frequency and damping ratio of the r-th mode, and A_r is the corresponding modal constant.

Increasing the modal order allows additional resonant features to be captured and may improve the agreement in frequency regions where multiple modes contribute simultaneously. However, modes identified in frequency ranges characterized by low coherence are poorly observable and tend to exhibit large uncertainty and sensitivity to noise. Moreover, adding modes primarily to improve local agreement can lead to overfitting and may degrade the accuracy of the parameters associated with the dominant modes [4]. For these reasons, increasing the modal order is justified only when additional modes are clearly supported by high-coherence experimental data.

2. Band-limited (local) modal fitting

An alternative strategy consists in performing band-limited fits around individual resonances instead of a single global fit over the entire frequency range. In a narrow frequency

band centered around a resonance, the FRF can be approximated by a single-mode model,

$$H_r(\omega) \approx \frac{A_r\, i\omega}{\omega_r^2 - \omega^2 + 2i\zeta_r\omega_r\omega}$$

This approach is particularly effective for estimating damping ratios, since the resonance bandwidth and shape are less affected by neighboring modes. Local fits are also less sensitive to modeling errors at distant frequencies. The main limitation of this method is that it does not provide a unified global representation of the system and may yield modal parameters that are not fully consistent across overlapping frequency ranges. Consequently, band-limited fitting is best used as a complementary tool for validating global estimates or refining damping values of individual modes [8].

3. Inclusion of zero dynamics to model anti-resonances

The measured FRF exhibits pronounced anti-resonances, which cannot be reproduced by a pole-only modal model. These features arise from transmission zeros associated with the spatial distribution of modes, the input–output locations, and dynamic coupling effects. A more general representation of the FRF can be written as

$$H(\omega) = K\frac{\prod_{j=1}^{M}(i\omega - z_j)}{\prod_{r=1}^{N}(i\omega - p_r)(i\omega - \bar{p}_r)}$$

where z_j are transmission zeros and p_r are the system poles.

Introducing zeros would allow the model to reproduce both resonances and anti-resonances more accurately [4]. However, the identification of zeros from experimental data is significantly more ill-conditioned than pole identification, especially in the presence of noise. Furthermore, without detailed spatial information, the physical interpretation of identified zeros remains ambiguous. For these reasons, zero-enriched models are not pursued when the primary objective is robust damping estimation.

4. Frequency-dependent and non-proportional damping models

The adopted model assumes classical (proportional) damping, characterized by constant modal damping ratios. In reality, several physical mechanisms may lead to frequency-dependent dissipation. In particular, the blades investigated in this study are impregnated with viscoelastic material, which is known to exhibit complex, frequency-dependent mechanical properties. In such materials, the elastic modulus can be expressed as a complex quantity,

$$E^*(\omega) = E'(\omega) + iE''(\omega)$$

leading to a frequency-dependent loss factor $\tan(\delta(\omega)) = E''(\omega)/E'(\omega)$ and, consequently, to an effective modal damping ratio that varies with frequency [11].

This behavior provides a physically consistent explanation for the progressive attenuation of the response and the reduced quality of the fit observed at higher frequencies, even in frequency regions where the coherence remains reasonably high. While more advanced formulations could explicitly incorporate frequency-dependent damping terms,

$$\omega_r^2 - \omega^2 + 2i\zeta_r(\omega)\,\omega_r\omega$$

or non-proportional damping models, such approaches would introduce a significant number of additional parameters and substantially increase the complexity of the identification problem. In the absence of independent material characterization data, these models risk attributing experimental noise or modeling errors to apparent damping effects.

For these reasons, frequency-dependent damping models are not pursued in the present work, and the reduced-order model with constant modal damping ratios is retained as a physically meaningful and robust approximation, particularly suited for the accurate estimation of damping of the dominant low-frequency mode.

5. Frequency weighting based on coherence

An effective compromise between model fidelity and robustness consists in introducing frequency-dependent weighting in the cost function used for parameter estimation. A weighted least-squares formulation can be written as

$$J = \sum_k w(\omega_k) \; |H_{\text{meas}}(\omega_k) - H_{\text{model}}(\omega_k)|^2$$

where the weighting function $w(\omega_k)$ may be chosen proportional to the coherence $\gamma^2(\omega_k)$ or to the inverse noise variance.

This approach naturally emphasizes frequency regions where the experimental data are reliable and de-emphasizes noise-dominated regions. It is particularly useful for preserving the accuracy of low-frequency modes while preventing high-frequency discrepancies from biasing the global fit [2].

6. Exclusion of low-coherence frequency bands

Finally, explicitly excluding frequency bands characterized by persistently low coherence represents a pragmatic and effective strategy. In these regions, the FRF estimate does not reliably represent the underlying system dynamics, and attempting to fit the model provides little benefit. Restricting the fitting range to frequency bands supported by high coherence ensures that the identified modal parameters are driven by physically meaningful information rather than by measurement noise or numerical artifacts [12].

Although several strategies are available to improve the visual agreement between measured and fitted FRFs, not all improvements correspond to better physical models. In the present work, priority is given to robustness, physical interpretability, and accurate damping estimation of the dominant mode. The adopted reduced-order modal model, combined with coherence-informed interpretation and careful modal selection, represents a deliberate and well-justified compromise aligned with the primary objectives of the analysis.

4.3 *Uncertainty Analysis and Reliability of Damping Estimates*

Despite the careful design of the experimental setup and the structured analysis procedures adopted for both free and forced oscillation tests, the identified dynamic parameters must be interpreted as approximations affected by multiple sources of uncertainty

distributed along the entire experimental chain. These sources of error originate at different stages, from mechanical design and excitation, through sensing and data acquisition, to signal processing and model identification. In this context, it is important to distinguish between uncertainties introduced by the experimental and identification chain and those potentially arising from specimen-specific manufacturing characteristics, such as impregnation-related internal defects.

A dedicated repeatability study was carried out by testing multiple blade pairs exhibiting different levels of impregnation-related internal defects (primarily bubbles and flakes), as well as blade pairs with visually reduced defect content. Within the investigated range, the identified damping ratio of the dominant low-frequency mode remained approximately invariant across specimens. This observation indicates that, for the considered excitation levels and measurement configuration, such internal defects do not represent a primary driver of the dominant energy dissipation mechanisms. In other words, the variability observed in the identified damping metrics is not governed by specimen-to-specimen manufacturing or impregnation quality (within the tested population), but is more strongly influenced by the test configuration and the identification methodology adopted in each experiment.

At the mechanical level, uncertainties arise from assembly tolerances, alignment imperfections, and boundary condition variability. Although the test bench was redesigned to improve stiffness and alignment repeatability small deviations in blade mounting, preload conditions, and contact interfaces inevitably affect the effective stiffness and damping perceived by the structure. In the forced oscillation tests, additional uncertainty is introduced by the coupling between the shaker, the decoupling rods, the suspension threads, and the test specimen. As a result, the experimentally measured frequency response function inherently reflects the dynamics of the coupled system rather than those of the blades alone, making a strict separation between specimen-dominated and rig-dominated modes intrinsically difficult.

Excitation-related uncertainties further contribute to the approximation of the results. In the free oscillation tests, the imposed displacement and release mechanism may introduce small variations in initial conditions between runs, while transient effects at the beginning of the decay can influence time-domain damping estimates if not carefully excluded. In the forced oscillation tests, the swept-sine excitation is subject to limitations imposed by shaker controllability, especially at high frequencies. Trade-offs between sweep rate, excitation amplitude, and stability necessarily constrain the achievable signal-to-noise ratio, particularly in frequency bands where the structural response is weak. These limitations lead to frequency regions where the measured response approaches the noise floor, reducing coherence and limiting the reliability of modal information.

Measurement and sensing introduce additional uncertainty. Although non-contact laser vibrometry minimizes mass loading and mechanical interference, it remains sensitive to alignment accuracy, and environmental disturbances. The accelerometer-based measurement of base excitation, followed by numerical integration to obtain velocity, introduces further sensitivity to low-frequency drift, bias, and filtering choices.

Signal processing and spectral estimation choices represent another critical source of

approximation. While the adoption of block-wise acquisition, band-pass filtering, Welch averaging, and H1 estimation significantly improves robustness compared to a single full-band FFT-based approach, these techniques involve parameter choices (segment length, windowing, overlap, frequency resolution) that influence variance, bias, and smoothness of the estimated spectra. Even when sensitivity analyses show limited dependence on these parameters within certain frequency bands, residual estimation uncertainty remains unavoidable, particularly near anti-resonances and in regions of low coherence.

Finally, modeling and identification assumptions introduce conceptual sources of error. The free oscillation analysis relies on the assumption of linear viscous damping and single-degree-of-freedom behavior, which is strongly supported by the data but remains an idealization. In the forced oscillation analysis, the use of a limited number of modes to represent a system that clearly exhibits multi-degree-of-freedom behavior, anti-resonances, and frequency-dependent damping necessarily results in a simplified representation of the true dynamics. At higher frequencies, where viscoelastic effects become more pronounced and damping is expected to be frequency-dependent, the assumption of constant modal damping further contributes to modeling error.

In summary, the experimental results presented in this chapter are the outcome of a carefully controlled but inherently imperfect measurement process. Each stage of the experimental chain contributes incremental uncertainty, and their combined effect defines the accuracy limits of the identified dynamic parameters. Recognizing and explicitly accounting for these sources of error is essential for a meaningful interpretation of the results and provides the necessary context for the subsequent quantitative comparison between free and forced oscillation tests, as well as for the validation of numerical models in the following chapters.

Although both experiments target the same structural system, the identified Q-factors are not expected to coincide, as they arise from fundamentally different excitation conditions, measurement configurations, and identification principles. Table 4.5 summarizes the Q-factors identified for the dominant low-frequency mode and for higher-frequency modes using the different experimental approaches.

Table 4.5: Comparison of Q-factors obtained from free and forced oscillation tests

Test type	f_n	**Method**	**Q-factor**
Free oscillation	∼14 Hz	−3 dB bandwidth	$Q \approx 17.7$
Free oscillation	∼14 Hz	Linear Fit	$Q \approx 18$
Free oscillation	∼14 Hz	Hilbert Envelope	$Q \approx 18.5$
Forced oscillation	∼14 Hz	FRF −3 dB bandwidth	$Q \approx 6.1$
Forced oscillation	∼14 Hz	Multi-mode fit (ζ estimate, 1st Mode)	$Q \approx 9.3$
Forced oscillation	∼38 Hz	Multi-mode fit (ζ estimate, 2nd Mode)	$Q \approx 24.3$
Forced oscillation	∼531 Hz	Multi-mode fit (ζ estimate, 3rd Mode)	$Q \approx 11.2$

The higher Q-factor consistently identified in free oscillation tests reflects the intrinsic damping of the dominant low-frequency mode under nearly ideal single-degree-of-freedom conditions. In free decay, damping is extracted directly from the exponential amplitude envelope, which is closely related to the energy dissipated per cycle and is only weakly affected by secondary dynamics or measurement noise. In contrast, forced oscillation tests measure the steady-state response of a coupled system comprising the blades, the

test bench, the suspension, and the shaker. As a result, the identified damping reflects not only the intrinsic dissipation of the blades, but also additional loss mechanisms associated with interfaces, joints, suspension elements, and shaker control dynamics. This inevitably leads to lower effective Q-factors in forced response, particularly for the first mode.

Furthermore, the commonly used -3 dB bandwidth method assumes an isolated second-order resonance. In the forced-response FRFs, the presence of anti-resonances, modal interaction, and non-negligible coupling violates this assumption and artificially broadens the apparent resonance bandwidth, resulting in an underestimation of Q. The multi-mode complex FRF fit partially mitigates this issue by separating overlapping modal contributions, which explains why the fitted Q-factor ($Q \approx 9.3$) is consistently higher than the raw bandwidth estimate but still lower than the free-decay value. At higher frequencies, the interpretation of Q becomes even more delicate. The response amplitude decreases, coherence degrades, and viscoelastic dissipation mechanisms become more pronounced. Under these conditions, the assumption of constant modal damping represents an effective approximation, and the identified Q-factors should be interpreted as frequency- and configuration-dependent effective values rather than intrinsic material properties.

Based on the experimental evidence and on the known limitations of each identification approach, a hierarchy is adopted for interpreting and using the estimated Q-factors. The intent is not to declare a single "correct" Q-value in absolute terms, but to assign each estimate to the context in which it is physically meaningful and most reliable.

The Q-factor identified from free oscillation decay for the first mode ($Q \approx 18$) is taken as the most reliable estimate of the intrinsic damping of the dominant SDOF-like response. This choice is motivated by three main reasons:

- Damping is extracted directly from the exponential decay physics (energy dissipation per cycle),
- The high repeatability of the decay response across runs and specimens,
- The consistency between independent estimators (time-domain and spectral).

For this reason, this Q-factor is used as the baseline for material/structural damping characterization and as the preferred reference input for low-frequency model calibration.

Q-factors obtained from complex multi-mode fitting of the forced-response FRF are interpreted as effective damping values of the coupled experimental system under shaker excitation. They are valuable for reproducing measured response levels under forced conditions and for validating numerical models in an operational sense (i.e., matching FRF shapes and amplitudes in the frequency ranges where coherence is high). However, they are not taken as purely intrinsic material damping parameters, because the forced test necessarily includes additional dissipation paths (interfaces, suspension elements, actuator dynamics) and may be affected by frequency-dependent viscoelastic contributions and reduced coherence at higher frequencies.

This interpretation framework provides a consistent and physically grounded basis for the use of the identified damping parameters throughout the remainder of this work.

By clearly distinguishing between intrinsic damping, primarily characterized through free oscillation tests on the first mode, and effective damping extracted from forced response measurements, the experimental results can be interpreted without conflating material dissipation mechanisms with excitation- and configuration-dependent effects.

Within this context, the damping parameters obtained from the experimental campaign are not intended to serve as exact modal descriptors across the entire frequency range, but rather as representative inputs for the validation and calibration of numerical models. In particular, the reliably identified low-frequency damping is employed to inform the numerical simulations presented in the following chapter, where the focus shifts toward reproducing the global dynamic response and assessing the influence of damping assumptions on predicted vibration levels. This approach ensures coherence between experimental observations and numerical analyses, while acknowledging the inherent limitations associated with high-frequency damping identification under forced excitation.

§ 5 FEM Model Development

In the previous project, the finite element simulations were limited to two configurations: 3D simulations of parallel metallic lattice blades without viscoelastic materials, and 2D simulations of plain sandwich blades (without lattice structures), where the elastomer layer was modeled as a purely linear elastic material, neglecting viscoelastic effects. For the present study, these simplifications were no longer sufficient, as the focus shifted toward accurately capturing the time-dependent mechanical behavior of the elastomeric core. Consequently, a fully three-dimensional finite element model incorporating viscoelastic material properties was developed to provide a more realistic representation of the system's dynamic response.

5.1 *Model Development*

The finite element simulations in this project were carried out using COMSOL Multiphysics 6.3. The FEM model reproduces the experimental setup of the free-oscillation test shown in Figure 4.1, enabling a direct comparison between the simulated and measured responses under identical loading conditions. In addition, an eigenfrequency study is conducted to verify whether the modes identified through the fitting procedure described in Section 4.2.2 are consistent with those obtained numerically in COMSOL.

5.1.1 Geometry and Constraints

The FEM model represents the 3D geometry of the lattice blade, as shown in Figure 5.1 (left). To reduce the computational domain and the associated number of nodes, only a single blade was modeled rather than the full parallelogram consisting of two blades. This simplification significantly decreased the computational effort while retaining the essential structural features of the system.

The viscoelastic material volume is constructed as a parallelepiped representing the full thickness of the sandwich blade. The volume occupied by the lattice metal blade is then subtracted using a Boolean difference operation, ensuring that the viscoelastic material accurately fills only the space between the metal lattice structures. The result is shown in Figure 5.1 (right).

Figure 5.1: Blade Geometry

The moving part that connects the two blades in the physical setup was represented using a rigid connector in COMSOL, treating it as a completely rigid link. A point mass of 111 g was assigned to this rigid link to account for the mass of components not explicitly meshed in the model. Since the motion of this part is purely translational, no rotational inertia was assigned.

5.1.2 Mesh

To optimize both accuracy and computational efficiency, several meshing strategies were developed for the titanium lattices and the viscoelastic components. For the titanium lattice structures (Figure 5.2), triangular surface elements were chosen and subsequently swept through the volume. In the initial configuration, two layers were used to minimize simulation time during preliminary analyses, enabling faster convergence checks and validation of boundary conditions (Figure 5.2(bottom)). Once the setup was verified, additional layers were introduced to refine the mesh and assess the sensitivity of the results to mesh density. Swept meshes were preferred because they provide a more structured and uniform element distribution through the thickness compared to fully free tetrahedral meshes. This uniformity ensures better element quality, reduces distortion, and improves convergence in regions where the geometry has a dominant direction (such as extruded lattices). Moreover, swept meshes typically require fewer elements to achieve the same level of accuracy, further reducing computational cost [17].

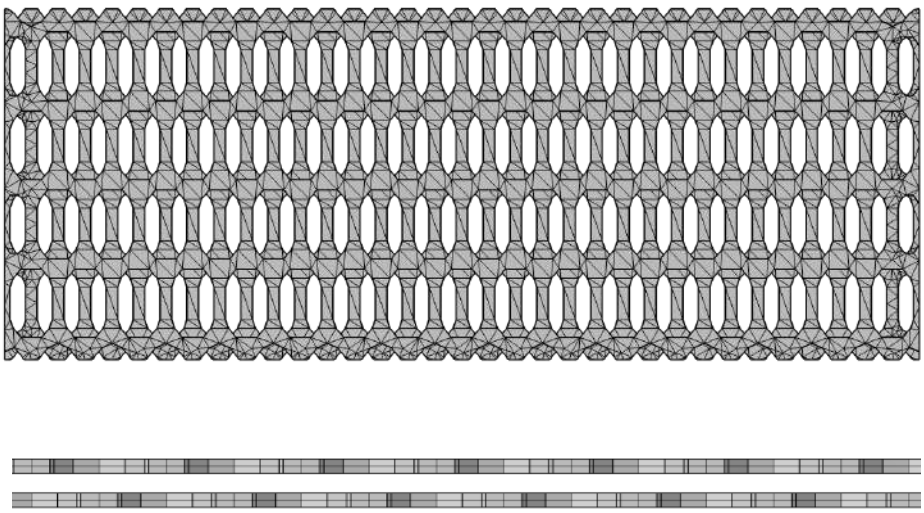

Figure 5.2: Titanium Lattices Meshing with swept triangular elements for the baseline configuration

The mesh density was intentionally varied throughout the titanium structure. Refinement was concentrated in areas where high stress gradients were expected, while coarser meshes were applied in regions subjected to minimal loads. This selective refinement strategy avoids unnecessary computational expense without compromising result accuracy. To achieve this, the clamp was divided into two sections using *work planes*, creating a partition that isolated a narrow stripe region (Figure 5.3) where stress concentrations were expected.

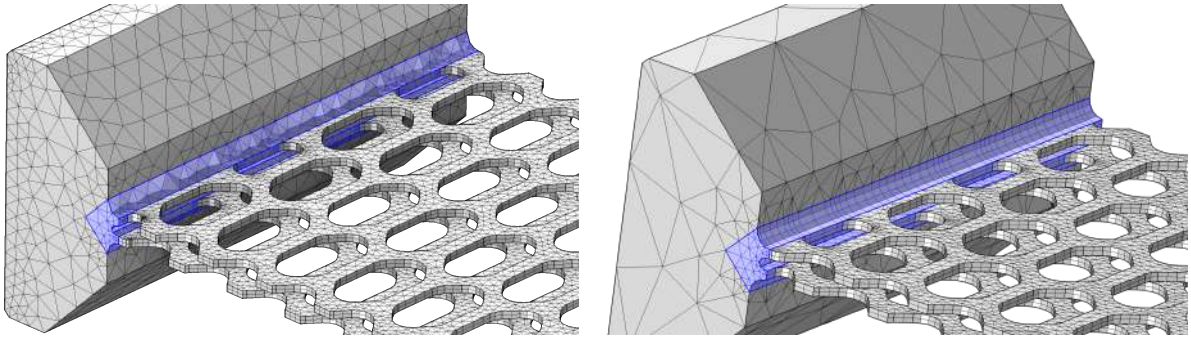

Figure 5.3: Titanium Stripe meshing with free tetrahedrals (left) and mapped (right)

For this "stripe", after several attempts using fine free-tetrahedral meshes, which resulted in distorted elements with inconsistent sizes (Figure 5.3 (left)), an alternative, tailor-made meshing strategy was implemented. This involved generating several points along the edges and constructing a mapped mesh on the side faces (Figure 5.3 (right)). This approach ensured a high-quality, well-aligned mesh through the fillet region, which was particularly important for capturing local stress gradients accurately. The mapped mesh also facilitated a smooth transition to the adjacent regions, which were meshed using free tetrahedral elements. For the remaining portions of the clamp, where mechanical stresses were negligible, an extremely coarse free-tetrahedral mesh was applied to further minimize computational effort.

The viscoelastic part of the model was also partitioned to allow a more controlled mesh generation and a consistent interface with the titanium lattices. It was divided into three main regions: two honeycomb-like sides interfacing with the lattices (Figure 5.4(left)) and a central parallelogram-shaped section (Figure 5.4(right)). The honeycomb regions were meshed with free triangular surface elements matching the element size of the titanium lattice surfaces, and then swept through the thickness to maintain mesh compatibility at the interface. The central parallelogram region was further subdivided along its mid-plane to facilitate the generation of free tetrahedral elements and ensuring an adequate number of layers. In future refinements, additional cuts can be introduced to increase the mesh resolution and evaluate convergence trends.

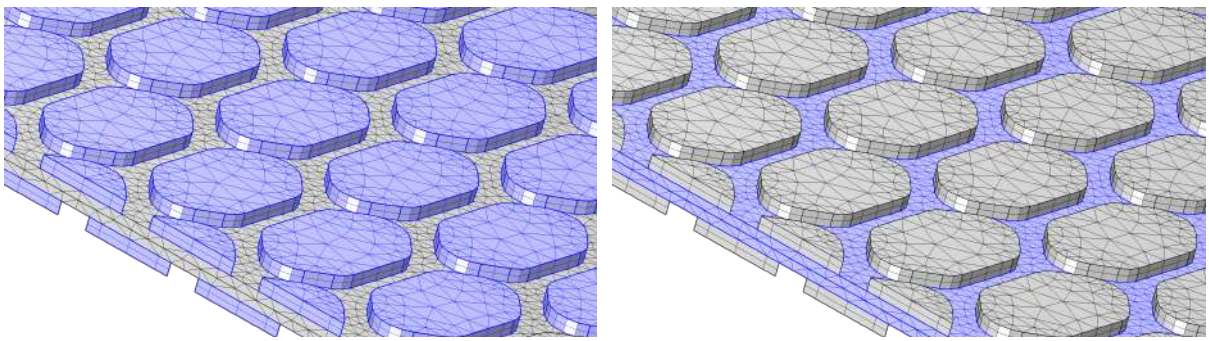

Figure 5.4: Elastomer Meshing

Overall, this multi-step meshing approach provides a balanced compromise between computational cost and model fidelity, ensuring precise stress prediction in critical regions while maintaining efficient runtime performance.

5.1.3 Material Model

Viscoelastic materials exhibit both elastic and viscous behavior, and their mechanical response depends on the rate or frequency of loading. This dual nature is typically characterized through the storage modulus (E') and the loss modulus (E''), which describe, respectively, the energy stored elastically and the energy dissipated as heat during cyclic deformation. These frequency-dependent moduli provide fundamental insight into the time-dependent mechanical behavior of polymers, rubbers, and other viscoelastic materials, and are therefore central to accurate numerical modeling.

The most common experimental technique used to determine E' and E'' is Dynamic Mechanical Analysis (DMA). In a DMA test, a small sinusoidal deformation is applied to a specimen, and the resulting stress response is measured. From the amplitude and phase shift between the stress and strain signals, the storage and loss moduli are derived. Depending on the configuration of the DMA instrument, such as tensile, bending, compression, or shear mode, different types of moduli can be obtained. The Lamé coefficients (λ and μ) can then be employed to convert between these moduli, enabling consistent comparison and use in multiphysics simulations.

In COMSOL Multiphysics, viscoelastic behavior can be defined by adding a *Viscoelasticity* subnode under the *Linear Elastic Material* node. Several viscoelastic constitutive models are available, including the *generalized Maxwell*, *generalized Kelvin–Voigt*, *Burgers*, and *user-defined* formulations. The user-defined option allows the direct input of storage and loss moduli as functions of frequency, which initially appeared to be the most straightforward approach for the present study, as it would permit direct use of the experimentally measured DMA data. However, this model is purely frequency-domain based and does not support time-dependent (transient) simulations in COMSOL. Since the aim of the present project was to simulate the material's time-domain response to dynamic loading, this limitation rendered the user-defined model unsuitable. To overcome this constraint, the *generalized Maxwell model* was adopted. This model can reproduce the experimentally measured viscoelastic behavior while remaining compatible with both frequency and time-domain simulations. Physically, it represents the material as a combination of multiple Maxwell elements, each consisting of a spring and a dashpot in series, arranged in parallel (see Figure 5.5).

Each element, or "branch", is characterized by a relaxation time (τ_i) and an associated shear modulus (G_i). The overall storage and loss moduli of the material are then expressed as:

$$G'(\omega) = G_0 + \sum_{i=1}^{N} \frac{G_i(\omega\tau_i)^2}{1+(\omega\tau_i)^2}\,, \qquad G''(\omega) = \sum_{i=1}^{N} \frac{G_i\omega\tau_i}{1+(\omega\tau_i)^2}$$

where G_0 denotes the equilibrium modulus (the long-term elastic response), $\omega = 2\pi\nu$ is the angular frequency, and N is the number of Maxwell branches. These equations describe how each branch contributes to the total viscoelastic response: branches with shorter relaxation times capture high-frequency (short-term) effects, while those with longer relaxation times capture low-frequency (long-term) behavior.

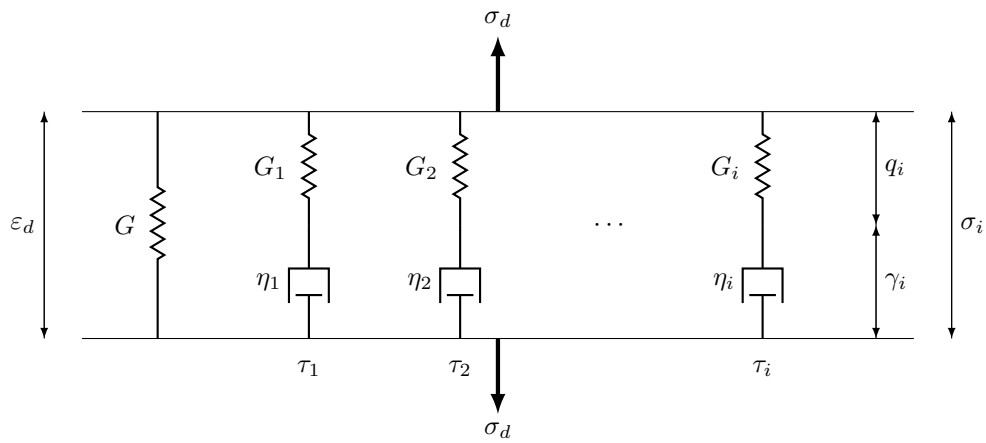

Figure 5.5: Rheological Scheme of the Generalized Maxwell Model

The parameters (τ_i, G_i) were determined by fitting the generalized Maxwell model to the experimental DMA data. To achieve this, a Python script was developed to perform nonlinear least-squares fitting of the measured storage and loss moduli. The fitting procedure minimizes the difference between the experimental data and the model predictions. This difference is expressed as a residual vector:

$$r = \begin{bmatrix} G'_{\text{exp}}(\omega) - G'_{\text{model}}(\omega) \\ w\left(G''_{\text{exp}}(\omega) - G''_{\text{model}}(\omega)\right) \end{bmatrix}$$

where w is a weighting factor (typically $w = 2$) used to give comparable importance to E' and E'' in the optimization because if we fit both curves simultaneously without weighting, the optimization will focus almost entirely on minimizing errors in the bigger term (E') because its residuals dominate the total squared sum.

Afterwards, the code employs nonlinear least-squares minimization to find the model parameters that minimize the sum of squared residuals:

$$\min_{\{G_0, G_i, \tau_i\}} \sum_{k=1}^{M} \left[\left(G'_{\text{exp}}(\omega_k) - G'_{\text{model}}(\omega_k)\right)^2 + w^2 \left(G''_{\text{exp}}(\omega_k) - G''_{\text{model}}(\omega_k)\right)^2 \right]$$

where M is the number of frequency points. The optimization is carried out using the Levenberg–Marquardt algorithm as implemented in `scipy.optimize.least_squares`. To determine the sufficient number of branches N, the code performs the fitting for increasing N and computes the Root Mean Square Error (RMSE) of each fit:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{k=1}^{M} r_k^2}$$

The smallest number of branches that produces a negligible improvement (set in this case at 5%) in RMSE is selected as the optimal model complexity. Figure 5.6 shows the comparison between the experimental DMA measurements and the generalized Maxwell model fitted to the data, illustrating how well the chosen number of branches captures both the storage and loss moduli across the frequency range.

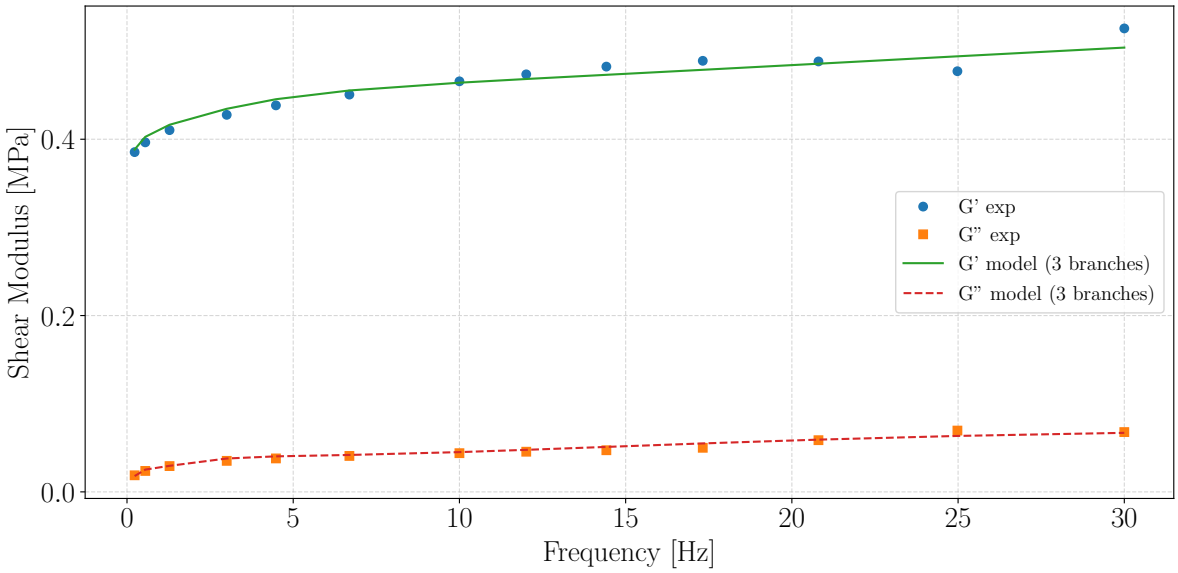

Figure 5.6: Comparison between experimental DMA data and the fitted generalized Maxwell model for the optimal number of branches

This adaptive fitting strategy ensured a balance between model accuracy and computational efficiency. Too few branches would result in an inadequate representation of the experimental behavior, while too many branches could lead to overfitting and increased numerical stiffness in time-domain simulations. The final set of branches chosen thus represented the simplest model capable of reproducing the measured viscoelastic response within the predefined accuracy threshold [7].

To convert the tensile moduli obtained from DMA measurements into shear moduli, the Poisson's ratio was set to 0.49. In viscoelastic modeling of polymers, materials are often assumed to be nearly incompressible because polymer chains in the rubbery or soft state can easily slide past one another, allowing shape deformations without significant changes in volume. A Poisson's ratio slightly below 0.5, such as 0.49, is typically used to capture this behavior in simulations. Setting the Poisson's ratio exactly to 0.5 would imply perfect incompressibility, corresponding to an infinite resistance to volume change (infinite bulk modulus), which is unphysical and can lead to numerical instabilities. Using a value of 0.49 preserves both physical realism and computational stability: it is high enough to reflect the negligible volumetric deformation characteristic of rubbers and polymers, yet finite enough to ensure the bulk modulus remains well-defined [7, 1].

Under this assumption, the time-dependent viscoelastic response is dominated by deviatoric strain, which corresponds to changes in the material's shape without volume change. The volumetric strain, which corresponds to uniform compression or expansion, occurs almost instantaneously and does not exhibit significant relaxation over time, so it can be treated as purely elastic. This allows the generalized Maxwell model to focus entirely on the deviatoric component of strain, which reflects the molecular motions and chain rearrangements responsible for energy dissipation in polymers. In practice, this means that when performing experiments such as DMA, which typically involve torsional or bending deformations, the measured storage and loss moduli correspond almost entirely to shear behavior, and the bulk response contributes negligibly to the observed viscoelastic effects.

5.2 *Free Oscillation Test Simulation*

In order to complement the experimental free oscillation tests (explained in Section 4.1) and to provide a quantitative interpretation of the observed damping behavior, a series of time-dependent finite element simulations were performed. The numerical campaign is structured around a reference (baseline) configuration, which is fully described in the following section, and a set of additional simulations in which selected geometrical, material, and modeling parameters are varied.

5.2.1 Baseline numerical configuration

The baseline numerical model is formulated within the Solid Mechanics interface under the assumption of small-strain kinematics. The titanium parts are modeled as Ti–6Al–4V Grade 23, assuming isotropic linear elastic behavior. The elastomeric component (MAPSIL QS 1123) is modeled as a nearly incompressible material, with its elastic modulus consistently derived from the instantaneous shear modulus G_0. The material properties adopted for the baseline configuration are summarized in Table 5.1.

Table 5.1: Material properties used in the baseline FEM configuration

Material	Young's modulus E [GPa]	Poisson's ratio ν [-]	Density ρ [$\mathrm{kg\,m^{-3}}$]
Ti–6Al–4V Grade 23	108	0.34	4430
MAPSIL QS 1123	$1.134{\cdot}10^{-3}$	0.49	1170

The time-dependent behavior of the elastomer is described using a generalized Maxwell model composed of three viscoelastic branches connected in parallel with an equilibrium elastic spring. Each branch is characterized by a shear modulus G_i and a relaxation time τ_i. The numerical values of these parameters were obtained following the identification methodology described in Section 5.1.3. The viscoelastic parameters used for the baseline configuration are reported in Table 5.2.

Table 5.2: Generalized Maxwell model parameters for the elastomer

Branch	Shear modulus G_i [MPa]	Relaxation time τ_i [ms]
1	0.1306	3.613
2	0.04946	45.14
3	0.03333	378.2

The finite element discretization consists of 66,544 three-dimensional domain elements, resulting in 417,246 mechanical degrees of freedom. In addition, the viscoelastic constitutive model introduces approximately 11×10^6 internal state variables associated with the generalized Maxwell formulation. Time integration was carried out using an implicit time-dependent solver with a variable time-step scheme. The solver was configured to store the solution at all time steps taken internally by the time integrator, resulting in a non-uniform temporal discretization with a maximum time-step size limited to $\Delta t_{\max} = 0.01$ s. This choice allowed the solver to adaptively resolve the oscillatory response while maintaining numerical stability in the presence of viscoelastic dissipation.

Table 5.3 collects the main geometric, material, discretization, and solver parameters used in the reference model, which serves as the starting point for the sensitivity analyses

discussed in the following sections. All post-processing quantities, including displacement histories, stress measures, stored elastic energy in the viscoelastic domains, and amplitude decay metrics, were extracted from probe points defined consistently across all simulations.

Table 5.3: Baseline numerical parameters of the free oscillation FEM simulation

Parameter	**Value**
Number of domain elements	66,544
Mechanical degrees of freedom	417,246
Internal DOFs	11,177,582
Maximum time step	0.01 s
Time integration scheme	Implicit, adaptive time stepping
Mesh Size	Fine
Computational time (Step 1: loading, 0–4 s)	8 min
Computational time (Step 2: free oscillation, 4–4.5 s)	17 min
Total computational time	25 min

To numerically reproduce the experimental free oscillation test while ensuring a stable and physically meaningful transient response, the simulation is divided into two consecutive time intervals. During the first phase, spanning from $t = 0$ to $t = 4$ s, a prescribed displacement is applied using a smooth ramp function. This loading strategy allows a gradual buildup of deformation and internal stresses in both the metallic and elastomeric components, avoiding the introduction of spurious high-frequency content that would arise from an abrupt displacement step. Figure 5.7 shows the prescribed displacement law adopted during this initial loading phase.

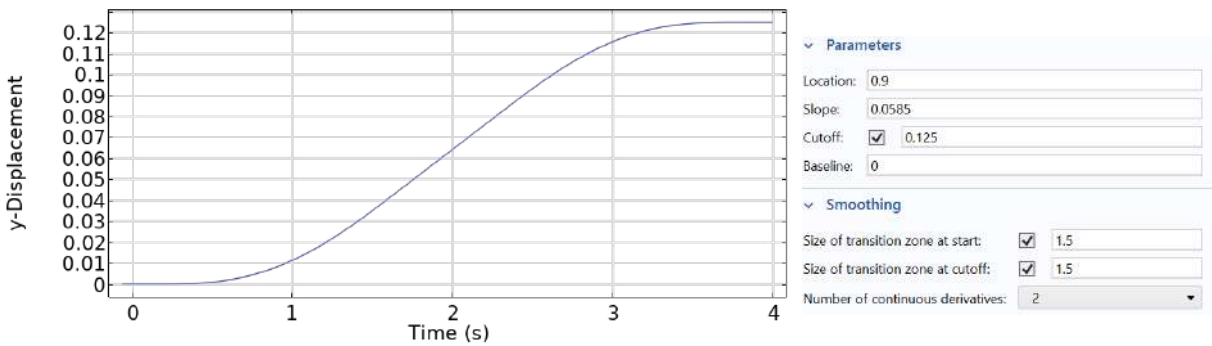

Figure 5.7: Ramp Function imposed before free oscillation

At the end of the ramp, the prescribed displacement is released and the structure is allowed to vibrate freely over a subsequent time interval, from $t = 4$ to $t = 4.5$ s. The resulting displacement decay, shown in Figure 5.8, constitutes the numerical free oscillation response. This signal is subsequently analyzed to extract the dominant oscillation frequency, the quality factor, and the amplitude-dependent damping characteristics. The adopted numerical procedure closely reproduces the experimental test conditions, thereby enabling a meaningful and direct comparison between simulated and measured free oscillation responses.

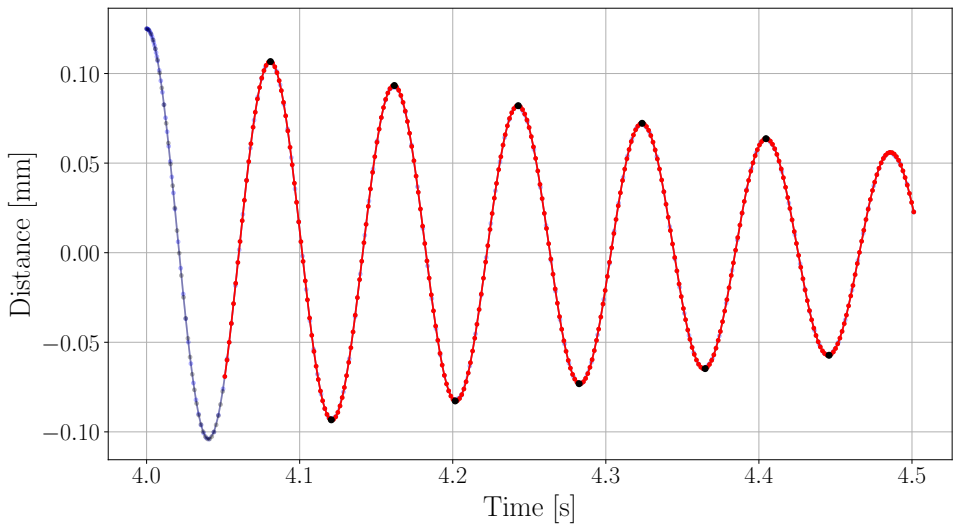

Figure 5.8: Time-domain displacement response obtained from the FEM free oscillation test after release

Free-oscillation experiments performed on multiple blade pairs and documented in Section 4.1 demonstrate a highly repeatable modal response, characterized by a well-defined resonance frequency and a consistent damping behavior across specimens. When transitioning to numerical simulations incorporating a viscoelastic constitutive description for the elastomeric core, the simulated response preserves the same global bending mode observed experimentally, confirming that the structural kinematics of the system are correctly captured. In the baseline numerical configuration, the predicted oscillation frequency is approximately 12.3 Hz, which is lower than the experimentally measured value of 14.4 Hz. This discrepancy indicates that, although the dominant deformation mode is correctly reproduced, the effective dynamic stiffness of the numerical model remains underestimated with respect to the physical system. The observed frequency shift could reflect modeling assumptions inherent to the viscoelastic representation of the elastomeric core and to the effective stiffness contribution of the composite structure. In particular, the frequency underestimation suggests that the elastic response of the viscoelastic material, as activated under dynamic loading conditions, is lower than that experienced in the experimental system.

More importantly, the temporal evolution of the simulated oscillation amplitude reveals a damping behavior that differs fundamentally from the experimental response. Unlike the experimental decay, which can be reasonably approximated by a single exponential envelope associated with a constant quality factor, the numerical amplitude decay does not follow a strictly exponential law. Instead, the rate of amplitude reduction varies throughout the response, indicating that the energy dissipated per oscillation cycle is not constant over time. This behavior is intrinsic to the viscoelastic formulation adopted in the numerical model, which explicitly resolves time-dependent material dissipation mechanisms rather than representing damping through a single phenomenological parameter. As a direct consequence, the quality factor computed from numerical data using the classical definition $Q = \pi A/\Delta A$ does not yield a unique scalar value when applied to the numerical response. Evaluating the quality factor at different stages of the decay leads to noticeably

different values, revealing a clear dependence of Q on the oscillation amplitude. In the baseline configuration, quality factor values extracted from early cycles at larger amplitudes differ significantly from those obtained at later stages of the decay, even though the oscillation frequency remains essentially constant. This demonstrates that, within the viscoelastic numerical framework, the quality factor cannot be interpreted as an intrinsic property of the structure independent of the oscillation state.

To address the intrinsic amplitude dependence of viscoelastic damping and enable a physically meaningful comparison between numerical models, the quality factor must be evaluated in a manner that is independent of time and local transient effects. The adopted strategy consists in treating the quality factor as an amplitude-dependent quantity Q(A), rather than as a single global constant. For each oscillation cycle, the instantaneous quality factor is computed using the physically grounded energy-based definition

$$Q(A_i) = \pi \frac{A_i}{\Delta A_i}$$

where A_i is the mean oscillation amplitude over one cycle and ΔA_i is the corresponding amplitude loss between consecutive cycles. This formulation preserves the physical interpretation of Q as the ratio between stored and dissipated energy per cycle, while explicitly accounting for the non-linear evolution of dissipation during the decay.

In order to compare different simulations characterized by different mesh resolutions and therefore different amplitude histories, a common amplitude interval shared by all models is first identified. This interval defines a range of oscillation amplitudes for which all simulations provide valid data. Restricting the analysis to this common domain ensures that the quality factor is evaluated at comparable physical states of the system, rather than at arbitrary times. Within this amplitude interval, a set of quality factor values $Q(A_i)$ is obtained for each model. Instead of selecting a single value at a specific amplitude, which would remain sensitive to numerical noise, cycle-to-cycle fluctuations, and local non-linearities, the representative quality factor is defined as the median of Q(A) over the selected amplitude band:

$$Q = \text{median}\left\{Q(A_i) | A_{min} \leq A_i \leq A_{max}\right\}$$

The use of the median, rather than the mean, is a deliberate and physically motivated choice. The median is robust against outliers arising from early transient cycles, numerical discretization effects, or local irregularities in the amplitude decay. It does not assume symmetry or linearity of the underlying distribution and therefore provides a stable estimate even when Q(A) varies significantly within the band. From a numerical standpoint, this approach minimizes sensitivity to the precise selection of the analysis start time and avoids bias introduced by isolated cycles with anomalous dissipation behavior. Applying this methodology to the baseline configuration yields quality factor values that are higher than those measured experimentally, thus indicating that the viscoelastic dissipation implemented in the current model is insufficient to fully reproduce the experimentally observed energy losses.

Additional insight into the dissipation mechanisms is obtained by examining the evolution of the elastic energy stored in the viscoelastic material during free oscillation. Figure 5.9 shows the stored elastic energy in the elastomer as a function of displacement over

successive oscillation cycles. The resulting curves form distinct hysteresis loops, directly evidencing energy dissipation at the material level and confirming the presence of a phase lag between displacement and stored energy. The progressive reduction of the hysteresis loop area as the oscillation amplitude decreases indicates that the energy dissipated per cycle diminishes during the decay. This energetic perspective provides a direct physical interpretation of the amplitude-dependent quality factor observed in the time-domain analysis. As the oscillation amplitude decreases, the viscoelastic material explores regions of its constitutive response associated with reduced dissipation, leading to an increase in the instantaneous quality factor. The stored energy–displacement representation therefore complements the amplitude-based analysis and confirms that the damping behavior predicted by the baseline numerical model is inherently amplitude dependent.

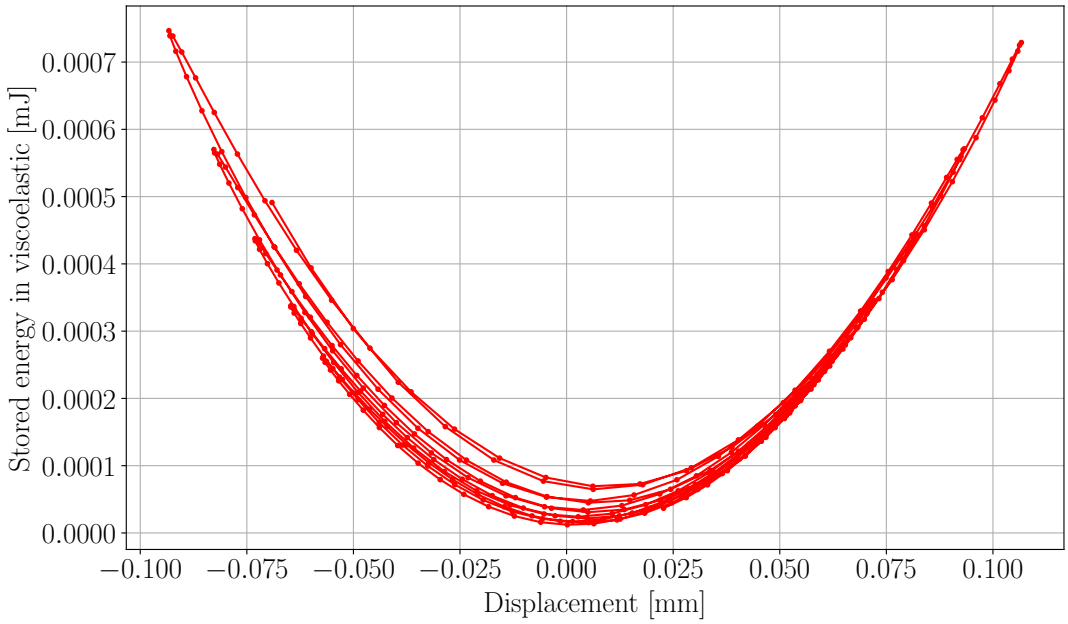

Figure 5.9: Stored elastic energy in the elastomer during free oscillation

Overall, the baseline numerical simulation captures the correct deformation mode and reproduces the qualitative features of viscoelastic damping, while underestimating the resonance frequency and overestimating the quality factor relative to experimental observations. These results establish a reference configuration for subsequent analyses and provide a clear physical framework for interpreting the effects of modeling choices and parameter variations explored in the following sections.

5.2.2 Higher Size Meshes

Besides the baseline mesh introduced in the previous section, two further discretizations, referred to as *finer* and *extremely fine* mesh, were considered. For the sake of conciseness, the results obtained with these refined meshes are discussed together in the present section, focusing on their impact on the predicted oscillation frequency, damping characteristics, and computational cost.

The differences between the considered mesh configurations are illustrated in Figure 5.10 and Figure 5.11, which show representative views of the finer and extremely fine meshes, respectively. Compared to the baseline discretization, both refined meshes exhibit a higher element density in the regions of stress concentration and within the viscoelastic core, allowing for a more accurate spatial resolution of deformation gradients. These visual comparisons provide a qualitative insight into the increasing level of geometric detail captured by the refined discretizations.

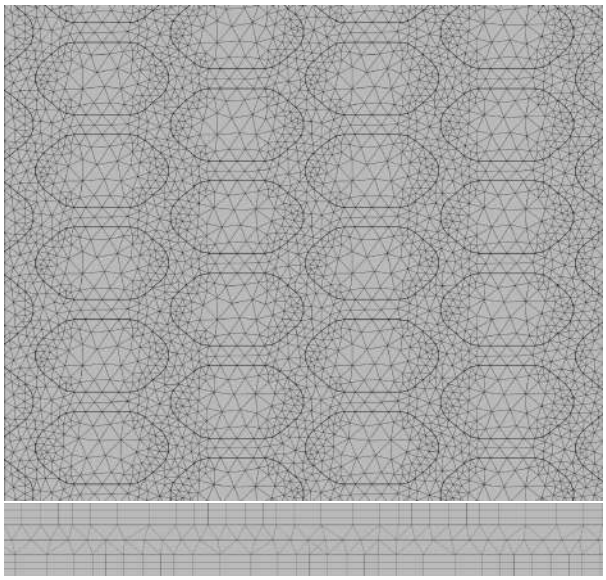

Figure 5.10: Finer Mesh Size

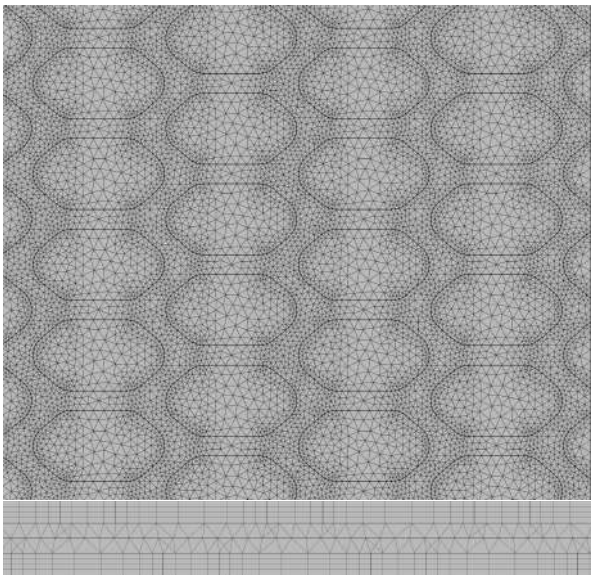

Figure 5.11: Extremely Fine Mesh Size

The comparison between the baseline, finer, and extremely fine mesh configurations reported in Table 5.4 demonstrates a highly consistent dynamic response across all investigated spatial discretizations. Both the oscillation frequency and the damping-related quantities exhibit only minor variations with mesh refinement, indicating that the numerical solution is already well converged at the baseline mesh level. In particular, the mean oscillation frequency shows a weak and monotonic dependence on mesh density, with values of 12.3 Hz, 12.12 Hz, and 12.06 Hz for the baseline, finer, and extremely fine meshes, respectively. The total variation remains below 2%, confirming that the global stiffness–inertia balance of the structure is accurately resolved even with the coarsest discretization considered.

Table 5.4: Comparison of numerical results and computational cost for the different mesh configurations

Parameter	**Fine mesh**	**Finer mesh**	**Extremely Fine mesh**
Number of domain elements	66,544	287,320	524,596
Mechanical DOFs	417,246	1,718,692	6,397,807
Internal DOFs	11,177,582	56,301,402	204,882,642
Maximum time step	0.01 s	0.01 s	0.01 s
Total computational time	26 min	2 h 22 min	12 h 34 min
Mean oscillation frequency [Hz]	12.3	12.12	12.06
Q factor (mins)	23.813	23.938	23.711
Q factor (maxs)	26.185	25.981	26.102

The absence of any pronounced frequency shift with increasing mesh resolution suggests that neither artificial stiffening nor numerical softening effects dominate the solution

in the investigated discretization range. This observation is particularly relevant given the presence of a nearly incompressible viscoelastic material coupled to stiff metallic components, a configuration that is often susceptible to volumetric locking or spurious compliance when inadequately discretized. The smooth convergence of the frequency further indicates that the coupling between elastic stiffness and viscoelastic dissipation does not introduce distortions in the frequency estimate extracted from the free-decay response.

A similar level of robustness is observed for the damping-related quantities. The quality factor values obtained from the baseline, finer, and extremely fine meshes are remarkably close: the variations across the three discretizations are well below 1%, indicating that the predicted dissipation is essentially mesh independent within the investigated resolution range. This result demonstrates that the viscoelastic energy dissipation mechanisms captured by the generalized Maxwell model are not significantly affected by the spatial discretization once a minimum level of mesh refinement is achieved.

The observed mesh independence of the quality factor is a particularly important result, as viscoelastic dissipation is inherently governed by local strain amplitudes, strain rates, and their temporal evolution within the elastomeric domains. The fact that these quantities lead to nearly identical damping predictions across all meshes indicates that the strain fields driving dissipation are already sufficiently resolved in the baseline configuration. Consequently, neither excessive smoothing of deformation gradients nor artificial localization effects play a dominant role in the computed energy loss per cycle.

Taken together, the results of the mesh refinement study confirm that both the structural response and the viscoelastic damping behavior of the model are numerically robust. The baseline mesh provides an accurate representation of the oscillation frequency and quality factor, while further mesh refinement does not lead to meaningful changes in the predicted dynamic response. This outcome allows subsequent analyses to focus on physical modeling aspects, such as material parameter identification and constitutive assumptions, without ambiguity regarding numerical discretization effects. The mesh study therefore establishes a solid numerical foundation for the sensitivity analyses presented in the following section.

A particularly insightful perspective on the interplay between mesh refinement, numerical dissipation, and the physical damping mechanisms of the system is provided by Figure 5.12, which compares the amplitude loss per oscillation period as a function of the oscillation amplitude for the different mesh configurations against the experimental reference curve. The experimental data display a nearly linear relationship between amplitude loss and amplitude over the investigated range, confirming that, within these strain levels, the dominant dissipation mechanism in the elastomer can be effectively described as linear viscoelastic. The numerical results show a high degree of consistency across the baseline, fine, and extremely fine discretizations, indicating that the amplitude-dependent dissipation predicted by the numerical model is largely insensitive to mesh refinement once a sufficient spatial resolution is reached. For all mesh sizes, the amplitude loss increases in an approximately linear manner with oscillation amplitude over the investigated range, reflecting a stable and physically coherent dissipation mechanism governed by the viscoelastic constitutive formulation.

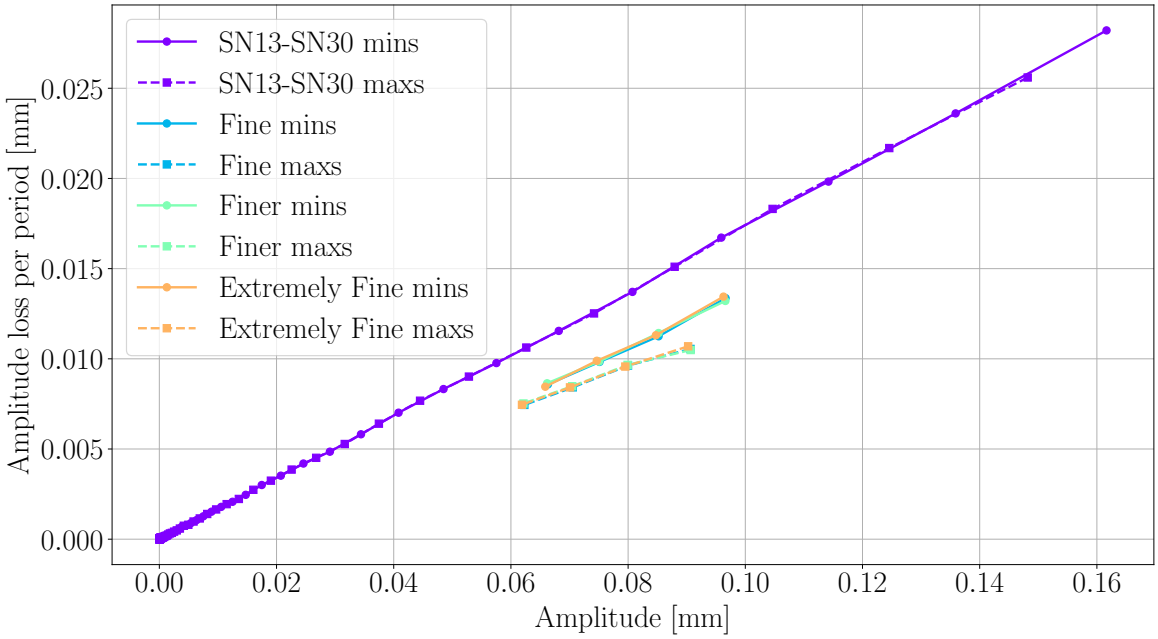

Figure 5.12: Amplitude Loss vs Amplitude: Mesh Size Comparison

The curves corresponding to the different discretizations collapse into a narrow band, with only minor deviations observable at the highest amplitudes. These differences do not follow a monotonic trend with mesh refinement and remain small compared to the overall dissipation level. This behavior demonstrates that the strain fields driving viscoelastic dissipation are already adequately resolved at the baseline mesh level and that further refinement does not fundamentally alter the activation of the internal variables associated with the generalized Maxwell model. As a consequence, the discrepancies observed between numerical and experimental dissipation levels cannot be attributed to insufficient spatial resolution, but must instead originate from other modeling aspects, such as material parameter identification or constitutive assumptions.

A persistent feature is the systematic separation between the curves obtained from minima-based and maxima-based evaluations of the oscillation cycles. For all mesh configurations, the amplitude loss extracted from maxima is consistently slightly higher than that obtained from minima at the same oscillation amplitude. Importantly, this asymmetry remains essentially unchanged with mesh refinement: the relative offset between minima and maxima is preserved across all discretizations and does not diminish as the mesh is refined. This observation confirms that the asymmetry is not a numerical artifact introduced by discretization, but rather an intrinsic feature of the modeled viscoelastic response. From a physical standpoint, this behavior is fully consistent with the adopted viscoelastic formulation. In a generalized Maxwell model, dissipation depends not only on the instantaneous amplitude but also on the strain history and strain-rate evolution within the material. During free oscillations, cycles corresponding to minima and maxima do not traverse identical paths in the internal-variable space, even when their mean amplitudes are comparable. The combination of time-dependent viscoelasticity, nearly incompressible behavior, and geometrical nonlinearity leads to slightly different internal states at opposite extrema of the motion, resulting in a small but systematic difference

in the energy dissipated per cycle. The fact that this asymmetry is stable across mesh refinements further supports its physical origin. At the same time, the magnitude of the minima–maxima asymmetry remains limited and does not dominate the global dissipation behavior. Both evaluations exhibit nearly parallel trends and comparable slopes in the amplitude loss versus amplitude representation, indicating that the overall damping mechanism remains governed by the same underlying viscoelastic processes. The asymmetry therefore represents a second-order effect superimposed on an otherwise well-defined and mesh-converged dissipation response.

Overall, the mesh-size study demonstrates that the numerical discretization is no longer a limiting factor for capturing amplitude-dependent viscoelastic dissipation. Both the global trend of amplitude loss and the finer features of the response, including the systematic minima–maxima asymmetry, are robust with respect to mesh refinement. This establishes a solid numerical foundation for the subsequent sensitivity analysis, which can therefore focus on material parameters and constitutive modeling choices, rather than on further spatial refinement, to explain the remaining discrepancies with experimental observations.

5.2.3 Sensitivity Analysis

The baseline simulations and the mesh-refinement study establish a numerically robust reference model: the global dynamic response is stable with respect to spatial discretization, while the remaining discrepancies with respect to experiments primarily concern the absolute stiffness level (as reflected by the oscillation frequency) and the magnitude of viscoelastic dissipation (as reflected by the quality factor and the amplitude-loss curves). Building on this baseline, the purpose of the sensitivity analysis is to systematically quantify how key modeling and material parameters influence the free-oscillation response, and to identify which parameters provide effective and physically interpretable "control knobs" for matching the experimental frequency and damping behavior. In practice, each parameter is varied around the baseline value while keeping all other settings unchanged, and the resulting response is evaluated using the same post-processing framework introduced previously; namely the oscillation frequency extracted from the time-domain decay, the amplitude-dependent quality factor, the amplitude-loss versus amplitude trends for minima and maxima, and the associated energetic indicators derived from the viscoelastic hysteresis loops. Particular attention is paid to separating effects that primarily shift the stiffness–inertia balance (e.g., parameters controlling the instantaneous shear modulus and compressibility) from those that primarily control dissipation within the relevant frequency range (e.g., the Maxwell branch moduli and relaxation times), while also tracking secondary signatures such as the persistent minima–maxima asymmetry. The following subsections report the results of these targeted parameter sweeps and discuss them in terms of both physical mechanisms and model calibration implications.

5.2.3.1 Real Dimensions

The first sensitivity case investigates the influence of geometric fidelity by replacing the nominal dimensions used in the baseline model with the real dimensions measured on the manufactured blades. The nominal dimensions correspond exactly to the CAD model employed for the actual manufacturing of the components. However, as is inherent to

any fabrication process, the final produced parts exhibit measurable deviations from the nominal design, resulting in effective geometric properties that differ from those assumed in the numerical model. Introducing the real dimensions leads to a pronounced increase in the predicted oscillation frequency, which rises from 12.30 Hz in the baseline configuration to 14.27 Hz. This adjustment brings the numerical frequency into excellent agreement with the experimentally measured value of 14.4 Hz, demonstrating that geometric deviations introduced during manufacturing represent one of the dominant contributors to the stiffness mismatch observed in the baseline simulations.

From a physical perspective, this strong sensitivity is expected. The dynamic response of the system is governed by bending-dominated deformation modes of the titanium blades (see Section 5.3), for which the effective stiffness depends nonlinearly on geometric parameters such as thickness, cross-sectional dimensions, and lever-arm length. Even modest deviations between nominal and manufactured geometries can therefore propagate into significant variations in bending stiffness and, consequently, in natural frequency. This result highlights that, for slender structures operating in bending, geometric fidelity is a prerequisite for any quantitative comparison between numerical predictions and experimental measurements.

The predicted quality factors indicate a relatively low level of effective dissipation, with a clear distinction between minima- and maxima-based evaluations. A clearer interpretation of these differences emerges from the analysis of the equivalent deviatoric strain field within the viscoelastic material. Figure 5.13 compares the deviatoric strain distributions obtained for the baseline configuration, based on nominal CAD dimensions, and for the model reconstructed using the real manufactured geometry. While the baseline model exhibits a relatively smooth and homogeneous strain field, the real-dimensions configuration shows substantially higher deviatoric strain levels, together with pronounced strain localization in specific regions of the elastomer.

In particular, increased strain concentrations are observed near the interfaces between the titanium blades and the viscoelastic core, as well as along the ligaments separating adjacent honeycomb cells. These localized deformation patterns indicate that the real geometry promotes stronger shear-dominated deformation within the elastomer, directly activating the dissipative mechanisms described by the generalized Maxwell model. As viscoelastic dissipation is intrinsically governed by deviatoric strain rather than by stress magnitude alone, these localized strain enhancements play a dominant role in modifying the dynamic response.

From a physical standpoint, these results demonstrate that the dynamic behavior of the system is not determined solely by material parameters, but is strongly influenced by the precise geometric pathways through which deformation is transmitted. Even small deviations from the nominal design can significantly modify strain redistribution within the viscoelastic phase, thereby altering both the effective stiffness and the energy dissipation per oscillation cycle. The observed increase in oscillation frequency and quality factor therefore arises naturally from geometry-induced changes in strain transfer, rather than from numerical artifacts or modeling inconsistencies.

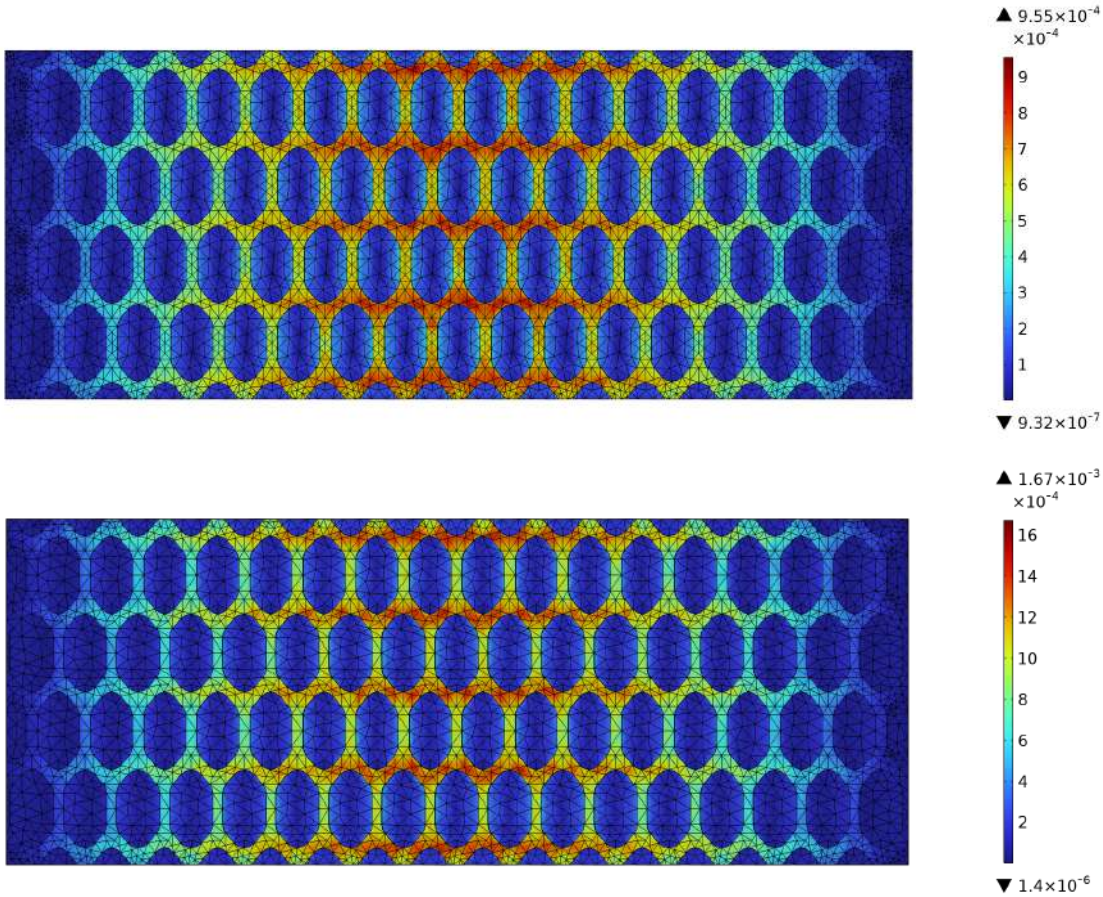

Figure 5.13: Equivalent deviatoric strain distribution in the viscoelastic core at t=4.5 s for Baseline configuration (top) and Real Dimensions configuration (bottom)

5.2.3.2 Viscoelastic Spillovers

The second sensitivity case addresses the influence of additional viscoelastic material that is present in the experimental specimens but was not included in the baseline numerical model (Figure 5.14).

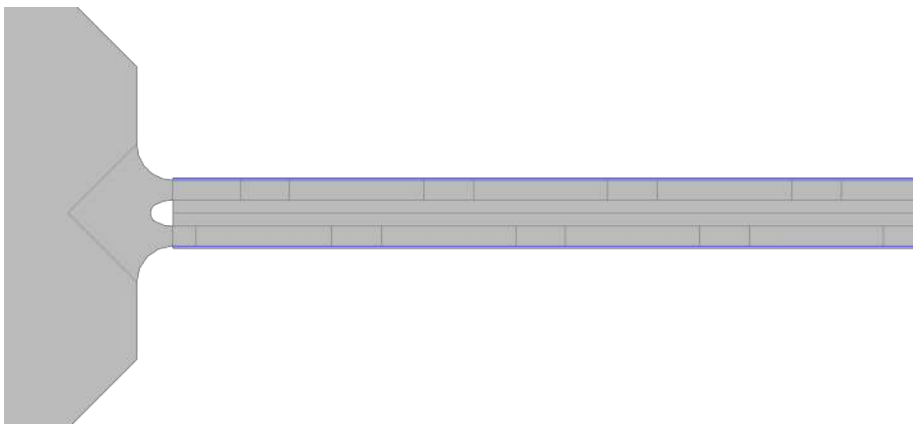

Figure 5.14: Detail of Viscoelastic Spillovers (in blue)

In the initial formulation, the viscoelastic domain was restricted to the elastomer located within the honeycomb core of the blade. However, during the impregnation process used in manufacturing, a non-negligible amount of elastomer is deposited on the outer surfaces of the titanium blades, forming a thin continuous layer that remains bonded to the metallic structure. This additional material, hereafter referred to as viscoelastic spillover, represents a physically real contribution to both stiffness and dissipation that is absent from the baseline configuration. To account for this effect, an additional viscoelastic layer was introduced in the numerical model using the same constitutive formulation adopted for the core material. This modification increases the total volume of viscoelastic material participating in the deformation and alters the mechanical coupling between the elastomer and the titanium blades.

The introduction of the spillover layers produces a modest increase in the predicted oscillation frequency, which rises from 12.30 Hz to 12.54 Hz. This limited variation confirms that the global stiffness of the system remains primarily governed by the titanium blades, while the additional elastomer contributes only marginally to the elastic response. This behavior is physically consistent, as the thin spillover layer mainly undergoes surface-level shear deformation and does not significantly modify the bending stiffness of the structure.

In contrast, the effect on damping-related quantities is more subtle. The average quality factor remains essentially unchanged with respect to the baseline configuration, with variations below 1%. This result indicates that the mere addition of viscoelastic volume does not automatically translate into increased energy dissipation. Instead, dissipation depends critically on the strain amplitudes and strain-rate distributions experienced by the viscoelastic material. Although the spillover layers increase the total amount of elastomer, they are subjected to relatively low strain levels during the dominant bending mode of the blade and therefore contribute only weakly to the overall energy loss per oscillation cycle. This observation provides an important physical insight: viscoelastic damping in the present system is not governed by the total amount of polymeric material alone, but by how effectively that material is mechanically engaged by the deformation mode.

Further insight is obtained by analyzing the spatial distribution of the stored energy density within the viscoelastic regions. Figure 5.15 (top) reports the stored energy density for the baseline configuration, while Figure 5.15 (bottom) shows the corresponding field for the spillover model. A direct comparison reveals a pronounced reduction of peak stored energy density when the spillover layer is included. In the baseline configuration, localized maxima reach values of approximately 0.56 J/m^3, whereas in the spillover configuration the maximum stored energy density decreases to about 0.12 J/m^3. This reduction indicates a substantial attenuation of local energy concentration within the viscoelastic material.

Beyond the reduction in peak values, the qualitative structure of the energy field also changes significantly. In the baseline configuration, the stored energy density is strongly localized in narrow regions of the viscoelastic core, reflecting pronounced strain concentration associated with the interaction between the honeycomb geometry and the bending deformation of the blades. In contrast, the spillover configuration exhibits a smoother and more spatially distributed energy field. The presence of the additional

elastomer layer promotes a more gradual transfer of deformation between the titanium blades and the viscoelastic material, effectively spreading the strain over a larger volume and reducing sharp localization effects.

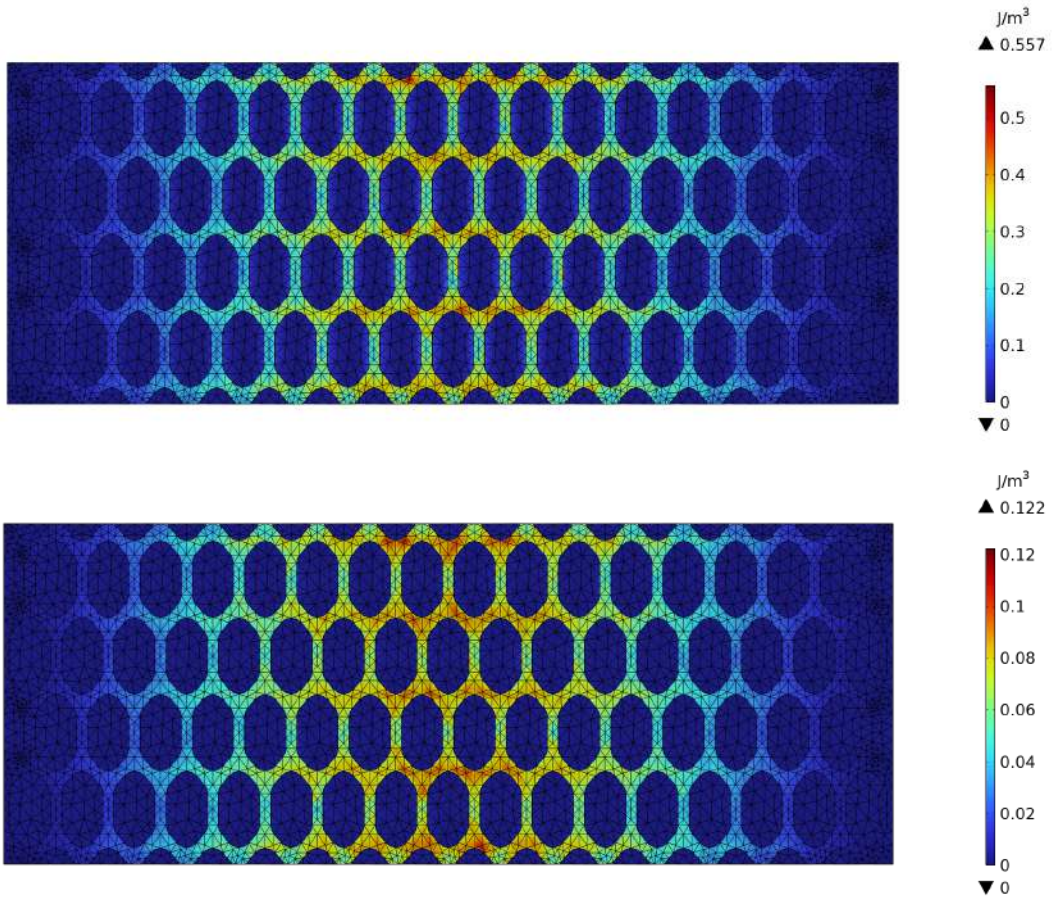

Figure 5.15: Stored Energy Density (J/m^3) in the viscoelastic core at t=4.5 s for Baseline configuration (top) and Spillover configuration (bottom)

This redistribution mechanism has important physical implications. Viscoelastic dissipation is inherently driven by local strain amplitudes and strain-rate histories. When deformation is highly localized, dissipation mechanisms are activated intensely but only within limited regions. Conversely, when deformation is distributed more uniformly, local strain peaks are reduced even though a larger portion of the viscoelastic domain participates in the mechanical response. The observed reduction in stored energy density peaks therefore suggests that the spillover layer acts primarily as a mechanical buffer, mitigating sharp strain localization rather than enhancing viscoelastic activation.

This behavior provides a consistent interpretation of the global dynamic results obtained for the spillover configuration. Although additional viscoelastic material is introduced, the reduced local strain intensity limits the amount of energy stored and subsequently dissipated per oscillation cycle at the material level. As a consequence, the spillover configuration does not lead to an increase in effective damping; instead, it tends to preserve, or slightly reduce, the overall dissipation relative to the baseline model. These findings highlight that viscoelastic damping in the present system is not governed solely

by material volume, but critically depends on the manner in which deformation is transmitted, concentrated, and distributed within the viscoelastic regions.

From a modeling perspective, the spillovers case highlights the importance of correctly representing not only the presence of viscoelastic material, but also its structural role and mechanical coupling. While including the spillover layers improve the geometric and physical completeness of the model, it does not resolve the discrepancy with experimental damping levels. This suggests that the experimentally observed dissipation is dominated by the viscoelastic material embedded within the load-bearing deformation path, namely the elastomer within the honeycomb core, rather than by peripheral elastomeric layers.

5.2.3.3 Instantaneous Shear Modulus G_0

The next sensitivity study investigates the influence of the instantaneous shear modulus G_0 of the viscoelastic material. This parameter defines the short-time elastic response of the elastomer and therefore directly controls both the initial stiffness of the viscoelastic phase and the magnitude of stresses and strains driving dissipation within the generalized Maxwell model. In the baseline configuration, was identified from material characterization and represents the reference stiffness level at very short loading times. To assess its impact on the global dynamic response, a reduced value of G_0 equal to -20% of the baseline was considered.

The effect of G_0 on damping-related quantities is significantly more pronounced. Both minimum- and maximum-based quality factors decrease substantially (9% approximately). This increase in effective damping indicates that lowering the instantaneous stiffness of the viscoelastic material enhances energy dissipation per oscillation cycle. This behavior can be explained by examining the strain response of the viscoelastic core: a lower G_0 allows larger shear strains to develop under the same global deformation of the blade. Since the dissipation mechanisms of the generalized Maxwell model are driven by strain amplitudes and strain rates, increased deformation of the elastomer leads to stronger activation of the internal viscous branches. As a result, a greater fraction of mechanical energy is dissipated during each cycle, yielding a lower quality factor.

Importantly, this trend demonstrates that viscoelastic dissipation in the present system is not primarily controlled by stiffness alone, but by the balance between stiffness and strain localization. A stiffer elastomer limits deformation and suppresses dissipation, whereas a more compliant material amplifies strain levels and thereby promotes energy loss. This finding is fully consistent with the amplitude-dependent damping behavior discussed in the baseline analysis and further confirms that damping is governed by local viscoelastic strain states rather than by global modal properties.

Despite the improved damping performance obtained by reducing G_0, the numerical quality factors remain higher than those measured experimentally. The gap with respect to experimental values is reduced but not eliminated, indicating that modifying the instantaneous shear modulus alone is insufficient to fully reproduce the observed dissipation levels. This suggests that additional parameters, such as the relaxation spectrum of the Maxwell branches or the nearly incompressible volumetric response, play a concurrent role in controlling energy dissipation.

5.2.3.4 Poisson's Ratio ν

The following sensitivity study examines the influence of the Poisson's ratio of the viscoelastic material, which was increased from the baseline value of $\nu = 0.49$ to $\nu = 0.499$, thereby moving the elastomer closer to the nearly incompressible regime. While this parameter does not directly modify the shear relaxation properties of the generalized Maxwell model, it plays a fundamental role in determining the kinematic constraints under which viscoelastic deformation occurs. In viscoelastic materials, the Poisson's ratio governs the relative contribution of volumetric and deviatoric deformation modes. As ν approaches 0.5, volumetric deformation is increasingly suppressed, forcing the material to accommodate imposed displacements primarily through shear. This effect is particularly relevant in bonded elastomer–metal assemblies, where lateral expansion is geometrically constrained and shear deformation becomes the dominant mechanism.

The numerical results reflect this behavior clearly. Increasing the Poisson's ratio leads to a moderate increase in the oscillation frequency, which rises from 12.30 Hz in the baseline configuration to 12.59 Hz. This trend indicates a slight stiffening of the global response, consistent with the reduced compressibility of the elastomer. As volumetric compliance is suppressed, the viscoelastic core offers greater resistance to deformation, increasing the effective dynamic stiffness of the structure. The impact on damping, however, remains comparatively limited. The average quality factor decreases only marginally, from 24.99 to 24.39, corresponding to a reduction of approximately 2.4%. This modest variation indicates that modifying the Poisson's ratio alone does not substantially enhance energy dissipation. Although the deformation mode of the elastomer shifts toward a more shear-dominated state, the absolute strain amplitudes and strain rates experienced by the material remain largely unchanged. This means that while near-incompressibility affects how deformation is partitioned between volumetric and deviatoric components, it does not automatically increase the magnitude of viscoelastic strain. Consequently, the activation level of the Maxwell branches, and therefore the dissipated energy per cycle, remains similar to that of the baseline case. In other words, Poisson's ratio controls the nature of deformation, but not necessarily its intensity.

The limited sensitivity of the quality factor to changes in ν suggests that volumetric effects are not the dominant source of dissipation in the present configuration. Instead, dissipation appears to be primarily governed by shear deformation amplitudes and by the relaxation behavior encoded in the deviatoric part of the viscoelastic model. This conclusion is consistent with the earlier observation that dissipation is concentrated in regions experiencing significant shear strains, such as within the honeycomb core, rather than in regions dominated by volumetric response. Nevertheless, the influence of Poisson's ratio remains relevant from a numerical standpoint. Nearly incompressible materials are known to increase the susceptibility of finite element formulations to volumetric locking and mesh-dependent behavior if not treated carefully. The relatively small changes observed here confirm that the chosen formulation and discretization are capable of handling near-incompressibility without introducing spurious stiffness or artificial damping. At the same time, they indicate that further increases in ν alone are unlikely to bridge the remaining gap with experimental damping levels.

This behavior becomes even clearer when examining the spatial distribution of volumetric strain within the viscoelastic material. Figure 5.16 compares the volumetric strain

fields obtained for the increased Poisson's ratio case and for the baseline configuration. In the higher-ν model, the magnitude of volumetric strain is markedly reduced across the entire viscoelastic domain, with peak values significantly lower than those observed in the baseline case. This reduction confirms that the material response approaches an increasingly incompressible condition, suppressing volumetric deformation modes almost entirely.

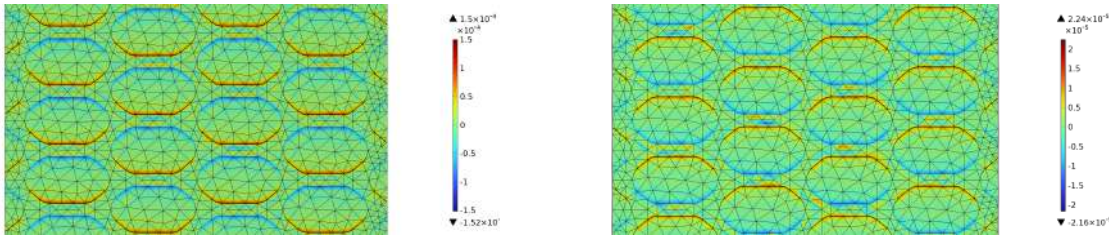

Figure 5.16: Volumetric strain in the viscoelastic core at t=4.5 s for Baseline configuration (left) and higher-ν configuration (right)

At the same time, the overall pattern of the strain field remains qualitatively similar between the two cases. The locations of strain concentration, primarily at the interfaces between the honeycomb geometry and the surrounding elastomer, are preserved. What changes is not the topology of the deformation field, but its nature: deformation becomes more strictly deviatoric as volumetric contributions are progressively constrained. As a consequence, the increase in Poisson's ratio does not lead to stronger dissipation, but instead redistributes deformation without significantly amplifying the activation of the viscoelastic internal variables.

5.2.3.5 Volumetric and Deviatoric Viscoelastic Dissipation

Following the analysis of the Poisson's ratio, which governs the kinematic partition between volumetric and shear deformation, the next sensitivity study investigates the role of volumetric dissipation within the viscoelastic constitutive model. In the baseline configuration, viscoelasticity is introduced exclusively through the deviatoric part of the stress tensor, such that time-dependent dissipation is associated only with shear deformation, while volumetric behavior remains purely elastic. This formulation is commonly adopted for elastomeric materials, where dissipation is typically dominated by shear mechanisms and volumetric losses are often assumed negligible.

To assess the validity of this assumption for the present configuration, an alternative model was considered in which viscoelasticity is applied to both the deviatoric and volumetric components of the stress response. In this formulation, not only shear strains but also volumetric deformation contribute to time-dependent energy dissipation through relaxation mechanisms. The numerical results show that introducing volumetric viscoelasticity produces only a minor effect on the global dynamic response. The oscillation frequency increases slightly from 12.30 Hz to 12.35 Hz, indicating a negligible modification of the effective stiffness of the system. This limited influence confirms that volumetric deformation plays a secondary role in the overall kinematics of the blade, which remains dominated by bending-induced shear in the elastomeric core. The impact on damping is similarly modest. The average quality factor decreases from 24.99 to 24.05. While this change is measurable, it remains small compared to the variations observed when

modifying shear-related parameters such as G_0 or the Maxwell relaxation spectrum. This indicates that, even when volumetric dissipation is explicitly enabled, the amount of energy dissipated through volumetric mechanisms remains limited.

This result can be understood by considering the deformation state of the viscoelastic material during free oscillations. Despite the nearly incompressible nature of the elastomer, the actual volumetric strain levels remain very small due to geometric constraints and the dominance of bending-induced shear deformation. Consequently, even when volumetric relaxation mechanisms are introduced, they are only weakly activated and contribute marginally to the total energy loss per cycle.

The close connection between this result and the Poisson's ratio sensitivity analysis is particularly noteworthy. Increasing the Poisson's ratio enforces near-incompressibility and suppresses volumetric strain, while introducing volumetric viscoelasticity allows volumetric strain (if present) to dissipate energy. The fact that both modifications produce only minor changes in the damping response provides strong evidence that volumetric deformation is not a governing mechanism in the present system. Instead, viscoelastic dissipation is overwhelmingly controlled by deviatoric shear processes within the elastomer.

From a modeling perspective, this finding justifies the use of a deviatoric-only viscoelastic formulation as a reasonable and physically consistent baseline choice. Including volumetric dissipation increases model complexity without yielding substantial improvements in predictive capability. At the same time, the small but non-zero influence observed here confirms that volumetric effects are not entirely irrelevant and may become more significant in alternative geometries or loading conditions where volumetric strain levels are higher.

5.2.3.6 Maxwell-branch Moduli

The present sensitivity case investigates the influence of the viscoelastic relaxation spectrum on the damping behavior of the system by simultaneously increasing both the shear moduli and the relaxation times of the generalized Maxwell branches by 20%. This combined variation modifies not only the intensity of the viscous stresses generated within the material, but also their characteristic temporal evolution. As a result, the analysis explores the coupled effect of stronger viscoelastic resistance and slower stress relaxation under cyclic deformation. From a constitutive standpoint, increasing the Maxwell shear moduli amplifies the magnitude of the internal viscous stresses activated by a given strain-rate history, while increasing the relaxation times prolongs the persistence of these stresses in time. Together, these two effects lead to a higher amount of mechanical energy stored transiently in the internal variables and subsequently dissipated during each oscillation cycle. Unlike modifications acting purely on stiffness or geometry, this parameter variation directly targets the dissipation mechanism encoded in the material model.

At the global level, the oscillation frequency remains only weakly affected by this modification, confirming that the dynamic stiffness of the system is still governed primarily by the elastic response of the titanium blades and by the equilibrium modulus G_0 of the elastomer. In contrast, the quality factor decreases substantially: Q_{min} drops to 19.22, Q_{max} to 21.20. This is the largest damping-related change observed among the material

sensitivity cases considered so far and demonstrates that the dissipation deficit identified in the baseline configuration can be effectively addressed by acting directly on the viscoelastic relaxation spectrum. Importantly, this modification reduces the discrepancy with experiments to approximately +10.7%, bringing the numerical damping level into a range that is no longer grossly inconsistent with the experimentally observed response.

Additional insight is obtained by examining the spatial distribution of the viscoelastic dissipation rate density. Figure 5.17 shows the dissipation rate maps for the baseline configuration and for the model with increased Maxwell moduli. A clear amplification of dissipation intensity is observed throughout the viscoelastic domain. While the baseline configuration exhibits peak dissipation values of approximately 13.6 W/m^3, the modified model reaches values close to 18.8 W/m^3, corresponding to an increase of nearly 40% in the locally dissipated power.

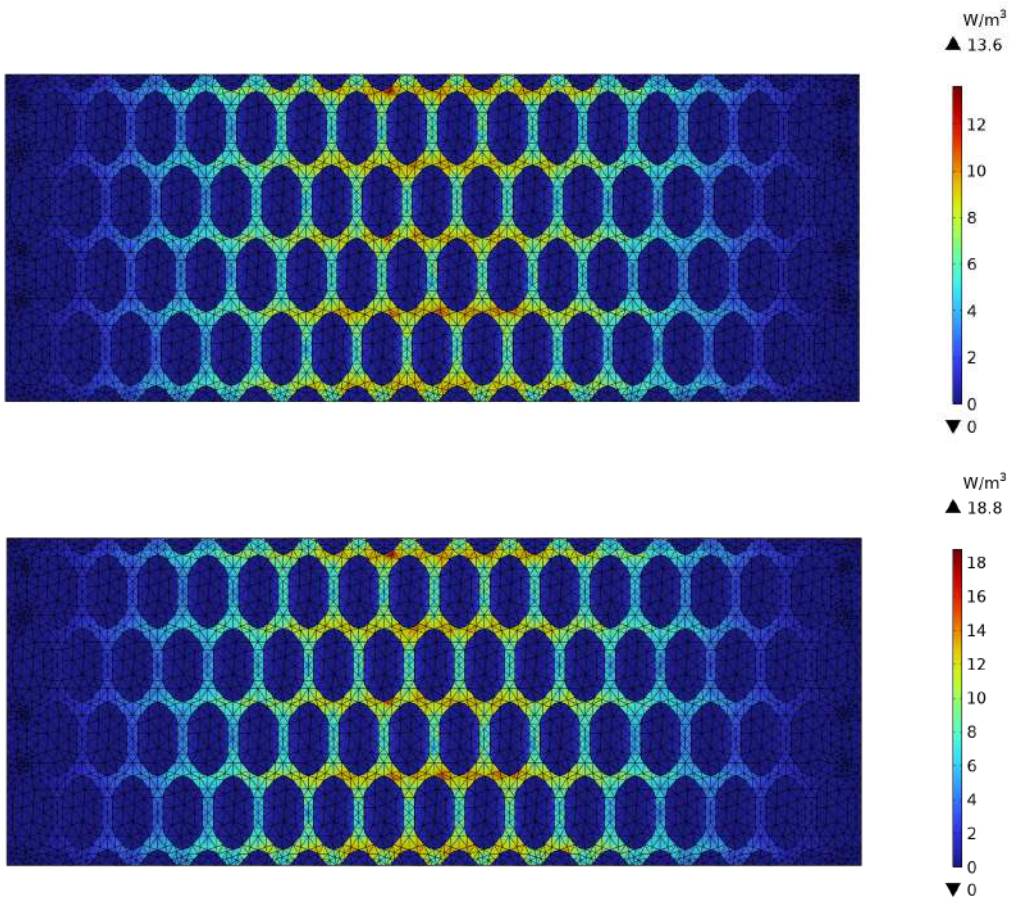

Figure 5.17: Viscoelastic dissipation rate density (W/m^3) at t=4.5 s for Baseline configuration (top) and higher Maxwell moduli configuration (bottom)

Importantly, the spatial pattern of dissipation remains essentially unchanged. The regions of highest dissipation continue to be localized along the ligaments of the honeycomb structure, where shear deformation and strain-rate amplitudes are maximal under the dominant bending mode. This indicates that the modification does not introduce new dissipative mechanisms or numerical artifacts, but rather intensifies the existing physical dissipation processes already active in the baseline model. The simultaneous increase of

shear moduli and relaxation times therefore leads to a twofold effect: on one hand, larger viscous stresses are generated for a given deformation; on the other hand, these stresses decay more slowly over time, increasing the energy dissipated during each oscillation cycle. This explains the strong sensitivity of the quality factor to the relaxation spectrum, in contrast with the comparatively weak influence of geometric or mesh-related parameters.

These results highlight the central role of the Maxwell parameters in controlling damping in viscoelastic lattice-based flexure systems. Unlike mesh refinement, which primarily affects the resolution of strain fields, or geometric modifications, which alter deformation pathways, the relaxation spectrum directly governs the conversion of mechanical energy into heat at the material level. Consequently, the Maxwell moduli and relaxation times emerge as the most influential parameters for tuning the numerical damping response and represent a key focus of the subsequent calibration and interpretation of the numerical model.

5.2.3.7 Large Displacements

The last sensitivity case investigates the influence of enabling large-displacement kinematics in the structural model. In the baseline configuration, the response is obtained under the small-strain/small-rotation assumptions typically associated with linearized kinematics, which are often sufficient for small-amplitude vibrations. However, the present free-oscillation test originates from a finite prescribed deflection followed by release, and the resulting motion can involve non-negligible rotations and geometric nonlinearities even if material strains remain moderate. Activating the large-displacement formulation therefore provides a direct way to assess whether geometric nonlinearities contribute meaningfully to the observed frequency and damping behavior, or whether the response remains effectively linear in kinematic terms.

The numerical results indicate that large-displacement effects have a negligible impact on the global dynamic response for the considered operating range. The mean oscillation frequency changes only marginally from 12.30 Hz to 12.34 Hz (+0.3%), confirming that the stiffness–inertia balance governing the dominant bending mode is essentially unchanged. This suggests that, despite the finite initial deflection, the subsequent free oscillations occur in a regime where geometric stiffening/softening mechanisms (such as mid-plane stretching or curvature-induced stiffening) do not significantly alter the effective tangent stiffness experienced during the decay. Similarly, damping-related quantities remain largely unaffected. The quality factor increases slightly indicating that the viscoelastic dissipation mechanisms are not strongly coupled to geometric nonlinearities in this configuration. In other words, enabling large displacements does not substantially modify the strain amplitudes and strain-rate histories within the elastomeric domains that drive the internal-variable evolution of the generalized Maxwell model. The slight increase in Q can be interpreted as a second-order effect associated with small shifts in the spatial distribution of deformation, e.g., a mild redistribution of strain away from the most dissipative regions, rather than as evidence of a fundamentally different damping regime.

From a modeling standpoint, this outcome is significant because it confirms that the baseline analysis and the amplitude-loss framework are not critically dependent on the

kinematic linearization. In particular, the near invariance of both frequency and quality factor implies that the observed amplitude-dependent dissipation and the minima–maxima asymmetry discussed previously are not artifacts of neglecting geometric nonlinearities. Instead, they remain intrinsic to the viscoelastic constitutive response and to the mechanical coupling between the elastomer and the titanium blades.

The results of all sensitivity cases discussed in the previous subsections are summarized in Table 5.5, which provides a consolidated overview of the influence of geometric, material, and numerical parameters on both the oscillation frequency and the damping response.

Table 5.5: Sensitivity analysis results with respect to the baseline configuration

Parameter	**Ref.**	**Var.**	f **[Hz]**	Δf	Q_{min}	Q_{max}	Q_{avg}	ΔQ_{FEM}	ΔQ_{exp}
Reference									
Baseline	–	–	12.30	0%	23.81	26.19	24.99	0%	+36.9%
Blades									
Real Dim.	–	–	14.27	+16%	28.98	31.21	30.09	+20.4%	+64.9%
Spillover	–	–	12.54	+1.9%	23.91	26.08	24.95	-0.16%	+36.7%
Viscoelastic material									
G_0	0.38 MPa	-20%	11.90	-3.2%	21.69	23.86	22.77	-8.9%	+24.8%
ν	0.49	0.499	12.59	+2.3%	23.30	25.48	24.39	-2.4%	+33.6%
Vol. + Dev.	Dev.	–	12.35	+0.4%	22.95	25.16	24.05	-3.7%	+31.8%
Maxwell Val.	–	+20%	12.44	+1.1%	19.22	21.20	20.21	-19.1%	+10.7%
Large Displ.	–	–	12.34	+0.3%	24.23	26.44	25.33	+1.3%	+38.8%
Mesh									
Finer	–	–	12.12	-1.4%	23.81	26.19	25	+0.04%	+37%
Extra Fine	–	–	12.06	-1.9%	23.71	26.10	24.91	-0.3%	+36.5%

The table highlights not only the relative variation with respect to the baseline configuration, but also the remaining discrepancy with experimental measurements, thereby offering a quantitative basis for identifying the parameters that most strongly govern the dynamic behavior of the system and for guiding subsequent model calibration efforts.

The analyses confirm that the oscillation frequency is primarily controlled by global stiffness-related parameters, such as the effective geometry of the blades and the elastic properties of the structural components. Variations in mesh resolution, viscoelastic formulation, and numerical settings lead only to limited frequency changes, indicating that the elastic response of the system is already well captured by the baseline configuration. In contrast, damping-related quantities exhibit a markedly higher sensitivity to both material modeling choices and the manner in which deformation is transmitted to the viscoelastic domains.

Material parameters associated with the generalized Maxwell model emerge as the dominant contributors to damping variations. In particular, the sensitivity to the Maxwell branch moduli highlights the critical role of the relaxation spectrum in controlling the rate at which mechanical energy is dissipated during cyclic loading.

Geometric refinements, including the use of real blade dimensions and the introduction of viscoelastic spillover, demonstrate that damping cannot be interpreted solely in terms

of material volume. Instead, the effectiveness of viscoelastic dissipation is governed by the spatial distribution of strain and strain-rate fields. Additional elastomer located in regions of weak deformation contributes marginally to dissipation, while modifications that alter strain localization patterns can significantly affect the energy exchanged within the viscoelastic core. These results emphasize that viscoelastic damping is fundamentally a deformation-driven phenomenon rather than a purely material-driven one.

At the same time, the analyses reveal that even when physically motivated parameter adjustments are introduced, residual discrepancies between numerical and experimental damping levels may persist. Part of this deviation may stem from modeling assumptions that are inherent to the current finite element formulation. In particular, the use of the *Form Union* approach in COMSOL implies a fully bonded interface between the titanium blades and the viscoelastic material, thereby suppressing any relative sliding or local contact compliance that may occur in the real system. In the experimental specimens, microscopic interfacial effects, partial debonding, or frictional sliding may contribute additional dissipation mechanisms that are not represented when contact interfaces are not explicitly modeled. The absence of such contact-driven phenomena may therefore lead to an underestimation or redistribution of energy dissipation in the numerical model.

5.3 *Eigenfrequency Study*

In order to assess whether the vibration modes identified experimentally in Section 4.2.2 correspond to physically meaningful structural modes, an eigenfrequency study was performed using a numerical model derived directly from the configuration employed in the forced oscillation simulations. The objective of this analysis was not to introduce an alternative representation of the system, but rather to verify the consistency between experimental observations and the dynamic characteristics predicted by a model that closely reproduces the actual test conditions.

Starting from the forced oscillation model of Figure 4.12, the boundary conditions were modified to reflect the suspension-based experimental setup more realistically. In particular, the fixed constraint previously applied at the upper end of the blade to model the free oscillation test, was removed. This constraint, while convenient for free oscillation analyses, artificially suppresses rigid-body and low-frequency compliant motions that are present in the experimental configuration. It was therefore replaced by a rigid connector imposing zero rotations about the x and z axes and zero displacement along the y direction, corresponding to the direction of displacement application. This choice enforces the dominant kinematic constraints introduced by the shaker interface while allowing the structure to deform consistently with the experimental setup. To further account for the compliant nature of the suspension threads supporting the blade, a linear spring element was introduced at the connection point. Translational and rotational stiffness values were both set to $k_u = 1$ N/m and $k_\theta = 1$ N m/rad, respectively. These values were selected to emulate the low but finite stiffness of the suspension system, ensuring that the model captures the essential low-frequency compliance without introducing artificial rigid-body modes. This modeling strategy allows the blade to exhibit global bending and higher-order deformation modes in a manner consistent with the experimental configuration, while avoiding over-constraining the system.

The resulting eigenfrequency analysis yielded two dominant complex eigenvalues, corresponding to modes at approximately 12 Hz and 443 Hz. A third mode previously observed at approximately 274 Hz in earlier, more constrained models was no longer present. The disappearance of this intermediate mode indicates that it was not associated with a physically meaningful deformation of the structure, but rather originated from modeling artifacts related to unrealistic boundary conditions or localized numerical degrees of freedom. The absence of a clearly identifiable and repeatable experimental resonance in the corresponding frequency range further supports this interpretation.

The first eigenmode is characterized by a complex eigenfrequency

$$f_1 = 11.994 + 0.4298\, i \text{ Hz}$$

The associated mode shape, shown in Figure 5.18 exhibits a smooth, global bending deformation along the entire blade length, with no localized curvature or higher-order features. This deformation pattern is consistent with a first bending mode and confirms that the low-frequency dynamic response of the system is dominated by a single structural degree of freedom. This observation is fully aligned with the experimental evidence obtained from free oscillation tests, which demonstrated a strongly SDOF-like behavior with highly repeatable resonance frequencies and damping estimates.

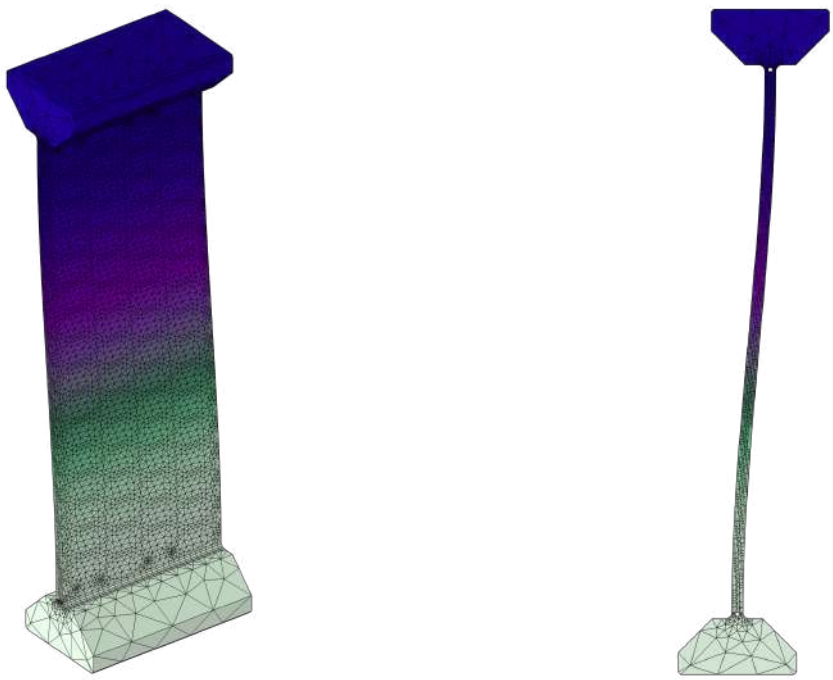

Figure 5.18: First Mode Shape

From the complex eigenfrequency, the modal damping ratio is computed as

$$\zeta_1 = \frac{\mathrm{Im}(f_1)}{\mathrm{Re}(f_1)} = \frac{0.4298}{11.994} \approx 0.0358$$

which corresponds to a quality factor

$$Q_1 = \frac{1}{2\zeta_1} \approx 14.0$$

This value lies between the damping ratios obtained from free oscillation tests and those identified through forced oscillation FRF fitting, indicating that the numerical model captures the dominant intrinsic dissipation mechanisms governing the first mode. The small discrepancy between numerical and experimental resonance frequencies, with experimental values typically around 14 Hz, is attributed to uncertainties in the effective stiffness of the suspension system, added mass contributions from the experimental apparatus, and unavoidable simplifications in the numerical representation of the boundary conditions.

The second retained eigenmode is characterized by a complex eigenfrequency

$$f_2 = 443.04 + 18.469\, i \text{ Hz}$$

The corresponding mode shape (Figure 5.19) shows a higher-order bending deformation with increased curvature and strain localization along the blade, particularly in regions away from the suspension points. This mode represents a physically meaningful structural deformation rather than a local or numerical artifact. Experimentally, a resonance-like feature was observed in the high-frequency range between approximately 530 Hz (modal fit FRF) and 580 Hz (raw-signal FRF) during forced oscillation tests. While the numerical eigenfrequency underestimates the experimental resonance frequency, the qualitative correspondence in deformation pattern and dynamic behavior supports the interpretation that both results refer to the same underlying structural mode.

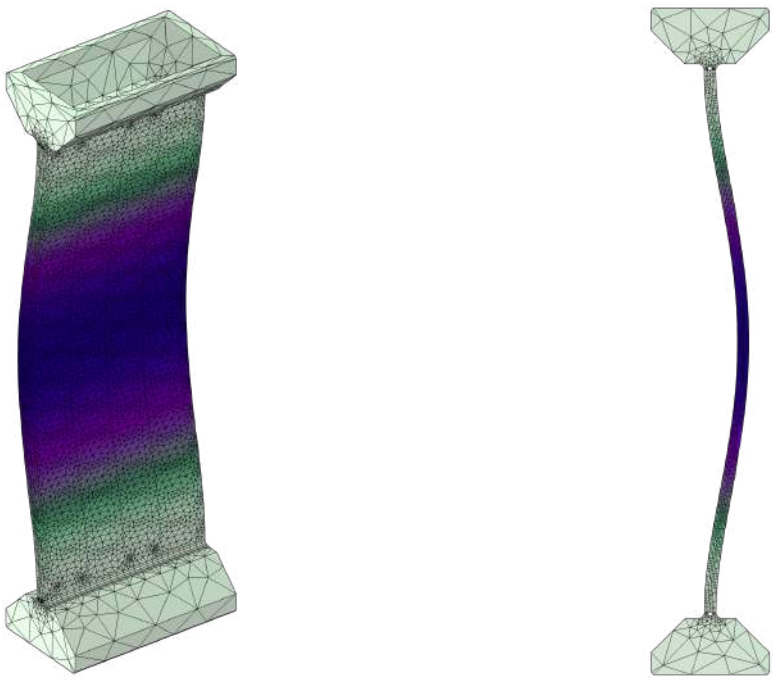

Figure 5.19: Second Mode Shape

The modal damping ratio associated with this mode is computed as

$$\zeta_2 = \frac{\mathrm{Im}(f_2)}{\mathrm{Re}(f_2)} = \frac{18.469}{443.04} \approx 0.0417$$

leading to a quality factor

$$Q_2 = \frac{1}{2\zeta_2} \approx 12.0$$

This value is in close agreement with the effective damping estimates obtained from the forced oscillation FRF analysis in the same frequency range. The larger discrepancy in resonance frequency could be attributed to the increasing influence of frequency-dependent viscoelastic material behavior, enhanced sensitivity to boundary condition compliance, and the reduced reliability of experimental identification at high frequencies due to modal interaction and anti-resonant effects. Importantly, while the precise resonance frequency is more difficult to predict accurately in this regime, the numerical model successfully reproduces the correct order of magnitude of the damping, indicating that the dominant energy dissipation mechanisms are appropriately captured.

It is worth noting that no eigenmode corresponding to the experimentally observed resonance around 38 Hz was identified in the numerical eigenfrequency analysis. This apparent discrepancy is not unexpected and does not indicate an inconsistency between experimental and numerical results. The resonance observed experimentally in this frequency range does not exhibit the characteristics of a classical structural eigenmode, but rather manifests as a weak, excitation-dependent feature with limited modal dominance. In the forced oscillation measurements, the response around 38 Hz is strongly influenced by the interaction between the global bending mode and the specific excitation and suspension configuration, leading to a localized amplification that does not correspond to a standalone deformation pattern. Such effects are commonly associated with dynamic coupling phenomena, boundary-condition-induced resonances, or forced-response features arising from the interaction between structural flexibility and compliant supports. Since eigenfrequency studies identify only the natural modes of the unconstrained or lightly constrained structure, response features that depend on the forcing location, direction, or boundary compliance are not expected to appear as independent eigenmodes. This interpretation is further supported by the experimental observations, as the resonance near 38 Hz exhibits lower modal robustness compared to the first bending mode and is more sensitive to excitation conditions and frequency resolution. Consequently, its absence in the eigenfrequency analysis confirms that this feature should be interpreted as an effective response amplification under forced excitation rather than as an intrinsic structural mode of the blade.

Overall, the eigenfrequency study confirms that the experimentally observed modes at low and high frequencies correspond to physically meaningful structural modes of the blade when realistic boundary conditions are applied. The first mode at approximately 12 Hz is conclusively identified as the dominant global bending mode and provides a robust basis for intrinsic damping characterization. The higher-frequency mode around 443 Hz represents a genuine structural deformation whose effective damping is strongly influenced by viscoelastic effects and experimental configuration.

5.4 *Design Limitations and Potential Improvements*

This chapter investigates the mechanical behavior of the blade assembly with the objective of identifying its design limitations and potential avenues for improvement. Stress and strain responses of the blades and the viscoelastic interlayer are analyzed to clarify load

transfer mechanisms, deformation modes, and critical regions. The results are then used to assess the effectiveness of the current design, highlight governing parameters that limit performance, and provide physically motivated guidance for design optimization.

5.4.1 Blades Mechanical Behavior

The deformation behavior of the blade assembly with an interposed viscoelastic layer can be interpreted by considering the interaction between bending of the steel blades and shear deformation within the compliant interlayer, as illustrated in Figure 5.20. Under the applied loading, the blades exhibit opposite tensile and compressive stress states across their thickness, which induces a bending-driven lateral displacement along their length. The presence of the viscoelastic material mechanically couples the two blades and modifies their kinematic response compared to an uncoupled configuration. Two limiting deformation scenarios can be identified to frame the system behavior. In the first scenario, the viscoelastic layer fully follows the deformation of the steel blades, accommodating the relative motion through large shear strains; this represents a highly compliant interlayer with minimal resistance to relative sliding. In the second scenario, the viscoelastic forces are sufficiently strong to enforce a coupled behavior between the blades, causing the assembly to respond as a single, effectively thicker blade with reduced relative displacement but increased global bending stiffness. These two idealized scenarios establish upper and lower bounds for shear deformation and bending stiffness. In the following sections, the results of the finite element simulations are analyzed and systematically positioned within these bounds to assess the actual deformation behavior of the assembly and to identify the implications for design limitations and potential improvements.

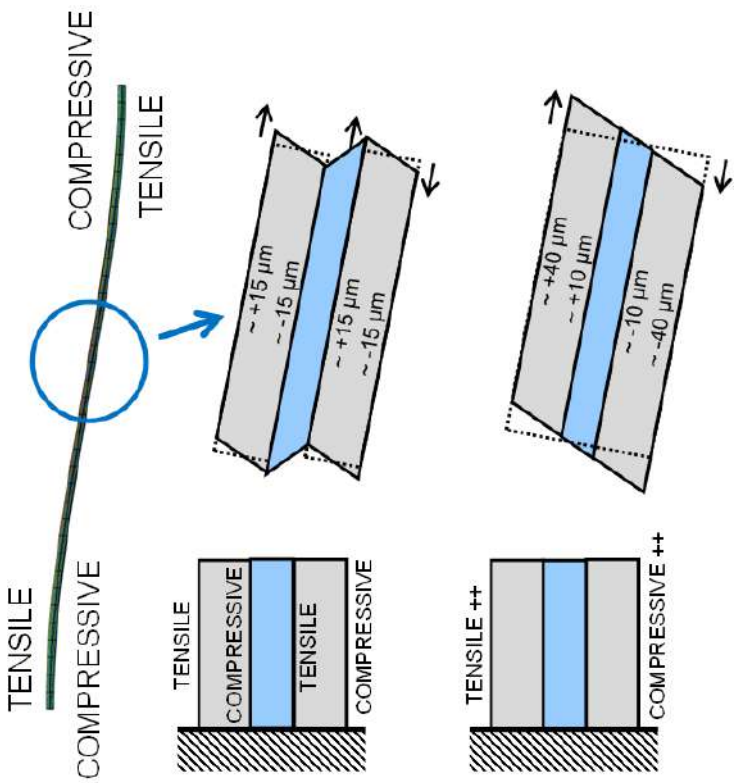

Figure 5.20: Extreme deformation scenarios of the viscoelastic interlayer (blue) within the blade assembly

5.4.2 Behavior from Simulations

The mechanical response of the blade assembly obtained from the numerical simulations reveals a deformation behavior that is significantly more complex than what would be expected from a simplified one-dimensional or planar bending interpretation. Although the prescribed displacement is applied exclusively along the longitudinal y-direction, the resulting deformation field is intrinsically three-dimensional, involving substantial displacement components in both the transverse (x) and out-of-plane (z) directions. This three-dimensional nature of the response plays a central role in the activation of viscoelastic dissipation mechanisms and must therefore be carefully analyzed before addressing design limitations and improvement strategies.

The out-of-plane displacement component, reported in Figure 5.21(left), highlights a secondary deformation mode superimposed on the primary bending response. The blade does not deform within a single plane, but instead exhibits measurable deflections normal to the bending plane, particularly in the central portion of the blade where curvature is highest. This behavior indicates a coupling between bending and torsional-like deformation, which introduces additional strain components within the viscoelastic interlayer.

The transverse displacement field, shown in Figure 5.21(right), reveals an alternating spatial pattern characterized by regions of positive and negative displacement along the blade span. This response indicates lateral motion of the titanium blades relative to one another, which directly promotes shear deformation within the viscoelastic material. Importantly, this shear is not uniformly distributed, but instead concentrates in specific regions dictated by the blade geometry and by the local bending curvature.

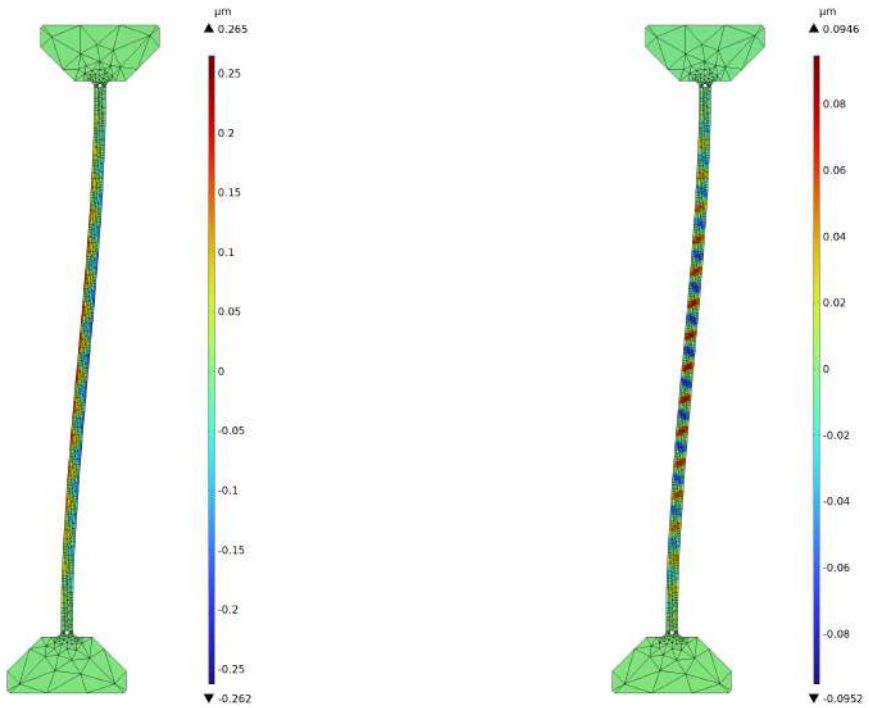

Figure 5.21: Displacement field obtained from numerical simulations: displacement along the z-direction (left) and displacement along the x-direction (right)

The simultaneous presence of transverse sliding and out-of-plane deformation implies that the viscoelastic layer is subjected to a multi-axial strain state rather than to pure shear alone. As a result, the internal variables of the generalized Maxwell model are activated through a combination of shear and deviatoric strain components, significantly influencing both the spatial distribution of stored energy and the local dissipation rate. This three-dimensional deformation mechanism explains why dissipation-related quantities observed in previous sections, such as amplitude loss and quality factor, are highly sensitive to geometric features and local strain distributions, while global quantities such as oscillation frequency remain comparatively robust. The viscoelastic damping behavior therefore emerges not merely from material properties, but from the manner in which the blade geometry channels deformation into the compliant interlayer.

From a design perspective, these results indicate that the current configuration does not exploit the full dissipation potential of the viscoelastic material. Since dissipation is governed by localized shear and multi-axial strain activation, improving damping performance requires promoting deformation modes that increase strain intensity within the elastomer without excessively increasing global stiffness.

The internal load transfer mechanisms within the blade assembly can be further clarified by examining the stress and strain fields developed under the imposed deformation. While the displacement analysis provides insight into the global kinematic response of the structure, the stress and strain distributions reveal how bending-induced loads are transmitted through the titanium blades and subsequently transferred to the viscoelastic interlayer. These fields are therefore essential for identifying mechanically critical regions and for understanding which portions of the structure effectively govern the system response. Figure 5.22 reports the von Mises stress distribution evaluated at the maximum imposed displacement. The stress field is predominantly localized within the titanium blades, confirming that the metallic components carry the majority of the bending load. Distinct stress bands develop along the blade length, with higher stress levels occurring in correspondence with regions where bending curvature is most pronounced. The stress distribution remains smooth and continuous, without the emergence of sharp local peaks, indicating that the current geometry does not introduce severe stress concentrations that could compromise structural integrity.

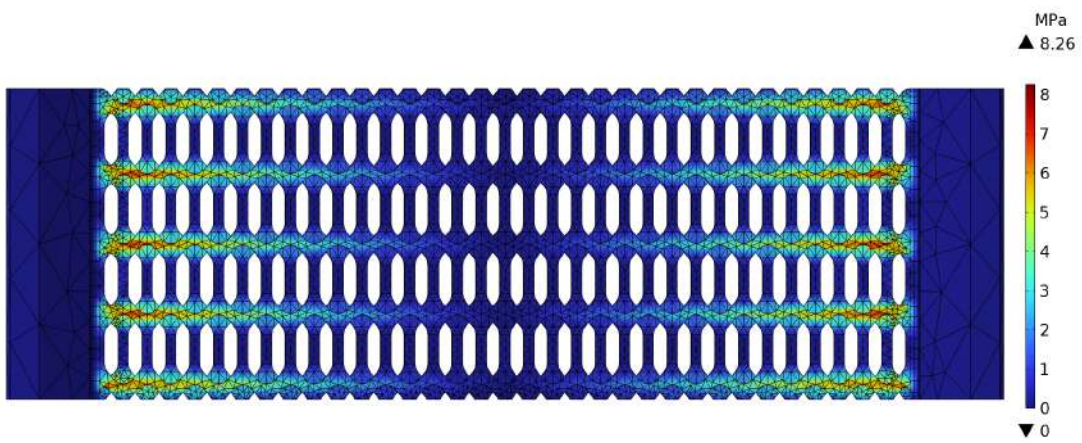

Figure 5.22: Von Mises stress distribution in the titanium blades at maximum displacement

A closer inspection reveals that stress intensification occurs primarily near the interfaces between the blades and the viscoelastic interlayer. These regions represent the locations where normal bending stresses in the titanium must be transferred into shear within the elastomer. The periodic nature of the stress pattern reflects the underlying honeycomb architecture, demonstrating that the geometric layout of the interlayer directly governs the spatial modulation of load transfer. This behavior highlights that the interlayer does not act as a uniform continuum, but rather as a mechanically structured medium that channels stresses through preferred paths.

Complementary information is provided by the equivalent deviatoric strain field shown in Figure 5.23. Unlike the stress distribution, which is dominated by the stiff metallic components, the deviatoric strain is almost entirely confined within the viscoelastic material. Elevated strain levels develop along the ligaments connecting adjacent honeycomb cells, particularly in regions aligned with the direction of relative blade motion. This localization clearly indicates that shear deformation within the elastomer constitutes the primary mechanism enabling compatibility between the bending blades.

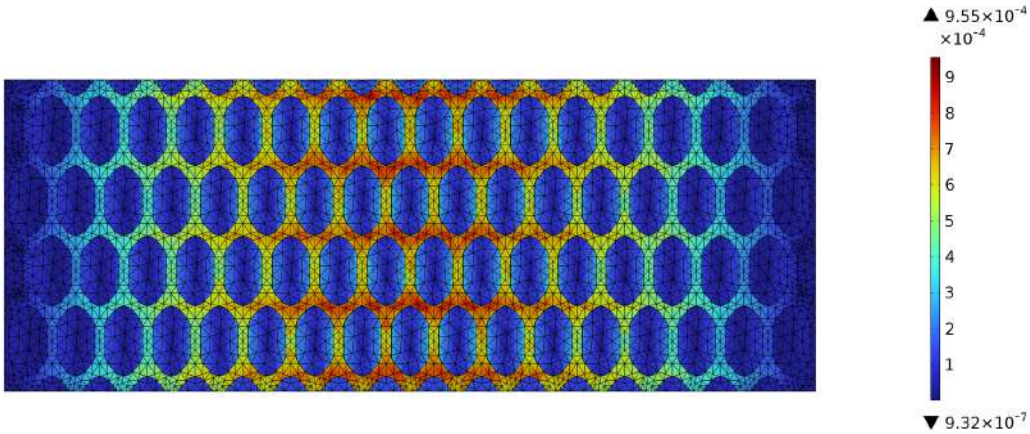

Figure 5.23: Equivalent deviatoric strain in the viscoelastic interlayer at t=4.5 s

The strong contrast between the stress-dominated response of the titanium blades and the strain-dominated response of the viscoelastic layer underscores the fundamentally different mechanical roles of the two materials. The blades behave as load-bearing elements, storing elastic energy associated with bending, whereas the viscoelastic interlayer functions mainly as a deformation mediator, accommodating relative motion through shear. Importantly, the strain field is highly non-uniform: only a subset of the viscoelastic volume experiences significant deviatoric deformation, while large regions remain weakly strained.

From a design perspective, this non-uniform strain engagement represents a key limitation of the current configuration. Since viscoelastic dissipation is directly linked to shear strain amplitude and strain-rate history, regions of the interlayer that remain weakly deformed contribute minimally to damping performance. The present strain distribution therefore suggests that the effectiveness of the viscoelastic layer is governed not by its total volume, but by the efficiency with which deformation is concentrated within mechanically active regions.

Further insight into the mechanical functioning of the blade assembly can be obtained by examining the energetic response of the viscoelastic interlayer. In particular, the spatial distributions of stored elastic energy density and viscoelastic dissipation rate density provide a direct link between local deformation mechanisms and the global damping behavior observed at the system level.

Figure 5.24 shows the elastic strain energy density stored within the viscoelastic material at t=4.5 s. The energy distribution closely reflects the deformation pattern induced by the bending of the titanium blades. Energy localization is not uniform throughout the elastomeric domain, but is strongly concentrated along narrow ligament regions connecting adjacent cavities of the honeycomb structure. These regions experience dominant shear deformation and therefore act as the primary carriers of elastic energy within the viscoelastic layer.

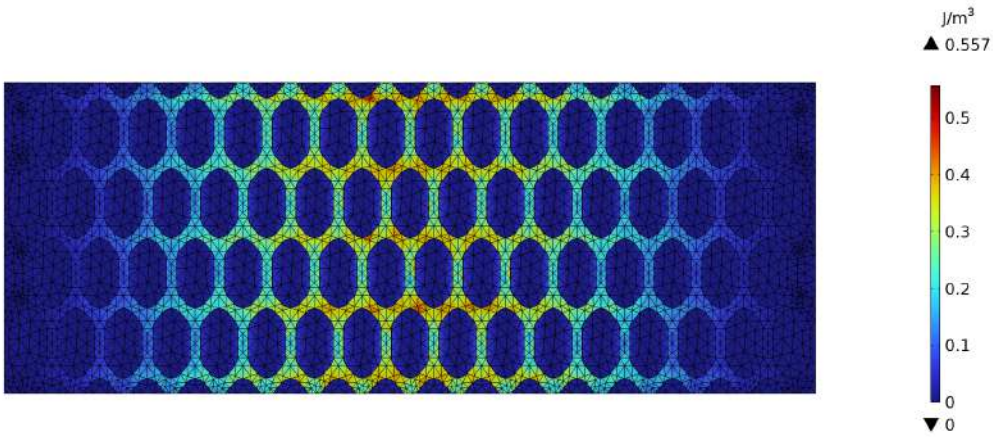

Figure 5.24: Spatial distribution of stored elastic energy density in the viscoelastic material at t=4.5 s

Several important features emerge from this map. First, the highest energy density does not occur within the bulk of the elastomer cells, but rather in the thin material bridges between them. Second, the spatial distribution of stored energy is highly periodic, mirroring the geometric repetition of the honeycomb pattern. This indicates that the local mechanical response is primarily geometry-driven, with each unit cell contributing similarly to the overall energy storage of the system.

While the stored energy density identifies where elastic energy accumulates during deformation, it does not directly quantify where energy is dissipated. For this purpose, Figure 5.25 reports the viscoelastic dissipation rate density computed at the same stage of the oscillation. The dissipation map reveals a distribution that is strongly correlated with the stored energy field, yet not identical. Regions of high stored energy generally coincide with elevated dissipation rates, confirming that dissipation is activated where cyclic shear strains are largest. However, the dissipation field exhibits sharper localization and more pronounced peaks, indicating that viscoelastic losses are particularly sensitive to strain-rate amplification rather than strain magnitude alone.

This distinction is physically significant. Whereas elastic energy reflects the instantaneous deformation state, the dissipation rate depends on the temporal evolution of that deformation through the internal variables of the generalized Maxwell model. As a result,

even moderate strain levels can produce substantial dissipation if associated with high strain rates, especially in regions where deformation gradients are spatially concentrated. The observed dissipation pattern therefore highlights the combined influence of geometry, kinematics, and material time-dependence. This confirms that the damping performance of the system is governed not by the total volume of viscoelastic material, but by the effectiveness with which the structural deformation activates shear-dominated mechanisms within these critical regions.

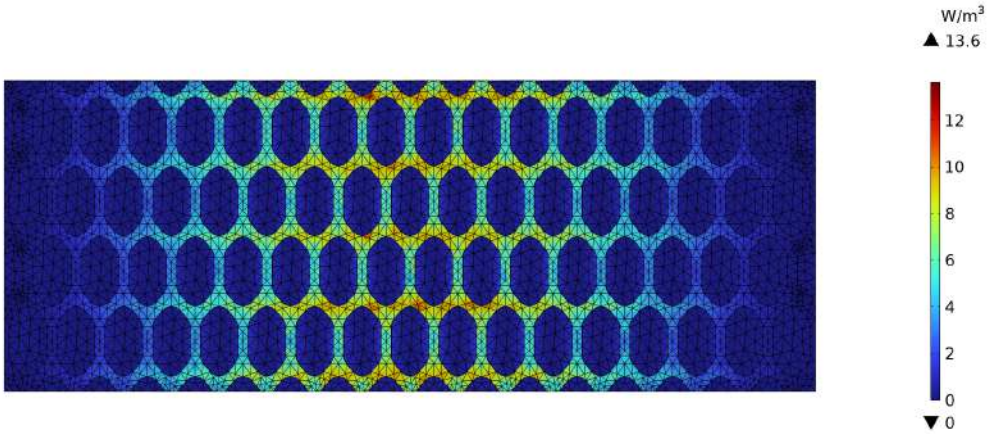

Figure 5.25: Viscoelastic dissipation rate density within the elastomeric core at t=4.5 s

From a design perspective, this observation carries important implications. Since both stored energy and dissipation are concentrated in specific geometric features of the honeycomb core, modifications aimed at increasing damping efficiency should focus on enhancing shear activation within these ligaments. Potential strategies include altering cell geometry, ligament thickness, or local compliance in order to increase strain amplitudes and strain-rate levels in the most dissipative regions, while avoiding unnecessary stiffening of the overall structure.

Building upon the kinematic, mechanical, and energetic analyses discussed in this section, the overall behavior of the blade assembly can now be interpreted in terms of its intrinsic design limitations and potential directions for optimization: the finite element results consistently indicate that the global dynamic response of the system is governed by a delicate balance between blade bending stiffness, viscoelastic shear activation, and geometric constraint imposed by the honeycomb architecture.

The displacement analysis demonstrated that the dominant deformation mode of the assembly is bending-driven, with the titanium blades undergoing opposite curvature and inducing relative motion across the viscoelastic interlayer. However, this relative motion is not uniformly converted into shear deformation within the elastomer. Instead, only specific regions of the honeycomb core, primarily the narrow ligaments between adjacent cavities, experience significant shear strains. Large portions of the viscoelastic material remain weakly engaged, undergoing limited deformation and therefore contributing marginally to energy dissipation.

The stress and strain analyses further clarified this behavior. While the titanium blades carry the majority of the bending-induced normal stresses, the viscoelastic layer is

primarily subjected to deviatoric deformation. Equivalent deviatoric strain maps reveal that shear activation is spatially localized and strongly geometry-dependent, rather than uniformly distributed across the interlayer. This confirms that the damping capability of the system is not controlled by the overall stiffness contrast between metal and polymer alone, but by the efficiency with which the structural configuration channels deformation into shear-dominated modes within the elastomer.

The energetic investigation provided a direct physical interpretation of these observations. Stored elastic energy density and viscoelastic dissipation rate density were both shown to concentrate in the same restricted regions of the honeycomb ligaments. These zones act as the true functional core of the damping mechanism, where energy is first accumulated during deformation and subsequently dissipated through time-dependent viscoelastic processes. Conversely, viscoelastic regions located in areas of low strain amplitude or low strain-rate contribute negligibly to damping, regardless of their volume.

Taken together, these results highlight a fundamental design limitation of the current configuration: the majority of the viscoelastic material is geometrically present but mechanically underutilized. The damping performance is therefore not limited by the intrinsic material properties alone, but by the inability of the structural layout to uniformly activate the viscoelastic layer under bending excitation.

From a design optimization standpoint, this insight suggests that effective improvements should prioritize deformation transfer rather than simply increasing viscoelastic volume. Strategies aimed at enhancing damping efficiency should focus on amplifying shear strains within the elastomer, especially in regions already identified as energetically active. Potential approaches include modifying the honeycomb topology, reducing ligament thickness to increase local compliance, altering cell aspect ratios to promote shear-dominated deformation, or introducing geometric asymmetries that prevent overly uniform strain redistribution. Such modifications would increase both stored energy and dissipation rates locally, thereby improving damping effectiveness without significantly penalizing global stiffness or mass. At the same time, the analyses indicate that excessive coupling between the blades, whether due to overly stiff viscoelastic properties or geometric constraints, may suppress relative motion and reduce shear activation, ultimately limiting dissipation. An optimal design therefore lies between the two idealized extremes discussed previously: neither a fully decoupled system dominated by large relative sliding, nor a fully coupled configuration behaving as a monolithic beam. Instead, maximum damping is achieved when controlled relative motion is preserved and efficiently converted into localized shear deformation within the viscoelastic interlayer.

In this context, the present numerical framework provides not only a validation tool, but also a physically interpretable design platform. By combining displacement fields, stress–strain distributions, and energy-based metrics, the model enables the identification of critical regions and governing mechanisms that directly inform design decisions. The insights obtained in this chapter thus establish a clear foundation for future optimization efforts, where geometry, material properties, and interface modeling can be systematically tuned to enhance damping performance while maintaining structural integrity and manufacturability.

§ 6 Conclusion

This chapter concludes the present work by synthesizing the main experimental and numerical findings, critically discussing the limitations of the adopted methodologies, and outlining perspectives for future developments. The objective is to place the obtained results within a broader engineering context, clarifying both the validated contributions of this study and the remaining challenges associated with modeling, characterization, and practical implementation. Finally, the applicability of the proposed damping concept to alternative geometries and real-world compliant systems is discussed, with particular emphasis on its relevance for precision engineering and watchmaking applications.

6.1 *Summary of Key Findings*

This work investigated the feasibility, performance, and modeling of additively manufactured damped flexure blades obtained by impregnating titanium lattice structures with a viscoelastic elastomer. The experimental campaign demonstrated that elastomer-impregnated lattice flexures constitute an effective and repeatable solution for introducing passive damping into compliant mechanisms while preserving their intrinsic advantages, such as frictionless motion and geometric simplicity. Dynamic Mechanical Analysis enabled a detailed characterization of the frequency-dependent viscoelastic behavior of the selected silicone material, providing the basis for implementing a generalized Maxwell formulation within a fully three-dimensional finite element model. Free oscillation experiments revealed highly repeatable responses across multiple blade pairs, with a dominant low-frequency mode exhibiting linear viscous damping behavior. The close agreement between damping estimates obtained from time-domain logarithmic decrement methods and frequency-domain half-power bandwidth analyses confirmed that, within the investigated amplitude range, the system can be accurately described by a linear SDOF approximation. Forced oscillation tests further extended the experimental validation over a wide frequency range, up to 750 Hz, highlighting the persistence of damping effects under harmonic excitation and enabling the identification of multiple resonant contributions.

A comparison between experimental results and numerical simulations showed good agreement in terms of resonance frequencies and qualitative damping trends; however, a quantitative mismatch remained, with the predicted quality factor differing from experimental values by approximately 30%. Extensive sensitivity analyses on geometrical parameters, viscoelastic properties, Poisson's ratio, mesh refinement, and energy dissipation mechanisms demonstrated that no single parameter uncertainty could account for this discrepancy. Aging effects of the viscoelastic material were clearly observed through DMA measurements, with progressive stiffening and reduction of loss factor over time, yet these effects alone were insufficient to explain the observed FEM–experiment mismatch. The results instead suggest that additional physical mechanisms not captured in the current numerical model, such as partial delamination at the metal–elastomer interface, microslip, or frictional dissipation, may play a significant role in the overall damping behavior. Furthermore, a frequency shift dependent on oscillation amplitude was experimentally observed, providing a plausible explanation for the broadening of resonance peaks identified in forced oscillation tests and indicating the presence of weak nonlinear effects.

6.2 *Limitations*

Despite the overall robustness of the results, the present work is subject to several experimental, numerical, and conceptual limitations that must be carefully acknowledged in order to properly frame the validity and scope of the conclusions. From an experimental standpoint, the identification of damping parameters at higher frequencies is inherently limited by a progressive reduction in response amplitude, signal-to-noise ratio, and coherence, particularly in forced-oscillation tests. As frequency increases, the measured frequency response functions increasingly reflect the dynamics of the coupled blade–fixture–actuator system rather than those of the flexure blades alone, making it difficult to unambiguously attribute observed resonances and dissipation levels to intrinsic blade behavior. Although non-contact laser vibrometry minimizes mass loading and mechanical interference, the measurement chain remains sensitive to alignment accuracy, environmental disturbances, and signal processing choices, such as filtering and numerical integration, which introduce unavoidable uncertainty in the extracted modal parameters already widely discussed in Section 4.3.

From a numerical perspective, the finite element model successfully captures the dominant dynamic trends observed experimentally; however, it systematically underestimates the energy dissipation measured during free oscillation tests, particularly with respect to its dependence on oscillation amplitude. This discrepancy persists even when adopting refined mesh configurations, indicating that numerical discretization effects alone are insufficient to explain the observed mismatch between simulations and experiments. A fundamental limitation lies in the modeling strategy adopted for the viscoelastic material. Although the generalized Maxwell formulation accurately reproduces the frequency-dependent behavior identified through DMA measurements, it inherently assumes homogeneous material properties, perfect interfacial bonding, and dissipation mechanisms governed exclusively by continuum viscoelasticity. In the present configuration, however, the elastomer is confined within a complex additively manufactured lattice and interacts with metallic surfaces characterized by intrinsic roughness, local geometric irregularities, and potential adhesion variability. These interface-level phenomena can introduce additional energy dissipation mechanisms, such as microslip, partial debonding, or frictional interactions at the metal–elastomer interface, which are not represented within a standard viscoelastic constitutive framework. As a result, the numerical model is intrinsically limited in its ability to reproduce the full amplitude-dependent damping observed experimentally, despite accurately capturing the global dynamic response.

At a conceptual level, the interpretation of damping parameters relies on simplified modeling assumptions that, while strongly supported by the data, remain approximations of a more complex reality. The free-oscillation analysis assumes linear viscous damping and single-degree-of-freedom behavior, which is valid for the dominant low-frequency mode but becomes increasingly questionable as higher-order modes and frequency-dependent viscoelastic effects emerge. Similarly, the damping parameters extracted from forced-response measurements represent effective values that depend on excitation conditions and experimental configuration, rather than intrinsic material properties. These limitations do not undermine the validity of the experimental observations or the numerical framework developed in this thesis; rather, they highlight the intrinsic com-

plexity of damped compliant systems and define clear boundaries within which the current results should be interpreted. Addressing these limitations requires more advanced interface modeling, refined experimental diagnostics, and targeted parametric studies, which are proposed as directions for future work.

6.3 *Future Work*

The results presented in this thesis open several promising directions for future work aimed at consolidating the proposed damping concept and extending its applicability toward robust engineering deployment. A first and essential step concerns the investigation of the long-term mechanical reliability of elastomer-impregnated lattice flexures through dedicated fatigue testing. While the present study focused on dynamic response and damping characterization, compliant mechanisms are often subjected to a large number of cyclic loading events over their operational lifetime. Systematic fatigue experiments, conducted under representative loading amplitudes and frequencies, would allow the assessment of damage accumulation, stiffness degradation, and potential changes in damping behavior over time, thereby providing critical data for lifetime prediction and safe design limits.

In parallel, the development of an analytical sizing model for damped flexure blades represents a key opportunity to complement the numerical framework established in this work. Such a model would aim to relate blade geometry, lattice parameters, and elastomer properties to global stiffness, natural frequency, and damping performance using closed-form or semi-analytical expressions. By extending classical beam theory and homogenization approaches to account for lattice compliance and viscoelastic shear dissipation, an analytical tool could provide rapid preliminary design guidance, reduce reliance on computationally intensive finite element simulations, and enable systematic trade-off studies during early design stages. This approach would be particularly valuable for defining admissible stroke, stress levels, and fatigue margins while simultaneously targeting specific damping and frequency requirements.

From a numerical standpoint, improvements to the finite element model remain a key objective. Future developments should aim at enhancing correlation between simulations and experiments by incorporating more advanced modeling strategies, including nonlinear viscoelastic behavior, interface elements, or contact-based formulations capable of representing frictional dissipation. Introducing amplitude-dependent material laws or cohesive interface models could significantly improve the prediction of damping levels and frequency shifts observed experimentally. Moreover, further efforts are required to improve the numerical reproduction of features observed during forced oscillation tests that are not captured by the current eigenfrequency analysis, including the experimentally observed response around 38 Hz. One promising direction consists in modeling the complete two-blade parallelogram configuration rather than a single flexure blade, thereby accounting for coupling effects, symmetry conditions, and load transfer mechanisms that may influence the forced dynamic behavior. Such an approach would enable a more realistic representation of the experimental setup and could improve correlation across the investigated frequency range.

Finally, improvements in the identification of multiple resonant modes during forced oscillation tests should be pursued. Refinement of multi-modal fitting strategies, improved

excitation schemes, and enhanced signal processing techniques could enable more accurate extraction of damping parameters across a broader frequency range, strengthening the experimental–numerical correlation and extending the applicability of the proposed methodology.

Taken together, these future research directions aim to transform the present framework from a validated proof of concept into a predictive and optimized design methodology for damped compliant mechanisms, fully leveraging the combined potential of additive manufacturing, viscoelastic materials, and advanced dynamic modeling.

6.4 *Applicability to Other Shapes and Configurations*

Although the present work focuses on a specific flexure blade geometry, the proposed damping concept is not intrinsically linked to a particular shape. The underlying dissipation mechanism relies on shear deformation of a viscoelastic material embedded within an additively manufactured metallic lattice, rather than on the global bending behavior of the flexure itself. As a result, the approach can be extended to a broad class of compliant mechanisms, provided that relative deformation within the structure induces shear strains in the impregnated elastomer. From this perspective, the concept is applicable to alternative flexure geometries, monolithic compliant structures, and flexure-based pivots, independently of their overall form.

This generality makes the approach particularly relevant to watchmaking applications, where compliant mechanisms are increasingly adopted to reduce friction, wear, and dependence on lubrication. In high-precision mechanical watches, flexure-based components offer excellent repeatability and long-term stability, but their inherently low damping makes them highly sensitive to external vibrations and shocks. Transient excitations can lead to prolonged oscillations and ringing, potentially affecting dynamic stability and timekeeping performance. Introducing passive damping without compromising precision therefore represents a key challenge in this field.

By integrating viscoelastic damping directly within the compliant structure through localized lattice regions, the proposed concept enables energy dissipation during dynamic excitation while preserving frictionless operation and monolithic design principles. The damping mechanism is activated primarily under cyclic deformation and does not significantly affect static positioning accuracy or quasi-static behavior, making it well suited for horological components operating at small amplitudes and low frequencies.

A particularly relevant and concrete extension of the present work concerns the approach developed at CSEM for watchmaking applications. FlexMEMS oscillators and pivots rely on monolithic silicon flexure structures to provide frictionless guidance, eliminating wear and lubrication while enabling highly stable kinematics. These characteristics have opened new possibilities in the design of advanced oscillators and escapements, as demonstrated by several FlexMEMS-based concepts developed in recent years (Figure 6.1). Despite these advantages, FlexMEMS structures inherently exhibit very low internal damping due to their monolithic elastic nature. As a consequence, they remain sensitive to external shocks and transient excitations, which can induce prolonged oscillations and dynamic ringing. This behavior is particularly critical in horological applications, where

robustness to handling shocks and environmental disturbances is a key requirement, especially for oscillators operating at relatively low amplitudes and high stiffness compared to classical balance-spring systems.

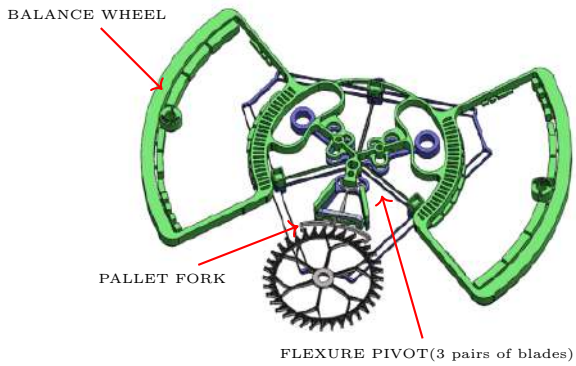

Figure 6.1: A simple escapment for FlexMEMS Oscilators (CSEM PATENT EP21200709.0)

The damping strategy investigated in this thesis provides a natural pathway to address this limitation without compromising the fundamental principles of the FlexMEMS approach. By locally integrating a lattice structure impregnated with a viscoelastic material within selected compliant regions of a FlexMEMS pivot or oscillator, controlled passive damping could be introduced through shear deformation of the elastomer during angular motion. Importantly, the damping mechanism would primarily act during transient excitations, such as shocks or sudden perturbations, while having a negligible influence on quasi-static behavior and steady-state oscillation. This is consistent with the operational requirements of FlexMEMS-based oscillators, where precision, repeatability, and long-term stability must be maintained. Notably, the characteristic frequency range of FlexMEMS oscillators and their supporting structures falls within the domain experimentally characterized and numerically validated in this work, making the proposed extension directly compatible with the material models and modeling framework developed.

From a design standpoint, the proposed integration of viscoelastic damping within FlexMEMS structures could be tailored through lattice topology, elastomer volume fraction, and material selection, allowing a trade-off between damping efficiency and stiffness to be optimized for specific horological requirements. This extension suggests a realistic and manufacturable evolution of existing FlexMEMS concepts, enhancing their robustness to dynamic disturbances while fully preserving the advantages that make flexure-based oscillators attractive for high-end watchmaking applications.

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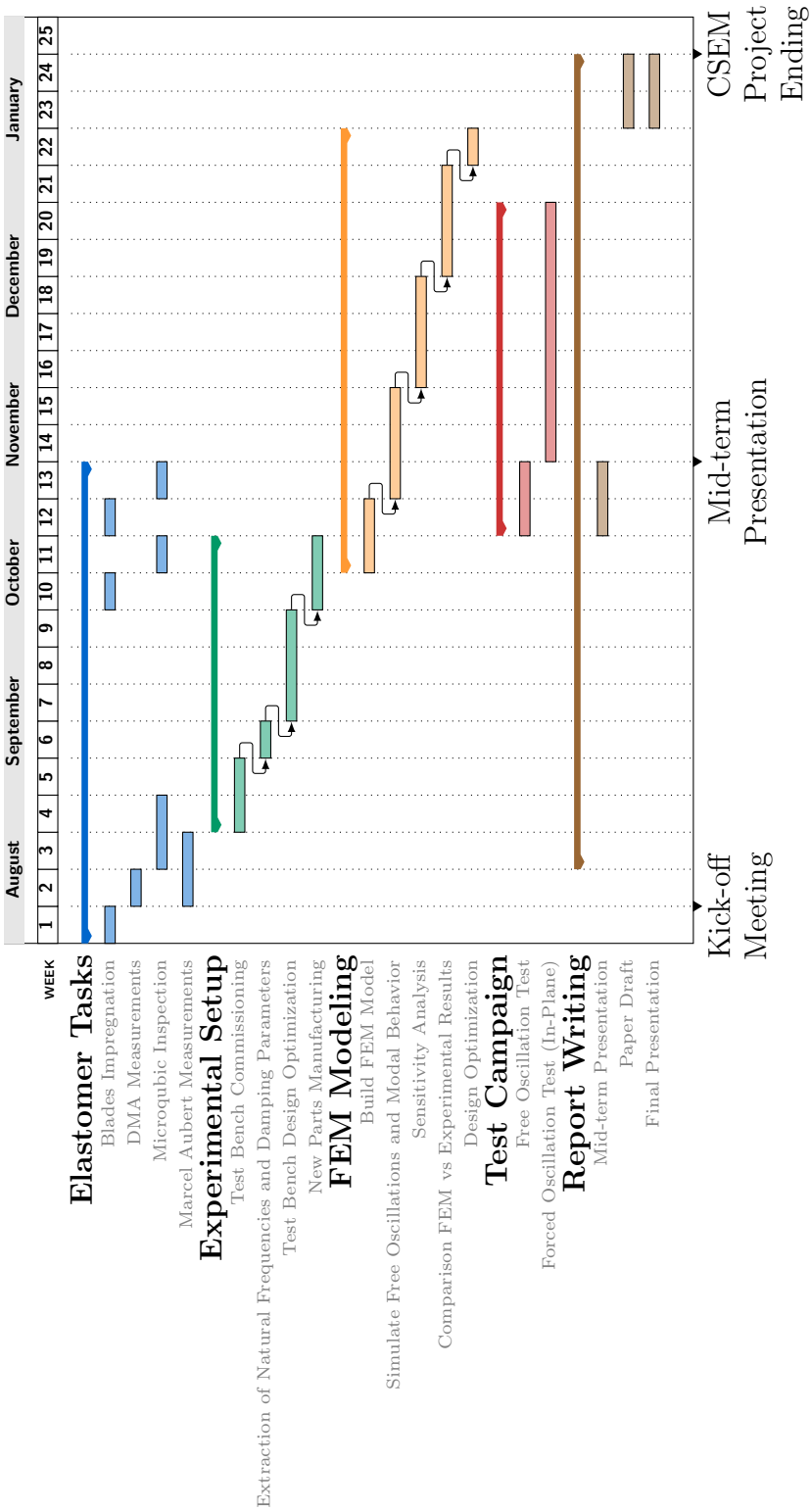
§ A Gantt Chart



§ B Technical Drawings

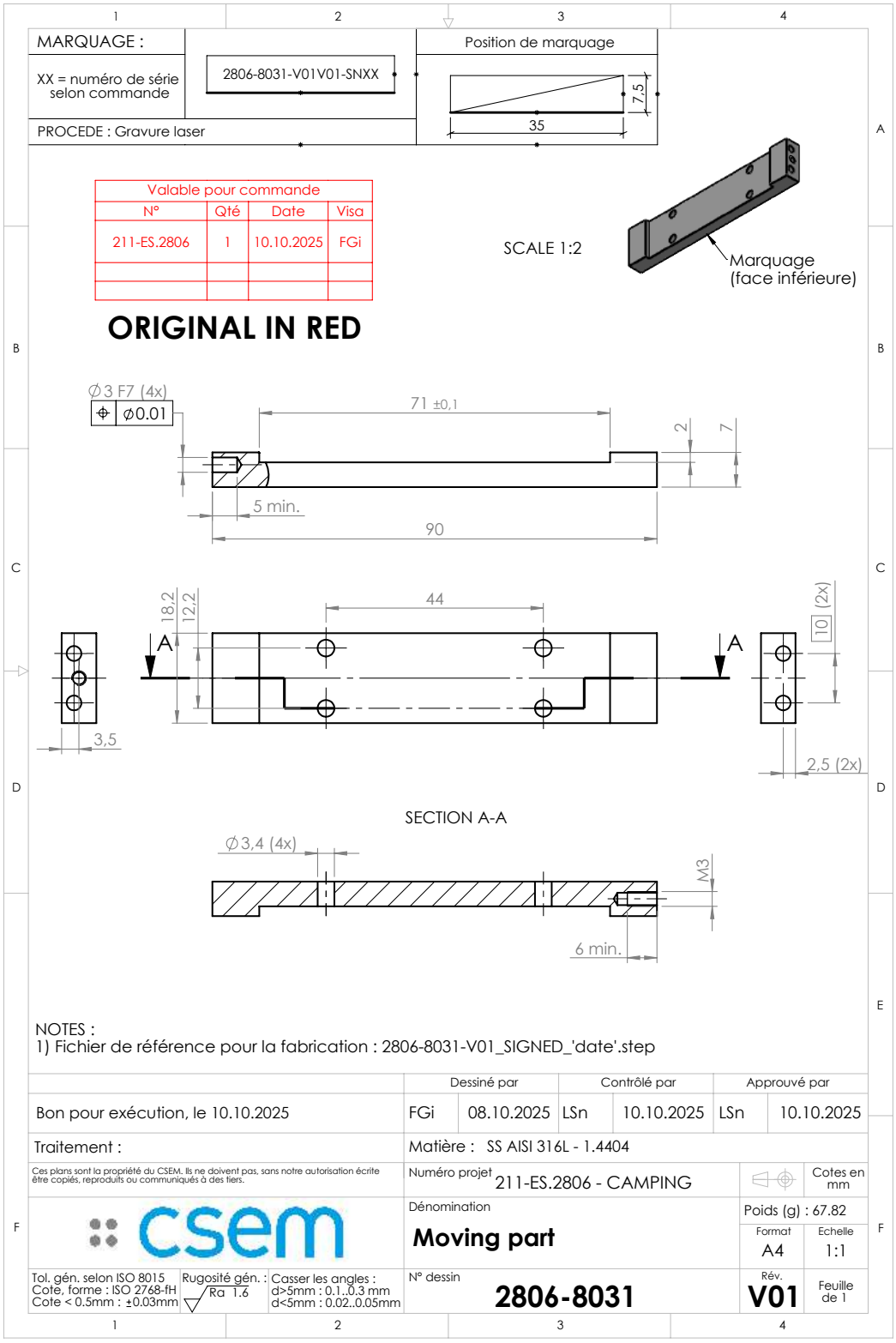


Figure B.1: Moving Part

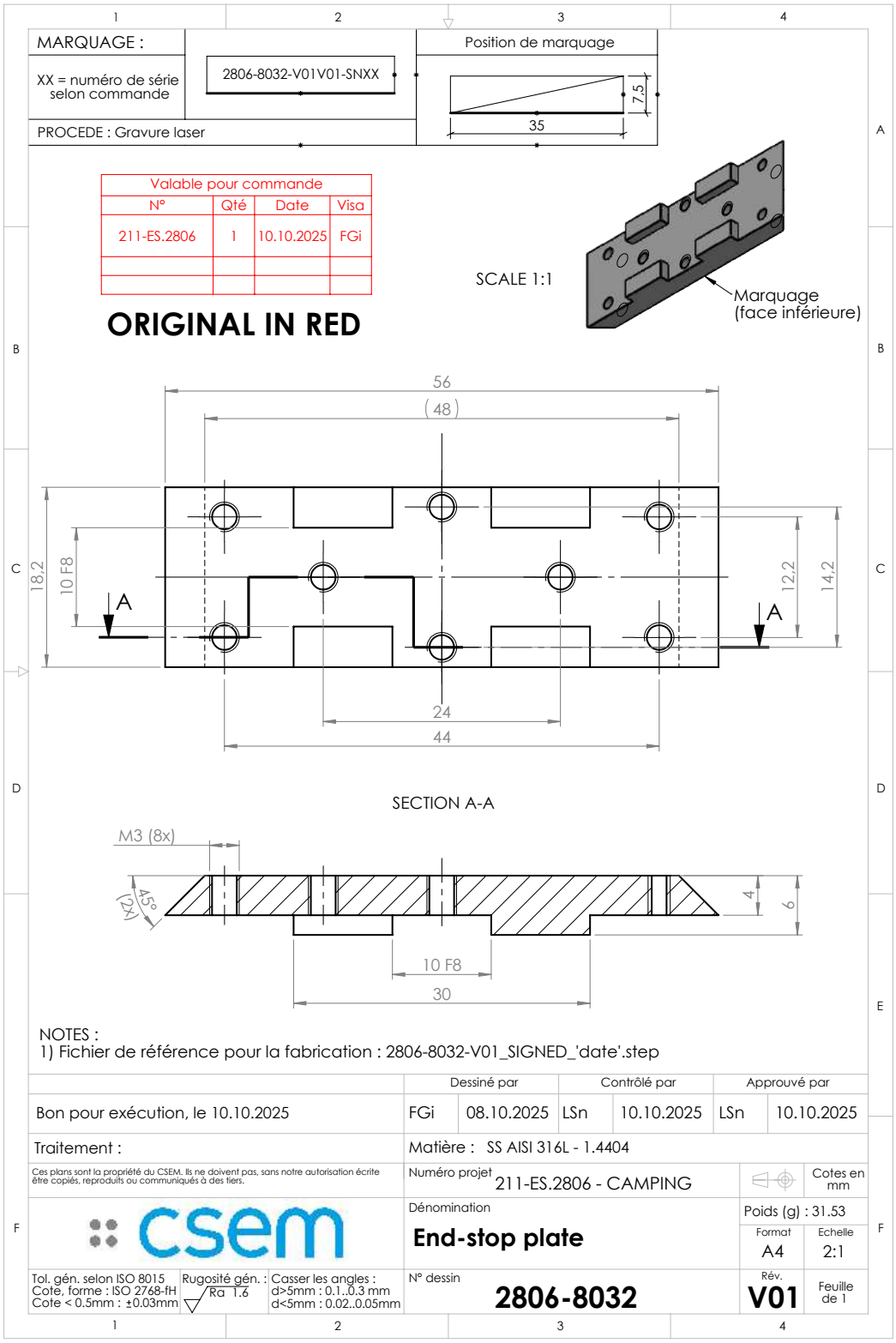


Figure B.2: End-stop Plate

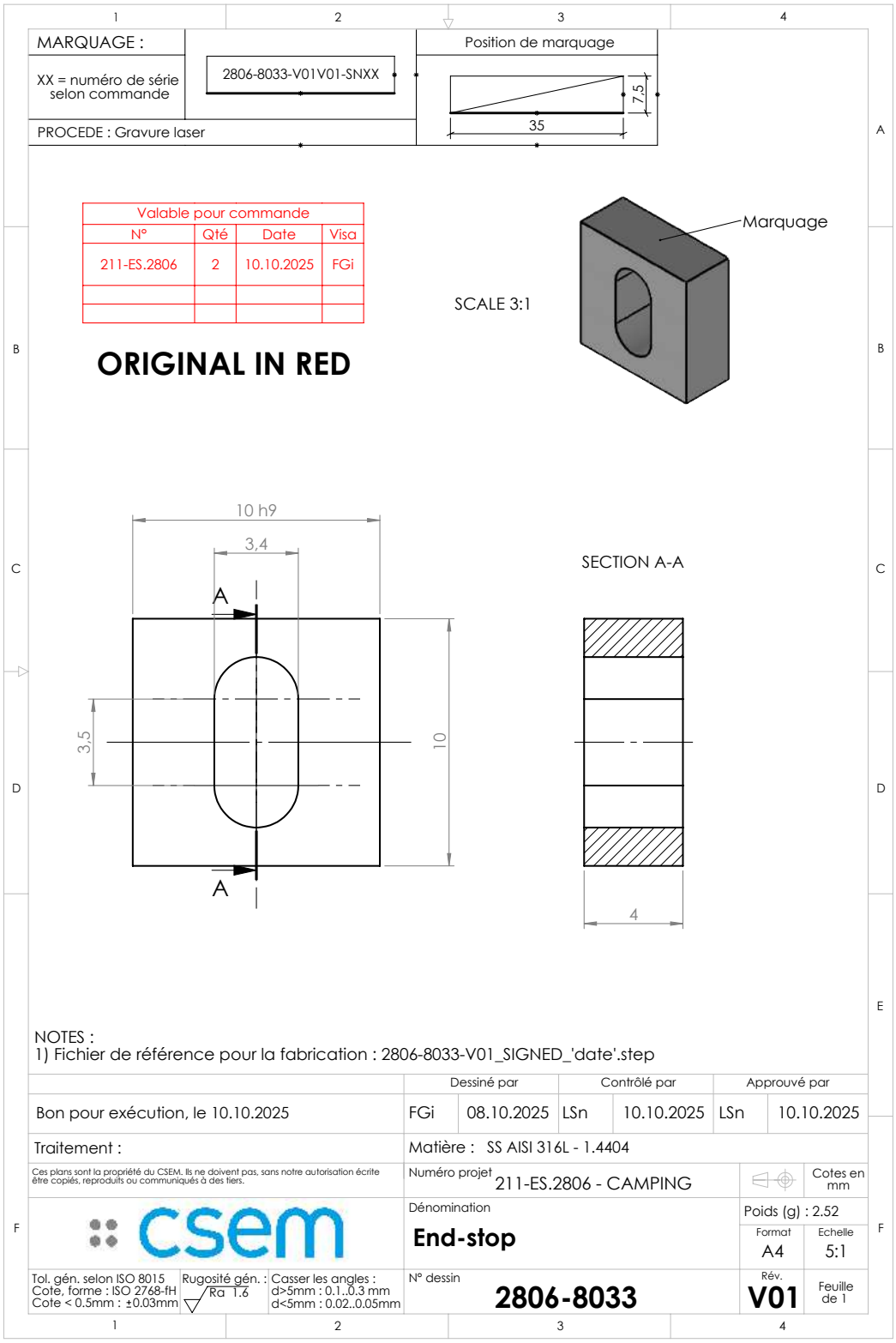


Figure B.3: End-stops

§ C Forced Oscillation Test Setup

The experimental setup required establishing a coherent and reliable measurement chain between the Medallion-II shaker controller, the Kistler accelerometer equipped with its charge/voltage amplifier, and the Polytec Vibrometer operated through VibSoft 6.2 software. Because the test bench did not provide a threaded mounting point or any dedicated mechanical interface, the accelerometer was rigidly bonded to the bench using a suitable adhesive. This ensures a stiff mechanical coupling, allowing the accelerometer to accurately track the inertial motion of the bench while minimizing any influence on the dynamic response of the flexure. After installation, the accelerometer was connected to its amplifier, and the amplifier's configuration parameters; specifically, the measurement range (±5 g) and sensitivity (1002 mV/g), were set using the Kistler LabAmp software. To access this configuration menu, the amplifier was connected to a laptop through an Ethernet-to-USB cable. Setting these parameters ensures that the amplifier's output voltage correctly represents the physical acceleration measured by the sensor.

Because the accelerometer has a nominal sensitivity of approximately 1 V per g, and 1 g corresponds to 9.81 m/s^2, the sensor's full measuring range of ±5 g corresponds to a physical acceleration range of ±49.05 m/s^2. The amplifier and the DAQ channel were configured with a ±10 V output/input range, meaning that the electrical span representing the entire ±5 g range is 20 V peak-to-peak (i.e., ±10 V). Therefore, the conversion factor between voltage and acceleration is obtained by dividing the full physical range by the full voltage range:

$$\frac{49.05\,\mathrm{m/s^2}}{10\,\mathrm{V}} = 4.905\,\frac{\mathrm{m/s^2}}{\mathrm{V}}$$

This explains why VibSoft 6.2 requires a scaling factor of 4.905 m/s^2/V: it directly reflects the ratio between the sensor's acceleration measuring range (set on the amplifier) and the ±10 V analog output range delivered to the DAQ.

Once the amplifier was calibrated, its two voltage outputs were routed to the rest of the system. One output was fed into the Medallion-II controller, which subsequently drives the PA-500L shaker power amplifier responsible for generating the physical excitation. The second output of the amplifier was connected to the vibrometer data-acquisition module via the "REF" input channel. Simultaneously, the "VELO" input was connected to the vibrometer controller's "CH1 OUT" port, which provides the velocity signal generated internally by the vibrometer; for completeness, "CH2 OUT" corresponds to the internally generated displacement signal and "CH3 OUT" corresponds to the acceleration signal. Routing these channels through the DAQ ensures that externally acquired signals (such as the accelerometer voltage) are perfectly time-aligned with the laser-derived velocity measurement, allowing synchronized acquisition and reliable computation of relative motion.

After completing the connections, the shaker chain was powered, and the PA-500L amplifier was set to "Control" mode. Once the over-current protection indicator reset to zero, the operating mode was switched to "Voltage", enabling the shaker to respond directly to the drive signal programmed in Medallion-2. Figure C.1 provides a complete overview of the final cabling layout, showing how accelerometer, amplifier, Vibrometer, and shaker controller are interconnected within the measurement chain.

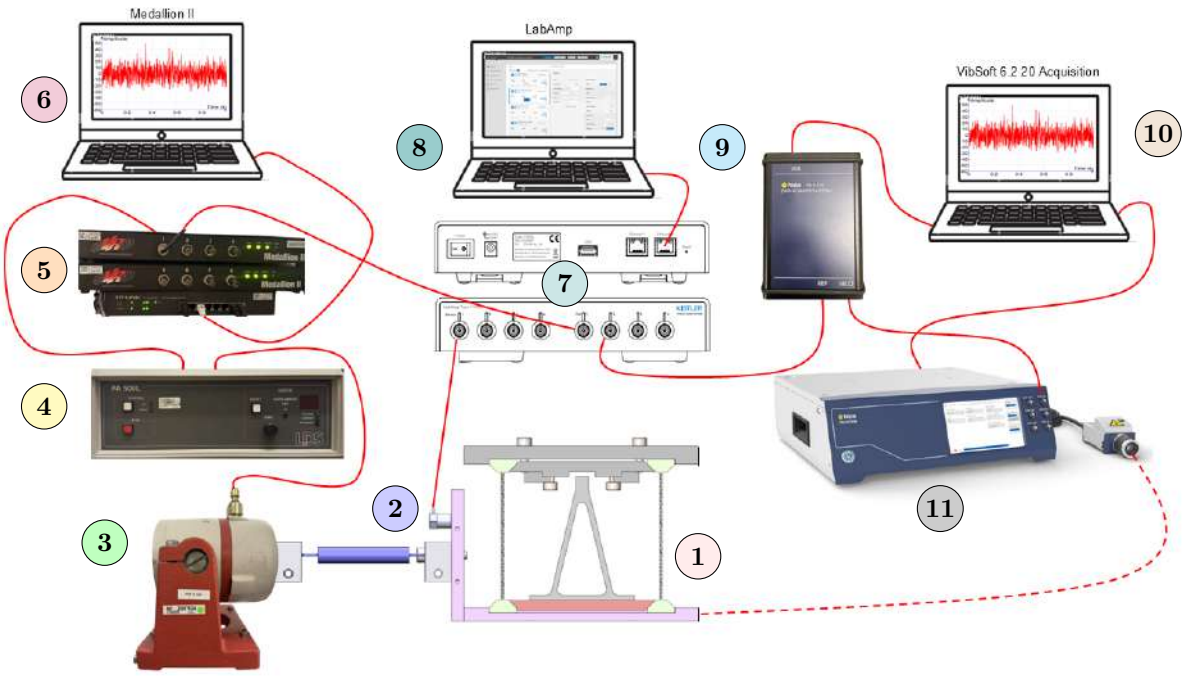


Figure C.1: Cabling Layout for Forced Oscillation Test

Table C.1: Legend of Figure C.1

1	Test Bench
2	Kistler Accelerometer
3	Shaker
4	PA-500L Amplifier
5	Shaker Controller
6	Shaker Software
7	Kistler Charge Amplifier
8	Kistler Charge Amplifier Software
9	Polytec Data Acquisition System (DAQ)
10	Polytec Software
11	Polytec Vibrometer

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